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ENDOSCOPIC MAGNETIC ANASTOMOSIS SYSTEM, DEVICES AND METHOD

Abstract

An endoscopic magnetic anastomosis system, devices and method for forming an anastomosis. The endoscopic magnetic anastomosis system includes an endoscopic assembly, a video processor console, a magnet detector, and a magnetic navigation console. The endoscopic assembly includes an endoscope, an adjustable snare mechanism, an outer sheath, and a magnetic implant assembly. The adjustable snare mechanism includes a snare assembly. The snare assembly passes through a snare channel in the endoscope and includes a snare catheter and a snare wire passing through the snare catheter. The snare is used to selectively tighten and release the magnetic implant assembly disposed at one end of the endoscope.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation-in-part of U.S. application Ser. No. 18/015,172 filed on Jan. 9, 2023, which is a 371 U.S. national phase application of the PCT Application No. PCT/CN2020/100662 filed on Jul. 7, 2020. The above-identified applications are hereby incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates generally to a robotic system, devices, and method for forming an anastomosis, and more particularly, to an endoscopic magnetic anastomosis system and devices for forming an anastomosis.

BACKGROUND

[0003] Diabetes is emerging as a primary cause of mortality, morbidity, disability, and discrimination in health care, education and employment. The International Diabetes Federation (IDF) estimates nearly half a billion people worldwide are presently living with diabetes. It is estimated that by 2045, the global prevalence will increase by 48%, swelling to an estimated 693 million individuals (aged 18-99 years) afflicted.

[0004] The socioeconomic burden associated with these conditions is immense, most of which can be attributed to Diabetes. Hyperglycemia, a high level of blood glucose, is the hallmark of diabetes. In Type 2 Diabetes (T2DM), hyperglycemia results from a varying combination of insulin resistance and inadequate insulin production. Chronic hyperglycaemia can cause damage to various organs, leading to the development of disabling and life-threatening complications such as cardiovascular disease, neuropathy, nephropathy, and eye disease leading to retinopathy and blindness. These complications contribute to more frequent hospitalization, increased medical care costs and lower quality of life, and often results in early death.

SUMMARY

[0005] Present example embodiments relate generally to and/or comprise systems, subsystems, processors, devices, logic, and methods for addressing conventional problems, including those described above.

[0006] In an exemplary embodiment, an endoscopic anastomosis system is described. The endoscopic anastomosis system includes a first main body assembly. The first main body assembly is an elongated body having a first end and a second end. At least a portion of the second end of the first main body assembly is controllable to bend in a plurality of directions. The endoscopic anastomosis system includes a head assembly. The head assembly includes a first end and a second end. The head assembly includes a head assembly body and a first expandable member. The head assembly body includes a first region and a second region. The first region includes a first end, a second end, and a midsection between the first and second ends of the first region. The first end of

the first region is secured to the second end of the first main body assembly. The second end of the first region includes a first section and a second section. The first section is secured to a first end of a second region of the head assembly. The second section includes a second main body assembly opening. The second region includes the first end and a second end. The second end of the second region includes a first pressure port. The first pressure port is configured to apply a negative pressure. The first expandable member is secured to at least a portion of the midsection of the first region of the head assembly body. The first expandable member is configured to expand radially away from the first region of the head assembly body. The endoscopic anastomosis system includes a second main body assembly. The second main body assembly is an elongated body having a first end and a second end. At least a portion of the second end of the second main body assembly is provided in and moveable through the second main body assembly opening of the second section of the first region of the head assembly body. The second main body assembly includes a second expandable member and a second pressure port. The second expandable member is secured to a portion of the second end of the second main body assembly. The second expandable member is configured to expand radially away from the second main body assembly. The second pressure port is configured to apply a negative pressure. The endoscopic anastomosis system includes a magnetic implant assembly. The magnetic implant assembly includes a magnetic body. The magnetic body is formed as a substantially flat body. The magnetic body includes a front wall, a rear wall opposite to the front wall, and an exterior circumferential sidewall. The front wall has a circular shape with a central axis. The exterior circumferential sidewall is formed around the magnetic body. The exterior circumferential sidewall defines a thickness of the magnetic body. The exterior circumferential sidewall is formed at a first radius from the central axis. The endoscopic anastomosis system includes a securing assembly. The securing assembly is secured to the second end of the second main body assembly. The securing assembly is actuatable between a locked configuration and an unlocked configuration. The locked configuration is a configuration in which the magnetic implant assembly is secured to the securing assembly. The unlocked configuration is a configuration in which the magnetic implant assembly is not secured to the securing assembly.

[0007] In another exemplary embodiment, a catheter system for an endoscopic anastomosis system is described. The endoscopic anastomosis system includes a main body and a head assembly. The head assembly is secured to a distal end of the main body. The head assembly includes a head assembly body, a first expandable member, and a first pressure port. The head assembly body includes a first region and a second region. The first region includes a first end and a second end. The first end of the first region includes a first section and a second section. The first section of the first region is secured to the second region. The second section of the first region includes a catheter body opening for at least a portion of the catheter assembly to be provided in and moveable through. The first expandable member is secured to a portion of the first region between the first and second end of the first region. The first pressure port is provided in the second region. The catheter assembly includes a catheter body, a magnetic implant assembly, and a securing assembly. The catheter body is an elongated body. The catheter body includes a first end and a second end. At least a portion of the second end of the catheter body is provided in and moveable through the catheter body opening of the first region of the head assembly body. The catheter body includes a second expandable member and a second pressure port. The second expandable member is secured to a portion of the second end of the catheter body. The second expandable member is configured to expand radially away from the catheter body. The second pressure port is configured to apply a negative pressure. The magnetic implant assembly includes a magnetic body. The magnetic body is formed as a substantially flat body. The magnetic body includes a front wall, a rear wall opposite to the front wall, and an exterior circumferential sidewall. The front wall has a circular shape with a central axis. The exterior circumferential sidewall is formed around the magnetic body. The exterior circumferential sidewall defines a thickness of the magnetic body. The exterior circumferential sidewall is formed at a first radius from the central axis. The securing

assembly is secured to the second end of the catheter body. The securing assembly is actuatable between a locked configuration and an unlocked configuration. The locked configuration is a configuration in which the magnetic implant assembly is secured to the securing assembly. The unlocked configuration is a configuration in which the magnetic implant assembly is not secured to the securing assembly.

[0008] In another exemplary embodiment, magnetic implant assembly for an endoscopic anastomosis system is described. The endoscopic anastomosis system includes a main body and one or more protruding portions formed on the main body. The main body includes a first end and a second end. The main body includes a securing assembly secured to the second end of the main body. The securing assembly is actuatable between a locked configuration and an unlocked configuration. The locked configuration is a configuration in which the magnetic implant assembly is secured to the securing assembly. The unlocked configuration is a configuration in which the magnetic assembly is not secured to the securing assembly. The magnetic implant assembly comprises a magnetic body. The magnetic body is formed as a substantially flat body. The magnetic body includes a front wall, rear wall opposite to the front wall, and an exterior circumferential sidewall formed around the magnetic body. The front wall includes a circular shape with a central axis. The exterior circumferential sidewall is formed around the magnetic body. The exterior circumferential sidewall defines a thickness of the magnetic body. The exterior circumferential sidewall is formed at a first radius from the central axis. The one or more protruding portions is formed on the front wall of the magnetic body. The one or more protruding portions includes a first ring shaped protrusion formed on the front wall of the magnetic body. The first ring shaped protrusion is centrally aligned with the central axis. The first ring shaped protrusion includes a first exterior radius from the central axis and a first interior radius from the central axis. The first exterior radius of the first ring shaped protrusion is less than the first radius. Alternatively or in addition, the first exterior radius of the first ring shaped protrusion is equal to the first radius and the magnetic implant assembly further includes an exterior ring shaped body. The exterior ring shaped body is formed around and fixedly secured to the exterior circumferential sidewall of the magnetic body. The exterior ring shaped body includes a front exterior ring shaped portion adjacent to the front wall of the magnetic body and a rear exterior ring shaped portion adjacent to the rear wall of the magnetic body. At least a portion of the front exterior ring shaped body is formed using a material other than a ferromagnetic or magnetic material.

[0009] In another exemplary embodiment, there is provided an endoscopic magnetic anastomosis system including at least one endoscopic assembly including: an endoscope having a first end and an opposite second end, the endoscope including a snare channel extending from the first end to the second end and a magnetic implant assembly disposed at the second end; and an adjustable snare mechanism including: a snare assembly passing through the snare channel and including a snare catheter and a snare wire passing through the snare catheter for selectively tightening and releasing the magnetic implant assembly.

[0010] In another exemplary embodiment, there is provided an outer sheath including: a body tube having a first end and an opposite second end; and a tube-locking mechanism mounted at the first end of the body tube, wherein the body tube further has an endoscopic channel extending from the first end of the body tube to the second end of the body tube for passage of the endoscope, the tube-locking mechanism is configured to be able to be in locking fit with the endoscope passing through the endoscopic channel to secure the endoscope with the body tube, and to disengage from being in locking fit with the endoscope passing through the endoscopic channel to enable the endoscope to slide and rotate relative to the body tube.

[0011] In another exemplary embodiment, there is provided a magnet detector for detecting a position of a magnet, the magnet detector including: a detector body; a first printed circuit board provided on a first end of the detector body and provided with a first group of magnetometers; a second printed circuit board provided on a second end of the detector body opposite to the first end,

and provided with a second group of magnetometers; and a processing unit configured to be able to receive measurement data from the first group of magnetometers and the second group of magnetometers and to determine a position of the magnet based on the received data.

[0012] In another exemplary embodiment, there is provided a magnetic navigation console for driving a magnet to move under action of a magnetic field, the magnetic navigation console including: a mounting bracket including a movable first mounting arm and a movable second mounting arm; a first magnetic actuator mounted to the first mounting arm; and a second magnetic actuator mounted to the second mounting arm, the first magnetic actuator and the second magnetic actuator are configured to be able to move to overlap in a vertical direction.

[0013] Other features and advantages of the present application will be set forth in the following specification.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] For a more complete understanding of the present disclosure, example embodiments, and their advantages, reference is now made to the following description taken in conjunction with the accompanying figures, in which like reference numbers indicate like features, and:

[0015] FIG. 1A is a perspective view illustration of an example embodiment of an endoscopic anastomosis system;

[0016] FIG. 1B is a side view illustration of an example embodiment of an endoscopic anastomosis system;

[0017] FIG. 1C is another side view illustration of an example embodiment of an endoscopic anastomosis system;

[0018] FIG. 2 is a perspective view illustration of an example embodiment of some of the elements of an endoscopic anastomosis system separated from one another;

[0019] FIG. 3A is a side view illustration of an example embodiment of a first main body assembly;

[0020] FIG. 3B is a side view illustration of an example embodiment of a first main body assembly and the first bendable section configured to bend;

[0021] FIG. 3C is another side view illustration of an example embodiment of a first main body assembly and the first bendable section configured to bend;

[0022] FIG. 4A is a perspective view illustration of an example embodiment of a head assembly with a magnetic implant assembly;

[0023] FIG. 4B is a perspective view illustration of an example embodiment of a head assembly without a magnetic implant assembly;

[0024] FIG. 4C is a side view illustration of an example embodiment of a head assembly with first expandable member not expanded;

[0025] FIG. 4D is a side view illustration of an example embodiment of a head assembly with first expandable member expanded;

[0026] FIG. 4E is a side view illustration of an example embodiment of a second main body assembly extended outward away from a head assembly;

[0027] FIG. 4F is a side view illustration of an example embodiment of a second main body assembly extended outward away from a head assembly, and a first expandable member configured to expand;

[0028] FIG. 4G is a side view illustration of an example embodiment of a second main body assembly extended outward away from a head assembly, and a second bendable section configured to bend;

[0029] FIG. 4H is a side view illustration of an example embodiment of a second main body

assembly extended outward away from a head assembly, a second bendable section configured to bend, and a second expandable member configured to expand;

[0030] FIG. 4I is a side view illustration of an example embodiment of a second main body assembly extended outward away from a head assembly and rotated so as to change an orientation of a magnetic implant assembly;

[0031] FIG. 5A is a perspective view illustration of an example embodiment of a magnetic implant assembly secured to a second main body assembly via a securing assembly;

[0032] FIG. 5B is a cross-sectional view illustration of an example embodiment of a magnetic implant assembly, a securing assembly, and a second main body assembly;

[0033] FIG. 5C is a perspective view illustration of another example embodiment of a magnetic implant assembly secured to a second main body assembly via a securing assembly;

[0034] FIG. 5D is a cross-sectional view illustration of another example embodiment of a magnetic implant assembly, a securing assembly, and a second main body assembly;

[0035] FIG. 5E is a perspective view illustration of another example embodiment of a magnetic implant assembly secured to a second main body assembly via a securing assembly;

[0036] FIG. 5F is a cross-sectional view illustration of another example embodiment of a magnetic implant assembly, a securing assembly, and a second main body assembly;

[0037] FIG. 5G is a perspective view illustration of another example embodiment of a magnetic implant assembly secured to a second main body assembly via a securing assembly;

[0038] FIG. 5H is a cross-sectional view illustration of another example embodiment of a magnetic implant assembly, a securing assembly, and a second main body assembly;

[0039] FIG. 6A is a cross-sectional view illustration of an example embodiment of a magnetic implant assembly and a second magnetic implant assembly;

[0040] FIG. 6B is a cross-sectional view illustration of an example embodiment of a magnetic implant assembly magnetically coupled to a second magnetic implant assembly;

[0041] FIG. 6C is a top view illustration of an example embodiment of a front wall of a magnetic implant assembly;

[0042] FIG. 6D is a top view illustration of another example embodiment of a front wall of a magnetic implant assembly;

[0043] FIG. 7A is a cross-sectional view illustration of another example embodiment of a magnetic implant assembly and a second magnetic implant assembly;

[0044] FIG. 7B is a cross-sectional view illustration of another example embodiment of a magnetic implant assembly magnetically coupled to a second magnetic implant assembly;

[0045] FIG. 7C is a top view illustration of an example embodiment of a front wall of a magnetic implant assembly;

[0046] FIG. 7D is a top view illustration of another example embodiment of a front wall of a magnetic implant assembly;

[0047] FIGS. 8A-S are illustrations of an example embodiment of a method of delivering a magnetic implant assembly rectally;

[0048] FIGS. 9A-I are illustrations of an example embodiment of a method of delivering a magnetic implant assembly orally;

[0049] FIG. 10 is a schematic structural diagram illustrating the usage state of an endoscopic magnetic anastomosis system according to an embodiment of the present application;

[0050] FIG. 11 is a schematic diagram of a structure of an endoscopic magnetic anastomosis system according to an embodiment of the present application;

[0051] FIG. 12A is a schematic structural diagram illustrating a process of using an adjustable snare mechanism of an endoscope magnetic anastomosis system according to an embodiment of the present application;

[0052] FIGS. 12B-12D are schematic cross-sectional structural diagrams illustrating a process of using an adjustable snare mechanism of an endoscopic magnetic anastomosis system according to

an embodiment of the present application;

[0053] FIG. **13A** is a schematic diagram of a structure of an outer sheath and an endoscope of an endoscope magnetic anastomosis system according to an embodiment of the present application, wherein a tube-locking mechanism is in a locked state;

[0054] FIG. **13B** is a schematic diagram of a structure of an outer sheath and an endoscope of an endoscope magnetic anastomosis system according to an embodiment of the present application, wherein the tube-locking mechanism is in an unlocked state;

[0055] FIG. **13C** is a schematic diagram of a structure of an outer sheath of an endoscopic magnetic anastomosis system according to an embodiment of the present application;

[0056] FIG. **13D** is a schematic diagram of a partial structure of a body tube of an outer sheath of an endoscopic magnetic anastomosis system according to an embodiment of the present application;

[0057] FIG. **13E** is a schematic diagram of a structure of a tube-locking mechanism of an outer sheath of an endoscope magnetic anastomosis system according to an embodiment of the present application;

[0058] FIG. **13F** is a partial cross-sectional schematic view of a structure of a distal end of an assembly structure of an outer sheath and an endoscope of an endoscope magnetic anastomosis system according to an embodiment of the present application;

[0059] FIG. **13G** is a partial cross-sectional schematic view of a structure of a proximal end of an assembly structure of an outer sheath and an endoscope of an endoscope magnetic anastomosis system according to an embodiment of the present application;

[0060] FIG. **13H** is a schematic cross-sectional structural diagram of an external pressure source of an outer sheath of an endoscopic magnetic anastomosis system according to an embodiment of the present application in different working states;

[0061] FIG. **13I** is a schematic cross-sectional structural diagram of a body tube of an outer sheath of an endoscopic magnetic anastomosis system according to an embodiment of the present application;

[0062] FIG. **14A** is a schematic diagram of a three-dimensional structure of a head assembly of an endoscope of an endoscope magnetic anastomosis system according to an embodiment of the present application;

[0063] FIG. **14B** is a schematic view of a front view structure of a head assembly of an endoscope of an endoscope magnetic anastomosis system according to an embodiment of the present application;

[0064] FIG. **14C** is a schematic diagram of a left side structure of a head assembly of an endoscope of an endoscope magnetic anastomosis system according to an embodiment of the present application;

[0065] FIG. **14D** is a schematic diagram of an assembly structure of a head assembly and a magnetic implant assembly of an endoscope of an endoscope magnetic anastomosis system according to an embodiment of the present application;

[0066] FIG. **14E** is a schematic diagram of a structure of a head assembly of an endoscope of an endoscopic magnetic anastomosis system in some cases;

[0067] FIG. **14F** is a schematic diagram of an assembly structure of a head assembly and a magnetic implant assembly of an endoscope of an endoscopic magnetic anastomosis system in some cases;

[0068] FIG. **15A** is a schematic diagram of an assembly structure of a video processor and a cable of an endoscope magnetic anastomosis system according to an embodiment of the present application;

[0069] FIG. **15B** is a schematic structural diagram illustrating an assembly process of a video processor and a cable of an endoscope magnetic anastomosis system according to an embodiment of the present application;

[0070] FIG. 15C is a schematic diagram of a three-dimensional structure of a main connector socket of a video processor of an endoscope magnetic anastomosis system according to an embodiment of the present application;

[0071] FIG. 15D is a schematic cross-sectional structural diagram of a main connector socket of a video processor of an endoscope magnetic anastomosis system according to an embodiment of the present application, wherein the main connector socket is in an initial state;

[0072] FIG. 15E is a schematic cross-sectional structural diagram of a main connector socket of a video processor of an endoscope magnetic anastomosis system according to an embodiment of the present application, wherein the main connector socket is in a locked state;

[0073] FIG. 15F is a schematic cross-sectional structural diagram of a main connector socket of a video processor of an endoscope magnetic anastomosis system according to an embodiment of the present application;

[0074] FIG. 15G is a schematic diagram of a structure of a main connector of a cable of an endoscope magnetic anastomosis system according to an embodiment of the present application;

[0075] FIG. 15H is a schematic cross-sectional structural diagram of a main connector socket of a video processor and a main connector of a cable of an endoscope magnetic anastomosis system according to an embodiment of the present application before plug-in connection;

[0076] FIG. 15I is a schematic cross-sectional structural diagram of a main connector socket of a video processor and a main connector of a cable of an endoscope magnetic anastomosis system according to an embodiment of the present application during plug-in connection;

[0077] FIG. 16A is a schematic diagram of a three-dimensional structure of a magnetic implant assembly of an endoscope of an endoscope magnetic anastomosis system according to an embodiment of the present application;

[0078] FIG. 16B is another schematic diagram of a three-dimensional structure of a magnetic implant assembly of an endoscope of an endoscope magnetic anastomosis system according to an embodiment of the present application;

[0079] FIG. 16C is a schematic cross-sectional structural diagram of a magnetic implant assembly of an endoscope of an endoscope magnetic anastomosis system according to an embodiment of the present application;

[0080] FIG. 16D is a schematic structural diagram illustrating a cooperative use of magnetic implant assemblies of two endoscopes of an endoscopic magnetic anastomosis system according to an embodiment of the present application;

[0081] FIG. 16E is a schematic cross-sectional structural diagram of a magnetic implant assembly of an endoscope of an endoscopic magnetic anastomosis system in some cases;

[0082] FIG. 16F is a schematic diagram of a magnetic field interaction between two magnetic implant assemblies according to an embodiment of the present application;

[0083] FIG. 16G is a schematic diagram of a magnetic field interaction between two magnetic implant assemblies in some cases;

[0084] FIG. 16H is a schematic diagram of a relationship between the magnetic field force and distance between two magnetic implant assemblies according to an embodiment of the present application;

[0085] FIG. 16I is a schematic diagram of a relationship between the magnetic field force and distance between two magnetic implant assemblies in some cases;

[0086] FIG. 17A is a schematic diagram of a structure of a magnet detector of an endoscopic magnetic anastomosis system according to an embodiment of the present application;

[0087] FIG. 17B is a schematic view of an exploded structure of a magnet detector of an endoscopic magnetic anastomosis system according to an embodiment of the present application;

[0088] FIG. 17C is a top view schematic diagram of a structure of the first printed circuit board of the magnet detector of an endoscopic magnetic anastomosis system according to an embodiment of the present application;

[0089] FIG. 17D is a bottom view schematic diagram of a structure of the first printed circuit board of the magnet detector of an endoscopic magnetic anastomosis system according to an embodiment of the present application;

[0090] FIG. 17E is a top view schematic diagram of a structure of the second printed circuit board of the magnet detector of an endoscopic magnetic anastomosis system according to an embodiment of the present application;

[0091] FIG. 17F is a bottom view schematic diagram of a structure of the second printed circuit board of the magnet detector of an endoscopic magnetic anastomosis system according to an embodiment of the present application;

[0092] FIG. 18A is a schematic diagram of a structure of a magnet detector of an endoscopic magnetic anastomosis system according to another embodiment of the present application;

[0093] FIG. 18B is a schematic diagram of the working principle of a magnet detector of an endoscopic magnetic anastomosis system according to another embodiment of the present application;

[0094] FIG. 19A is a schematic diagram of a three-dimensional structure of a magnetic navigation console of an endoscopic magnetic anastomosis system according to an embodiment of the present application;

[0095] FIG. 19B is a schematic diagram of a top view structure of a magnetic navigation console of an endoscopic magnetic anastomosis system according to an embodiment of the present application;

[0096] FIG. 19C is a schematic structural diagram illustrating a working process of a push rod mechanism of a magnetic navigation console of an endoscope magnetic anastomosis system according to an embodiment of the present application;

[0097] FIG. 19D is a schematic structural diagram illustrating a push rod mechanism of a magnetic navigation console of an endoscope magnetic anastomosis system according to an embodiment of the present application in different usage states;

[0098] FIG. 19E is a schematic diagram of a structure of a first magnetic actuator of a magnetic navigation console of an endoscopic magnetic anastomosis system according to an embodiment of the present application; and

[0099] FIG. 19F is a schematic diagram illustrating a magnetic field effect of a first magnetic actuator of a magnetic navigation console on a magnetic implant assembly of an endoscopic magnetic anastomosis system according to an embodiment of the present application.

[0100] The drawings are used for providing an understanding of technical solutions of the present application, and constitute a part of the specification. They are used together with embodiments of the present application to explain the technical solutions of the present application, and do not constitute a restriction on the technical solutions of the present application.

[0101] Although similar reference numbers may be used to refer to similar elements in the figures for convenience, it can be appreciated that each of the various example embodiments may be considered to be distinct variations.

[0102] Example embodiments will now be described with reference to the accompanying figures, which form a part of the present disclosure and which illustrate example embodiments which may be practiced. As used in the present disclosure and the appended claims, the terms “embodiment,” “example embodiment,” “exemplary embodiment,” and “present embodiment” do not necessarily refer to a single embodiment, although they may, and various example embodiments may be readily combined and/or interchanged without departing from the scope or spirit of example embodiments. Furthermore, the terminology as used in the present disclosure and the appended claims is for the purpose of describing example embodiments only and is not intended to be limitations. In this respect, as used in the present disclosure and the appended claims, the term “in” may include “in” and “on,” and the terms “a,” “an,” and “the” may include singular and plural references. Furthermore, as used in the present disclosure and the appended claims, the term “by”

may also mean “from,” depending on the context. Furthermore, as used in the present disclosure and the appended claims, the term “if” may also mean “when” or “upon,” depending on the context. Furthermore, as used in the present disclosure and appended claims, the words “and/or” may refer to and encompass any or all possible combinations of one or more of the associated listed items.

DETAILED DESCRIPTION

[0103] Along with obesity, diabetes is emerging as a primary cause of mortality, morbidity, disability, and discrimination in health care, education and employment. The International Diabetes Federation (IDF) estimates that at present, nearly half a billion people worldwide are living with diabetes, of which low income and middle income countries carry close to 80% of the burden. Moreover, worldwide growth in prevalence is projected to only further worsen over the coming years. It is estimated that by 2045, the global prevalence will increase by 48%, swelling to an estimated 693 million individuals (aged 18-99 years) afflicted. Likewise, there has been a closely paralleled marked increase in obesity. The World Health Organization (WHO) estimates that in 2016, more than 1.9 billion adults, 18 years and older, were overweight. Of these, over 650 million were obese.

[0104] The socioeconomic burden associated with these conditions is immense, most of which can be attributed to Diabetes. Hyperglycemia, a high level of blood glucose, is the hallmark of diabetes. In Type 2 Diabetes (or “T2DM”), hyperglycemia results from a varying combination of insulin resistance and inadequate insulin production. Chronic hyperglycaemia can cause damage to various organs, leading to the development of disabling and life-threatening complications such as cardiovascular disease, neuropathy, nephropathy, and eye disease leading to retinopathy and blindness. These complications contribute to more frequent hospitalization, increased medical care costs and lower quality of life, and often results in early death. In fact, the IDF estimates that approximately four million people aged between 20 and 79 years died from diabetes in 2017, which equates to about one death every eight seconds. Diabetes accounted for 10.7% of global all-cause mortality among people in this age group, which is higher than the combined number of deaths from infectious diseases (HIV/AIDS, Tuberculosis, and Malaria). Moreover, around 46.1% of deaths due to diabetes in this age bracket were in individuals under the age of 60. In addition to the human burden of diabetes, characterized by premature mortality and low quality of life, there is also a significant economic burden imposed by the disease and its complications. The brunt of this burden is bore upon countries, healthcare systems, and more importantly, directly and direly by inflicted individuals and their families. The global projection for annual healthcare expenditure on diabetes in 2017 was USD 727 billion, corresponding to one for every eight dollars spent on healthcare. These economic costs are ever-increasing, a trend which has been best chronicled and analyzed in the United States (US). The American Diabetes Association has estimated that, after adjusting for inflation, the economic costs of diabetes increased by 26% (from USD 188 billion to USD 237.3 billion) between 2012 to 2017 in the US, which can be attributed to both an increased prevalence of diabetes, as well as an increased cost per person with diabetes. Moreover, after adjusting for both inflation and growth in diabetes prevalence, the excess medical cost per person with diabetes grew by 14% (from USD 8,417 to USD 9,601) in the same 5-year time frame.

[0105] Notwithstanding lifestyle and behavioural modification, which have been deemed largely ineffective in the treatment of T2DM, there remains a substantive treatment gap in conventional approaches (surgery and medication) to T2DM treatment that has not been adequately addressed. For surgery, this treatment gap refers to several different cohorts of people, including a large subgroup of individuals that are overweight and diabetic but not considered to be severely obese and therefore do not qualify for traditional surgical procedures. The treatment gap also refers to individuals who do qualify for these procedures but who are unreceptive or unwilling to undergo invasive, anatomy-changing and irreversible operations. Further, the treatment gap also refers to individuals who qualify for surgery, but that are unable to access treatment due to barriers

associated with high cost, lack of insurance coverage, and/or available surgical skill. For medication, barriers to access that are commonly cited are high cost, marginal efficacy, and high incidences of side effects. All of these factors put together contribute to the fact that existing surgical procedures, interventions, and medication are used by less than 1% of those eligible worldwide.

[0106] Systems, devices, and methods are described herein for use in delivering and magnetically coupling magnetic implant assemblies so as to create an anastomosis in adjacent points of the digestive tract (between the duodenum and ileum, or the jejunum and the ileum). It is recognized in the present disclosure that such resulting alternate “pathway” or “short-cut” (i.e., the anastomosis) may provide an alternate pathway for nutrient-rich chyme to enter the ileum more quickly and distally, which may result in avoiding absorption in the foregut, triggering early satiety, and/or improving glucose metabolism (e.g., by mediating a supposed “incretin effect”, which is characterized by an increased secretion of Glucagon-like peptide (GLP-1), a gut hormone that stimulates insulin secretion, gene expression, and β -cell growth). In this regard, diabetes control may result from such expedited delivery of nutrient-rich chyme to the distal intestines, and may result in the emanation of a physiological signal leading to improved glucose metabolism. It is to be understood that the principles described in the present disclosure may be applied outside of the context of endoscopic anastomosis procedures, such as performing diagnostic procedures, surgical or therapeutic procedures, scientific experiments, and/or other procedures in the same and/or other environments, cavities, and/or organs not described in the present disclosure without departing from the teachings of the present disclosure.

[0107] Example embodiments will now be described below with reference to the accompanying figures, which form a part of the present disclosure.

Example Embodiments of the Endoscopic Anastomosis System (e.g., Endoscopic Anastomosis System **100**).

[0108] FIG. 1A, FIG. 1B, and FIG. 1C illustrate different views of an example embodiment of an endoscopic anastomosis system (e.g., system **100**). The system **100** may be configurable or configured to be inserted through a natural orifice (e.g., rectum) of a patient to deliver a first magnetic implant body (or magnetic body) **430** into a cavity (e.g., colon or ileocecal valve) of the patient. A complimentary, corresponding, or associated second system **100** adapted to be inserted through another natural orifice (e.g., mouth) of the patient delivers a second magnetic implant body **430** into an adjacent cavity (e.g., duodenum or jejunum) of the patient, which is then magnetically coupled through one or more cavity walls to the first magnetic implant body **430**. Together, the first and second magnetic implant bodies **430** are configured to form an anastomosis through the one or more cavity walls.

[0109] Each system **100** includes one or more elements. For example, as will be further described in the present disclosure, each system **100** includes a first main body assembly **200** (or first main body **200**). Each system **100** also includes a head assembly **300**. The head assembly **300** is secured to an end of the first main body assembly **200** (e.g., referred to herein as the second end **203** of the first main body assembly **200**, distal end of the first main body assembly **200**, or end that is inserted into a cavity of a patient). Although the head assembly **300** may be referred to in the present disclosure as being a separate element from (and secured to) the first main body assembly **200**, it is to be understood that the head assembly **300** may also be considered as an element or part of the first main body assembly **200** without departing from the teachings of the present disclosure. Each system **100** also includes a second main body assembly **400**. At least a portion of the second main body assembly **400** is housed in an interior of the head assembly **300**, and at least a portion of the second main body assembly **400** is provided/inserted through an opening **318** (referred to herein as a “second main body assembly opening” **318**, “catheter body opening” **318**, or the like) of the head assembly **300**. Furthermore, at least a portion of the second main body assembly **400** is housed in the first main body assembly **200**. In example embodiments, the first and second main

bodies **200**, **400** are slidable relative to one another. Each system **100** also includes a magnetic implant assembly **430**. Each system **100** also includes a securing assembly **440**. Although the magnetic implant assembly **430** and/or the securing assembly **440** may be referred to in the present disclosure as being separate element(s) from (and secured to) the second main body assembly **400**, it is to be understood that the magnetic implant assembly **430** and/or the securing assembly **440** may also be considered as an element or part of the second main body assembly **400** without departing from the teachings of the present disclosure. For ease of reference, FIG. 2 illustrates a view of these elements separated from one another.

[0110] As used in the present disclosure, when applicable, one or more elements of each system **100** may be controlled, in part or in whole, directly or indirectly, by or in cooperation with one or more processors, controllers, computing devices, processors, servers, systems, cloud-based computing, artificial intelligence (AI), or the like (referred to herein as a “controller”, “processor”, or the like) (not shown) and/or one or more surgeon consoles (not shown, which may be any console, or the like, for one or more surgeons to perform one or more actions described in the present disclosure). Such controller and/or surgeon console may be in communication with and/or control one or more external systems/devices (e.g., external pressure source for providing negative and/or positive pressure, etc.) (not shown). Such controller may be any processor, server, system, device, computing device, controller, microprocessor, microcontroller, microchip, semiconductor device, or the like, configurable or configured to perform, among other things, a processing and/or managing of information, searching for information, identifying of information, data communications, processing information and/or making one or more decisions via artificial intelligence, machine learning, deep learning, or the like, and/or any one or more other actions described in the present disclosure. Alternatively or in addition, such controller (and/or its elements) may include and/or be a part of a virtual machine, processor, computer, node, instance, host, or machine, including those in a networked computing environment. As used in the present disclosure, a communication channel, or the like, may be or include a collection of devices and/or virtual machines connected by communication channels that facilitate communications between devices and allow for devices to share resources. Such resources may encompass any types of resources for running instances including hardware (such as servers, clients, mainframe computers, networks, network storage, data sources, memory, central processing unit time, scientific instruments, and other computing devices), as well as software, software licenses, available network services, and other non-hardware resources, or a combination thereof. A communication channel may include, but is not limited to, the internet, intranets, WiFi systems, GPS systems, location systems, location-based service systems, computing grid systems, peer to peer systems, mesh-type systems, distributed computing environments, cloud computing environment, telephony systems, voice over IP (VOIP) systems, etc. Such communication channels may include hardware and software infrastructures configured to form a virtual organization comprised of multiple resources which may be in geographically disperse locations. Communication channel may also refer to a communication medium between processes on the same device or system.

[0111] These and other elements of the system **100** will now be described with reference to the accompanying figures.

The First Main Body Assembly (e.g., First Main Body Assembly **200**).

[0112] As illustrated in at least FIG. 3A and FIG. 3B, each system **200** includes a first main body assembly **200**. The first main body assembly **200** may include an elongated tubular structure having a first end **201** (or “proximal end” **201**) and a second end **203** (or “distal end” **203**). The first main body assembly **200** may include a flexible body or tube having one or more internal channels (not shown). For example, the one or more internal channels may be provided for a plurality of actuation control members (e.g., cables, wires, tendons, or the like) to extend from the controller and/or surgeon console (at or near the first end **201**) to a portion of the second end **203** (e.g., to the first bendable section **210**). As another example, the one or more internal channels may be provided

to house the second main body assembly **400** (which may extend from the controller and/or surgeon console (at or near the first end **201**) and through the second main body assembly opening **318** of the head assembly **300**. In this regard, the second main body assembly **400** and the first main body assembly **200** may be configured to slide relative to one another. As another example, the one or more internal channels may be provided to enable negative pressure and/or positive pressure to be supplied from one or more external pressure sources (not shown) to the one or more pressure openings **332**, **334**, **422**. As another example, the one or more internal channels may be provided for positive pressure and/or negative pressure to be supplied from one or more external pressure sources (not shown) to the first and second expandable members **320**, **420**. As another example, the one or more internal channels may be provided for washing fluid, positive pressure, and/or negative pressure to be supplied from one or more external pressure sources (not shown) to the first cleansing assembly **338**. As another example, the one or more internal channels may be provided for electrical and/or data cables to extend to the first image capturing assembly **336**. As another example, the one or more internal channels may be provided for cables to extend to one or more sensors (e.g., for haptic feedback, temperature sensor, pressure sensor, etc., not shown), etc. Other internal channels for other purposes are also contemplated in the present disclosure. It is to be understood that an internal channel of the first main body assembly **200** may be any channel (including those that are wholly or partially within the first main body assembly), and may include a smaller tube, or the like, provided in a larger channel or tube. It is also to be understood that an internal channel of the first main body assembly **200** may extend beyond the first end **201** and/or second end **203** of the first main body assembly **200**.

[0113] As illustrated in at least FIGS. **1A** and **1B**, the second end **203** of the first main body assembly **200** may be securable or secured to (and in example embodiments, detachable from) a connector portion **302** of the head assembly **300**.

[0114] The first main body assembly **200** includes a first bendable section (e.g., first bendable section **210**) at the second end **203** of the first main body assembly **200**. Although not illustrated in the Figures, the second end **203** of the first main body assembly **200** may also include one or more expandable members (e.g., similar to the first expandable member **320** and/or second expandable member **420**) and/or one or more pressure openings (e.g., similar to the first pressure opening **332** and/or second pressure opening **422**). Such one or more pressure openings may be provided before and/or after such one or more expandable members, and such one or more pressure openings and/or one or more expandable members may be provided before and/or after the first bendable section **210**. In some example embodiments, such expandable member(s) and/or pressure opening(s) of the first main body assembly **200** may be in addition to or replace the first expandable member **320** and/or the first pressure opening **332** of the head assembly **300**.

[0115] In an example embodiment, the first bendable section **210** is configurable or configured to guide, turn, bend, and/or steer (referred to herein as “bend”, “bending”, or the like) the system **100** in any one or more of a plurality of available directions and/or one or more of a plurality of locations along the first bendable section **210**. This may be desirable when the system **100** is being advanced forward into a body cavity, such as a colon or small bowel, and the system **100** reaches a bend, turn, or the like in the body cavity. Alternatively or in addition, such bending may be desirable when a particular area of the interior wall of the body cavity needs to be viewed and/or actioned (e.g., delivering of the magnetic implant body **430**). Such bending of the first bendable section **210** may be achievable or achieved by selectively configuring one or more locations along the first bendable section **210** to bend (e.g., away from a center axis formed by the first bendable section **210**). Such selective configuring may include selecting one or more locations along the first bendable section **210** to bend from among a plurality of bendable location(s) along the first bendable section **210**. FIG. **3B** illustrates an example of bending of the first bendable section **210**. Selective configuring may also include selecting, for each location along the first bendable section **210**, a degree of curvature for the bending from among a plurality of available degrees of curvature.

Selective configuring may also include selecting, for each location along the first bendable section **210**, one or more directions for the bending from among a plurality of available directions, etc. [0116] The bending of the first bendable section **210** may be selectively controllable by controlling an amount of force (e.g., tension via pulling or pushing) applied (increased, decreased, maintained, or not applied) to one or more actuation control members (not shown) and/or selecting one or more of the actuation control members to selectively control (i.e., which actuation control member will receive an increase in applied force, decrease in applied force, no change in applied force, and/or no applied force). In an example embodiment, the first bendable section **210** may include a serially (or linearly) connected arrangement of a plurality of bendable subsections (not shown). Each bendable subsection may include one or more distal termination points for receiving, securing, terminating, and/or connecting one or more actuation control members.

[0117] Each of the bendable subsections may include one or more internal cavities or channels for, among other things, enabling one or more actuation control members to extend through, enabling negative pressure and/or positive pressure to be provided to the one or more pressure openings **332**, **422**, enabling positive pressure and/or negative pressure to be provided to the first expandable member **320**, enabling positive pressure and/or negative pressure to be provided to the second expandable member **420**, enabling fluid and/or positive pressure (and/or negative pressure) to be provided to the first cleansing assembly **338**, enabling electrical and/or data cables to extend to the first image capturing assembly **336**, etc.

[0118] The distal termination points may be provided in any shape or form so long as it enables the receiving, connecting, terminating, and/or securing of the distal end of one or more actuation control members. For example, the distal termination point may be an opening, connector, termination, hook, etc. A degree of bending of one or more of the bendable locations of the first bendable section **210** may be between about 0 to 210 degrees from a center axis in example embodiments.

[0119] In an example embodiment, the first main body assembly **200** may have a length between about 1600 mm to about 2200 mm, and a diameter between about 12 mm to about 18 mm. The first main body assembly **200** may be formed having one or more of a plurality of cross-sectional shapes, including a circular cross-section, elliptical cross-section, etc. Other dimensions and shapes are also contemplated without departing from the teachings of the present disclosure. A length of the first bendable section **210** may be between about 70 mm to about 130 mm, and a diameter of the first bendable section **210** may be between about 12 mm to about 18 mm in example embodiments. Other dimensions are also contemplated without departing from the teachings of the present disclosure.

The Head Assembly (e.g., Head Assembly **300**).

[0120] As illustrated in at least FIG. 2, FIG. 4A, and FIG. 4B, each system **200** includes a head assembly **300**. The head assembly **300** includes a head assembly body **300**. The head assembly **300** also includes one or more first expandable members **320**. The head assembly **300** also includes one or more first pressure ports **332**. The head assembly **300** also includes a second main body opening **318**. The head assembly **300** also includes one or more insufflation ports **334**. The head assembly **300** also includes one or more first image capturing assemblies **336**. The head assembly **300** also includes one or more first cleansing assemblies **338**.

[0121] These and other elements of the head assembly **300** will now be described with reference to the accompanying figures.

The Head Assembly Body (e.g., Head Assembly Body **300**).

[0122] As illustrated in at least FIGS. 2 and 4A-B, the head assembly **300** includes a head assembly body **300**. The head assembly body **300** includes a connector portion **302** at a first end of the head assembly body **300** for securing to the second end **203** of the second main body assembly **200**. The head assembly body **300** also includes a first region, portion, or the like (referred to herein as the “first region” **310**) and a second region, portion, or the like (referred to herein as the “second

region” **330**). A first end **311** of the first region **310** is secured to the second end **203** of the second main body assembly **200** via the connector portion **302**, and at least a portion of a second end **313** of the first region **310** is secured to the first end **331** of the second region **330**.

[0123] In example embodiments, the second end **313** of the first region **310** of the head assembly body **300** includes a first section **316** (as illustrated in at least FIG. **4B**) and a second section (not shown) adjacent to the first section **316**. The second section of the first region **310** is secured to the first end **331** of the second region **330** of the head assembly body **300**. The first section **316** of the first region **310** includes a second main body assembly opening **318** (or second main body opening **318** or catheter body opening **318** or catheter opening **318**). The second main body assembly opening **318** is configured to house at least a portion of the second main body assembly **400**. That is, at least a portion of the second main body assembly **400** is provided/inserted through the second main body assembly opening **318**. In this regard, the second main body **402** is moveable/slidable relative to the head assembly body **300**. The first region **310** also includes one or more other pressure openings (not shown) for providing positive and/or negative pressure to an interior portion of the first expandable member **320** (e.g., to expand, maintain, or contract a volume of the first expandable member **320**). The first region **310** also includes one or more internal cavities or channels. For example, the one or more internal cavities or channels may house at least a portion of a second end of the second main body **402** of the second main body assembly **400**. The one or more internal cavities or channels may also house the plurality of actuation control members (e.g., cables, wires, tendons, or the like, as described in the present disclosure) that control the second bendable section **410** of the second main body assembly **400**. As another example, the one or more internal cavities or channels may be provided to enable negative pressure and/or positive pressure to be supplied from one or more external pressure sources (not shown) to one or more pressure openings **332**, **334**, **422**. As another example, the one or more internal cavities or channels may be provided for positive pressure and/or negative pressure to be supplied from one or more external pressure sources (not shown) to the first and second expandable members **320**, **420**. As another example, the one or more internal cavities or channels may be provided for washing fluid, positive pressure, and/or negative pressure to be supplied from one or more external pressure sources (not shown) to the first cleansing assembly **338**. As another example, the one or more internal cavities or channels may be provided for electrical and/or data cables to extend to the first image capturing assembly **336**. As another example, the one or more internal cavities or channels may be provided for cables to extend to one or more sensors (e.g., for haptic feedback, temperature sensor, pressure sensor, etc., not shown), etc. Other internal cavities or channels for other purposes are also contemplated in the present disclosure. It is to be understood that an internal cavity or channel of the first region **310** of the head assembly body **300** may be any cavity or channel (including those that are wholly or partially within the head assembly body **300**), and may include a smaller tube, or the like, provided in a larger channel or tube. It is also to be understood that an internal cavity or channel of the first region **310** of the head assembly body **300** may extend beyond the first region **310** of the head assembly body **300**. The first region **310** of the head assembly body **300** may have a length between about 12 mm to about 20 mm, and a diameter between about 12 mm to about 20 mm. The first region **310** of the head assembly body **300** may be cylindrical in shape and/or formed having one or more of a plurality of cross-sectional shapes, including a circular cross-section, elliptical cross-section, etc. Other dimensions and shapes are also contemplated without departing from the teachings of the present disclosure.

[0124] In example embodiments, the first end **331** of the second region **330** of the head assembly body **300** is secured to the second section of the first region **310** of the head assembly body **300**. As will be further described in the present disclosure, the second region **330** includes one or more first pressure ports **332**. The second region **330** also includes one or more first insufflation ports **334**. The second region **330** also includes one or more first image capturing assemblies **336**. The second region **330** also includes one or more first cleansing assemblies **338**. The second region **330** also

includes one or more internal cavities or channels. For example, the one or more internal cavities or channels may house at least a portion of a second end of the second main body **402** of the second main body assembly **400**. The one or more internal cavities or channels may also house the plurality of actuation control members (e.g., cables, wires, tendons, or the like, as described in the present disclosure) that control the second bendable section **410** of the second main body assembly **400**. As another example, the one or more internal cavities or channels may be provided to enable negative pressure and/or positive pressure to be supplied from one or more external pressure sources (not shown) to one or more pressure openings **332**, **334**, **422**. As another example, the one or more internal cavities or channels may be provided for positive pressure and/or negative pressure to be supplied from one or more external pressure sources (not shown) to the second expandable member **420**. As another example, the one or more internal cavities or channels may be provided for washing fluid, positive pressure, and/or negative pressure to be supplied from one or more external pressure sources (not shown) to the first cleansing assembly **338**. As another example, the one or more internal cavities or channels may be provided for electrical and/or data cables to extend to the first image capturing assembly **336**. As another example, the one or more internal cavities or channels may be provided for cables to extend to one or more sensors (e.g., for haptic feedback, temperature sensor, pressure sensor, etc., not shown), etc. Other internal cavities or channels for other purposes are also contemplated in the present disclosure. It is to be understood that an internal cavity or channel of the second region **330** of the head assembly body **300** may be any cavity or channel (including those that are wholly or partially within the head assembly body **300**), and may include a smaller tube, or the like, provided in a larger channel or tube. It is also to be understood that an internal cavity or channel of the second region **330** of the head assembly body **300** may extend beyond the second region **330** of the head assembly body **300**. The second region **330** of the head assembly body **300** may have a length between about 14 mm to about 25 mm. The second region **330** of the head assembly body **300** may be semi-cylindrical in shape and/or formed having one or more of a plurality of cross-sectional shapes, including a semi-circular cross-section, semi-elliptical cross-section, etc. Other dimensions and shapes are also contemplated without departing from the teachings of the present disclosure.

The First Expandable Member (e.g., First Expandable Member **320**).

[0125] As illustrated in at least FIGS. **2** and **4A-D**, the head assembly **300** includes one or more first expandable members (e.g., first expandable member **320**). As illustrated in at least FIG. **4C**, the first expandable member **320** is secured to at least a portion of the first region **310** of the head assembly **300**. More specifically, the first expandable member **320** is secured to an exterior portion of the head assembly body **300** between the first and second ends **311**, **313** of the first region **310** of the head assembly body **300**. The first expandable member **320** is configured to transition between a normal or unexpanded configuration (e.g., as illustrated in FIG. **4C**) and an expanded configuration (e.g., as illustrated in FIG. **4D**) in which the first expandable member **320** expands radially outward or away from the head assembly body **300**.

[0126] When the first expandable member **320** is controlled to be in the expanded configuration while in a cavity of a patient, the controller (not shown) is configured to control the first expandable member **320** to expand radially outward or away from the head assembly body **300** towards and/or to the cavity wall of the patient. When expanded, the first expandable member **320** may or may not reach the cavity wall of the patient. In situations where the first expandable member **320** (when expanded) reaches the cavity wall of the patient, the first expandable member **320** may encourage or push outward the cavity wall of the patient. However, in example embodiments, the first expandable member (when expanded) may stop just short (may not reach) or may not push outward (if it reaches) the cavity wall of the patient. In either of these situations, it is recognized that the first expandable member **320** in cooperation (or in combination) with one or more of the first pressure ports **332** (as further described in the present disclosure, which is configured to encourage, bring in, suction inward, and/or collapse a portion of the cavity wall of

the patient) enables the system **100** to anchor, grip, and/or otherwise secure to the cavity wall of the patient.

[0127] The first expandable member **320** may be formed completely or partially around the first region **310** of the head assembly body **300**. The first expandable member **320** may resemble a balloon, or the like, and may include one or more openings (not shown) to allow positive pressure (e.g., passage of gas and/or liquid, and/or allow a manipulation of pressure within the first expandable member **320**) to be introduced, controlled, and/or reduced in the first expandable member **320**. Each such opening may be connected to one or more of the pressure cavities, which are in turn connected to one or more external pressure sources (not shown). Alternatively, the first expandable member **320** may be formed via one or more membranes (e.g., rectangular sheets) of expandable material, and the opposing long sides of such membranes may be secured (e.g., via an overmolding process) to the first and second ends **311**, **313** of the first region **310** of the head assembly body **300**.

[0128] When in the expanded configuration, which may be a state in which the external pressure source provides a positive pressure to the first expandable member **320**, the first expandable member **320** may be configurable to expand radially outward (e.g., resembling a balloon, tire, or the like) with an overall diameter of the first expandable member **320**, when in the expanded configuration, between about 25 mm to 40 mm. Other dimensions are also contemplated without departing from the teachings of the present disclosure.

The First Pressure Port (e.g., First Pressure Port **332**).

[0129] As illustrated in at least FIGS. **2** and **4A-B**, the head assembly **300** includes one or more first pressure ports (e.g., first pressure port **332**; also referred to herein as the “first pressure opening” **332**). The one or more first pressure ports **332** may be provided at the second end **333** of the second region **330**, and configured to provide negative pressure (and/or positive pressure) to an exterior of the head assembly body **300** (e.g., to a cavity of a patient). For example, as illustrated in at least FIG. **4A**, the one or more first pressure ports **332** may be provided on (or through) the head assembly body **300** at a most distal wall or face of the head assembly body **300** in such a way that a negative pressure applied by the one or more first pressure ports **332** is directed in a direction in which the head assembly body **300** is pointed (or advanced) (e.g., parallel to a central axis formed through the head assembly body **300**). Alternatively or in addition, in example embodiments in which the one or more first pressure ports **332** are provided on (or through) the most distal wall or face of the head assembly body **300**, one or more of the first pressure ports **332** may be oriented, configured, directed, or pointed in such a way that a negative pressure applied by such one or more first pressure ports **332** is directed in a direction that is not parallel to a central axis formed through the head assembly body **300** (e.g., 10-80 degrees from the central axis formed through the head assembly body **300**). Alternatively or in addition, in example embodiments where more than one first pressure ports **332** are provided on (or through) the head assembly body **300**, such first pressure ports **332** may be provided around a circumference of the head assembly body **300** (e.g., if a portion of the head assembly body **300** has a circular circumference, such as the first region **310**; or otherwise distributed around the head assembly body **300**). Alternatively or in addition, in example embodiments where more than one first pressure port **332** is provided on (or through) the head assembly body **300**, such first pressure ports **330** may be provided at one or more different locations along the head assembly body **300** (e.g., in the first region **310** and/or the second region **330**). The one or more first pressure openings **332** may be configured to provide a negative pressure so as to encourage, bring in, suction inward, and/or collapse a portion of cavity wall of a patient toward the head assembly body **300**. It is recognized in the present disclosure that such encouraging, bringing in, suctioning inward, and/or collapsing of a portion of the cavity wall of the patient in combination with an expansion radially outward of the first expandable member **320** towards and/or to a portion of the cavity wall of the patient enables the system **100** to anchor, grip, and/or otherwise secure to the cavity wall of the patient.

[0130] Although the Figures illustrate that the first pressure port **332** is provided at the second end **333** of the second region **330** of the head assembly body **300**, it is to be understood that other configurations are also contemplated in the present disclosure. For example, in addition to or in replacement of the one or more first pressure ports **332** provided at the second end **333** of the second region **330**, the one or more first pressure ports **332** may be provided on a side portion (e.g., between the first and second ends **331**, **333**) of the second region **330** of the head assembly body **300**. Alternatively or in addition, the one or more first pressure ports **332** may be provided at the second end **313** of the first region **310** of the head assembly body **300**. Alternatively or in addition, the one or more first pressure ports **332** may be provided at the first end **311** of the first region **310** of the head assembly body **300**.

The First Insufflation Port (e.g., First Insufflation Port **334**).

[0131] As illustrated in at least FIGS. **2** and **4A-B**, the head assembly **300** includes one or more first insufflation ports (e.g., first insufflation port **334**; also referred to herein as the “first insufflation opening” **334**). The one or more first insufflation ports **334** may be provided at the second end **333** of the second region **330**, and configured to provide positive pressure to an exterior of the head assembly body **300** (e.g., to a cavity of a patient to insufflate the cavity of the patient). For example, the one or more first insufflation openings **334** may be configured to provide a positive pressure so as to encourage, push outward, and/or expand a portion of cavity wall of a patient away from the head assembly body **300**.

[0132] Although the Figures illustrate that the first insufflation port **334** is provided at the second end **333** of the second region **330** of the head assembly body **300**, it is to be understood that other configurations are also contemplated in the present disclosure. For example, in addition to or in replacement of the one or more first insufflation ports **334** provided at the second end **333** of the second region **330**, the one or more first insufflation ports **334** may be provided on a side portion (e.g., between the first and second ends **331**, **333**) of the second region **330** of the head assembly body **300**. Alternatively or in addition, the one or more first insufflation ports **334** may be provided at the second end **313** of the first region **310** of the head assembly body **300**. Alternatively or in addition, the one or more first insufflation ports **334** may be provided at the first end **311** of the first region **310** of the head assembly body **300**.

The First Image Capturing Assembly (e.g., First Image Capturing Assembly **336**).

[0133] As illustrated in at least FIGS. **2** and **4A-B**, the head assembly **300** includes one or more first image capturing assemblies (e.g., first image capturing assembly **336**). The first image capturing assembly **336** may be any image and/or video capturing device including, but not limited to, a 2-D video camera and/or a 3-D stereoscopic or autostereoscopic video camera. The first image capturing assembly **336** may also include one or more illumination sources (not shown), or the like, such as one or more LED lights.

[0134] Although the Figures illustrate that the first image capturing assembly **336** is provided at the second end **333** of the second region **330** of the head assembly body **300**, it is to be understood that other configurations are also contemplated in the present disclosure. For example, in addition to or in replacement of the one or more first image capturing assemblies **336** provided at the second end **333** of the second region **330**, the one or more first image capturing assemblies **336** may be provided on a side portion (e.g., between the first and second ends **331**, **333**) of the second region **330** of the head assembly body **300**. Alternatively or in addition, the one or more first image capturing assemblies **336** may be provided at the second end **313** of the first region **310** of the head assembly body **300**. Alternatively or in addition, the one or more first image capturing assemblies **336** may be provided at the first end **311** of the first region **310** of the head assembly body **300**.

The First Cleansing Assembly (e.g., First Cleansing Assembly **338**).

[0135] As illustrated in at least FIGS. **2** and **4A-B**, the head assembly **300** includes one or more first image cleansing assemblies (e.g., first cleansing assembly **338**). The first cleansing assembly **338** may be configured to direct a fluid (e.g., water, non-toxic washing fluid, etc.), positive

pressure, and/or negative pressure to the first image capturing assembly **336** so as to clean, unblock, and/or otherwise improve visibility and/or image capturing quality of the first image capturing assembly **336**.

[0136] Although the Figures illustrate that the first cleansing assembly **338** is provided at the second end **333** of the second region **330** of the head assembly body **300**, it is to be understood that other configurations are also contemplated in the present disclosure (so long as it is nearby the first image capturing assembly **336**). For example, in addition to or in replacement of the one or more first cleansing assembly **338** provided at the second end **333** of the second region **330**, the one or more first cleansing assembly **338** may be provided on a side portion (e.g., between the first and second ends **331**, **333**) of the second region **330** of the head assembly body **300**. Alternatively or in addition, the one or more first cleansing assembly **338** may be provided at the second end **313** of the first region **310** of the head assembly body **300**. Alternatively or in addition, the one or more first cleansing assembly **338** may be provided at the first end **311** of the first region **310** of the head assembly body **300**.

The Second Main Body Assembly (e.g., Second Main Body Assembly **400**).

[0137] As illustrated in at least FIG. **2** and FIGS. **4E-I**, each system **200** includes a second main body assembly **200**. The second main body assembly **400** includes one or more elements. For example, the second main body assembly **400** includes a second main body **402**. The second main body assembly **400** also includes a second bendable section **410**. The second main body assembly **400** also includes one or more second expandable members **420**. The second main body assembly **400** also includes one or more second pressure openings **422**. The second main body assembly **400** also includes a magnetic implant assembly **430**. The second main body assembly **400** also includes a securing assembly **440**.

[0138] These and other elements of the second main body assembly **400** will now be described with reference to the accompanying figures.

The Second Main Body (e.g., Second Main Body **402**).

[0139] As illustrated in at least FIGS. **4E-I**, an example embodiment of the second main body assembly **400** includes a second main body **402**. The second main body **402** may include an elongated tubular structure, or the like, having a first end (or “proximal end”) and a second end (or “distal end”, which is the end nearest and/or secured to the securing assembly **440**). The second main body **402** may include a flexible body having one or more internal channels (not shown). For example, the one or more internal channels may be provided for the plurality of actuation control members (e.g., cables, wires, tendons, or the like) to extend from the controller and/or surgeon console to a portion of the second end of the second main body **402**. As another example, the one or more internal channels may be provided to enable negative pressure and/or positive pressure to be supplied from one or more external pressure sources (not shown) to the one or more second pressure openings **422**. As another example, the one or more internal channels may be provided for positive pressure and/or negative pressure to be supplied from one or more external pressure sources (not shown) to the second expandable member **420**. As another example, the one or more internal channels may be provided for cables to control the securing assembly **440** (e.g., control the securing and unsecuring of the magnetic implant assembly **430**). Other internal channels for other purposes are also contemplated in the present disclosure. It is to be understood that an internal channel of the second main body **402** may be any channel (including those that are wholly or partially within the first main body assembly), and may include a smaller tube, or the like, provided in a larger channel or tube. It is also to be understood that an internal channel of the second main body **402** may extend beyond the first end and/or second end of the second main body **402**.

[0140] As illustrated in at least FIGS. **4E-I**, the second end of the second main body **402** may be securable or secured to (and in example embodiments, detachable from) the securing assembly **440**. The second end of the second main body **402** may also be securable or secured to (and in example embodiments, detachable from) the magnetic implant assembly **430** (e.g., via the securing assembly

440).

[0141] In an example embodiment, the second main body **400** may have a length between about 1800 mm to about 2500 mm, and a diameter between about 2 mm to about 4 mm. The second main body **400** may be formed having one or more of a plurality of cross-sectional shapes, including a circular cross-section, elliptical cross-section, etc. Other dimensions and shapes are also contemplated without departing from the teachings of the present disclosure.

The Second Bendable Section (e.g., Second Bendable Section **410**).

[0142] As illustrated in at least FIG. **4G** and FIG. **4H**, the second main body assembly **400** includes one or more second bendable sections (e.g., second bendable section **410**). The one or more second bendable sections **410** may be provided at the second end of the second main body assembly **400**.

[0143] In an example embodiment, the second bendable section **410** is configurable or configured to bend the second main body assembly **400** in any one or more of a plurality of available directions and/or one or more of a plurality of locations along the second bendable section **410**. This may be desirable when the second main body assembly **400** is being advanced forward into a body cavity, such as a colon or small bowel, and the system **100** reaches a bend, turn, or the like in the body cavity. Alternatively or in addition, such bending may be desirable when a particular area of the interior wall of the body cavity needs to be viewed and/or actioned (e.g., delivering of the magnetic implant body **430**). Such bending of the second bendable section **410** may be achievable or achieved by selectively configuring one or more locations along the second bendable section **410** to bend (e.g., away from a center axis formed by the second bendable section **410**). Such selective configuring may include selecting one or more locations along the second bendable section **410** to bend from among a plurality of bendable location(s) along the second bendable section **410**. FIGS. **4G-H** illustrate an example of bending of the second bendable section **410**. Selective configuring may also include selecting, for each location along the second bendable section **410**, a degree of curvature for the bending from among a plurality of available degrees of curvature. Selective configuring may also include selecting, for each location along the second bendable section **410**, one or more directions for the bending from among a plurality of available directions, etc.

[0144] The bending of the second bendable section **410** may be selectively controllable by controlling an amount of force (e.g., tension via pulling or pushing) applied (increased, decreased, maintained, or not applied) to one or more actuation control members (not shown) and/or selecting one or more of the actuation control members to selectively control (i.e., which actuation control member will receive an increase in applied force, decrease in applied force, no change in applied force, and/or no applied force). In an example embodiment, the second bendable section **410** may include a serially (or linearly) connected arrangement of a plurality of bendable subsections (not shown). Each bendable subsection may include one or more distal termination points for receiving, securing, terminating, and/or connecting one or more actuation control members.

[0145] Each of the bendable subsections may include one or more internal cavities or channels for, among other things, enabling one or more actuation control members to extend through, enabling negative pressure and/or positive pressure to be provided to the one or more pressure openings **422**, enabling positive pressure and/or negative pressure to be provided to the second expandable member **420**, enabling cables to extend to the securing assembly **440**, etc.

[0146] The distal termination points for the one or more actuation control members may be provided in any shape or form so long as it enables the receiving, connecting, terminating, and/or securing of the distal end of one or more actuation control members. For example, the distal termination point may be an opening, connector, termination, hook, etc. A degree of bending of one or more of the bendable locations of the second bendable section **410** may be between about 0 to 210 degrees from a center axis in example embodiments.

[0147] In an example embodiment, a length of the second bendable section **410** may be between about 5 mm to about 50 mm, and a diameter of the second bendable section **410** may be between about 2 mm to about 4 mm in example embodiments. Other dimensions are also contemplated

without departing from the teachings of the present disclosure.

[0148] Although the Figures illustrate that the sequence or order of the elements (e.g., when moving towards the securing assembly **440** or the magnetic implant assembly **430**) are such that the second expandable member **420** is provided between the second bendable section **410** and the one or more second pressure ports **422**, it is to be understood that other configurations are also contemplated in the present disclosure. For example, the one or more second pressure ports **422** may be provided between the second bendable section **410** and the second expandable member **420**. As another example, the bendable section **410** may be provided between the second expandable member **420** and the one or more second pressure ports **422**. The sequence or order of the elements may also be changed from the sequence shown in the Figures (which illustrates a sequence of the second bendable section **410**, followed by the second expandable member **420**, followed by the one or more second pressure ports **422**). For example, the sequence may be the one or more second pressure ports **422**, followed by the second expandable member **420**, followed by the second bendable portion **410**. As another example, the sequence may be the one or more second pressure ports **422**, followed by the second bendable portion **410**, followed by the second expandable member **420**. As another example, the sequence may be the second expandable member **420**, followed by the one or more second pressure ports **422**, followed by the second bendable portion **410**. As another example, the sequence may be the second expandable member **420**, followed by the one or more second pressure ports **422**, followed by the second bendable portion **410**, followed by the one or more second pressure ports **422**.

[0149] As illustrated in FIG. **4I**, an example embodiment of the second main body assembly **400** is also configurable or configured to rotate relative to a central axis formed by the second main body **402**. Such rotation enables an orientation of the magnetic implant assembly **430** to be selectively change, which, along with the bending of the second bendable section **410**, can assist with the positioning of the magnetic implant assembly **430** for magnetically coupling to another magnetic implant assembly **430'**.

The Second Expandable Member (e.g., Second Expandable Member **420**).

[0150] As illustrated in at least FIGS. **2** and FIGS. **4E-I**, the second main body assembly **400** includes one or more second expandable members (e.g., second expandable member **420**). As illustrated in at least FIG. **4E**, the second expandable member **420** is secured to a portion of the second end of the second main body assembly **400**. The second expandable member **420** is configured to transition between a normal or unexpanded configuration (e.g., as illustrated in FIG. **4E**) and an expanded configuration (e.g., as illustrated in FIG. **4F** and FIG. **4H**) in which the second expandable member **420** is expanded radially outward or away from the second main body **402** (or towards or to a cavity wall of the patient).

[0151] When the second expandable member **420** is controlled to be in the expanded configuration while in a cavity of a patient, the controller (not shown) is configured to control the second expandable member **420** to expand radially outward or away from the second main body **402** and towards and/or to the cavity wall of the patient. When expanded, the second expandable member **420** may or may not reach the cavity wall of the patient. In situations where the second expandable member **420** (when expanded) reaches the cavity wall of the patient, the second expandable member **420** may encourage or push outward the cavity wall of the patient. However, in example embodiments, the second expandable member **420** (when expanded) may stop just short (may not reach) or may not push outward (if it reaches) the cavity wall of the patient. In either of these situations, it is recognized that the second expandable member **420** in cooperation (or in combination) with one or more of the second pressure ports **422** (as further described in the present disclosure, which is configured to encourage, bring in, suction inward, and/or collapse a portion of the cavity wall of the patient) enables the system **100** to anchor, grip, and/or otherwise secure to the cavity wall of the patient.

[0152] The second expandable member **420** may be formed completely or partially around the

second main body **402**. The second expandable member **420** may resemble a balloon, or the like, and may include one or more openings (not shown) to allow positive pressure (e.g., passage of gas and/or liquid, and/or allow a manipulation of pressure within the second expandable member **420**) to be introduced, controlled, and/or reduced in the second expandable member **420**. Each such opening may be connected to one or more of the pressure cavities, which are in turn connected to one or more external pressure sources (not shown). Alternatively, the second expandable member **420** may be formed via one or more membranes (e.g., rectangular sheets) of expandable material, and the opposing long sides of such membranes may be secured (e.g., via an overmolding process) circumferentially around the second main body **402**.

[0153] When in the expanded configuration, which may be a state in which the external pressure source provides a positive pressure to the second expandable member **420**, the second expandable member **420** may be configurable to expand radially outward (e.g., resembling a balloon, tire, or the like) with an overall diameter of the second expandable member **420**, when in the expanded configuration, between about 10 mm to 30 mm. Other dimensions are also contemplated without departing from the teachings of the present disclosure.

[0154] Although the Figures illustrate that the sequence or order of the elements (e.g., when moving towards the securing assembly **440** or the magnetic implant assembly **430**) are such that the second expandable member **420** is provided between the second bendable section **410** and the one or more second pressure ports **422**, it is to be understood that other configurations are also contemplated in the present disclosure. For example, the one or more second pressure ports **422** may be provided between the second bendable section **410** and the second expandable member **420**. As another example, the bendable section **410** may be provided between the second expandable member **420** and the one or more second pressure ports **422**. The sequence or order of the elements may also be changed from the sequence shown in the Figures (which illustrates a sequence of the second bendable section **410**, followed by the second expandable member **420**, followed by the one or more second pressure ports **422**). For example, the sequence may be the one or more second pressure ports **422**, followed by the second expandable member **420**, followed by the second bendable portion **410**. As another example, the sequence may be the one or more second pressure ports **422**, followed by the second bendable portion **410**, followed by the second expandable member **420**. As another example, the sequence may be the second expandable member **420**, followed by the one or more second pressure ports **422**, followed by the second bendable portion **410**. As another example, the sequence may be the second expandable member **420**, followed by the second bendable portion **410**, followed by the one or more second pressure ports **422**.

The Second Pressure Port (e.g., Second Pressure Port **422**).

[0155] As illustrated in at least FIGS. 2 and 4E-I, the second main body assembly **400** includes one or more second pressure ports (e.g., second pressure port **422**; also referred to herein as the “second pressure opening” **422**). The one or more second pressure ports **422** may be provided at the second end of the second main body assembly **400**, and configured to provide negative pressure (and/or positive pressure) to an exterior of the second main body **402** (e.g., to a cavity of a patient). In example embodiments where more than one second pressure ports **422** are provided on (or through) the second main body **402**, such second pressure ports **422** may be provided around a circumference of the second main body **402** (e.g., if the second main body **402** has a circular circumference; or otherwise around the head assembly body **300**). Alternatively or in addition, in example embodiments where more than one second pressure port **422** is provided on (or through) the second main body **402**, such second pressure ports **422** may be provided at one or more different locations along the second main body **402**. For example, as illustrated in at least FIG. 4E, one or more second pressure ports **422** may be provided on (or through) the second main body **402** at a location between the securing assembly **440** (or most distal part of the second main body **402**) and the second expandable member **420** (or most distal second expandable member **420** if there are

more than one second expandable members **420**). As another example (not shown), one or more second pressure ports **422** may be provided on (or through) the second main body **402** in such a way that the second expandable member **420** is provided between the one or more second pressure ports **422** and the securing assembly **440** (or most distal part of the second main body **402**). As another example (not shown), one or more second pressure ports **422** may be provided on (or through) the second main body **402** before and after (or distal and proximal to) the second expandable member **420**. As another example (not shown), in example embodiments in which there are two or more second expandable members **420** secured to the second main body **402**, a plurality of second pressure ports **422** may be provided on (or through) the second main body **402** in such a way that second pressure ports **422** are provided before and after (or distal and proximal to) each of the second expandable members **420**. The one or more second pressure ports **422** may be configured to provide a negative pressure so as to encourage, bring in, suction inward, and/or collapse a portion of cavity wall of a patient toward the second main body **402**. It is recognized in the present disclosure that such encouraging, bringing in, suctioning inward, and/or collapsing of a portion of the cavity wall of the patient in combination with an expansion radially outward of the one or more second expandable members **420** towards and/or to a portion of the cavity wall of the patient enables the second main body assembly **400** to anchor, grip, and/or otherwise secure to the cavity wall of the patient.

[0156] Although the Figures illustrate that the sequence or order of the elements (e.g., when moving towards the securing assembly **440** or the magnetic implant assembly **430**) are such that the second expandable member **420** is provided between the second bendable section **410** and the one or more second pressure ports **422**, it is to be understood that other configurations are also contemplated in the present disclosure. For example, alternatively or in addition, the one or more second pressure ports **422** may be provided between the second bendable section **410** and the second expandable member **420**. As another example, alternatively or in addition, the bendable section **410** may be provided between the second expandable member **420** and the one or more second pressure ports **422**. The sequence or order of the elements may also be changed from the sequence shown in the Figures (which illustrates a sequence of the second bendable section **410**, followed by the second expandable member **420**, followed by the one or more second pressure ports **422**). For example, the sequence may be the one or more second pressure ports **422**, followed by the second expandable member **420**, followed by the second bendable portion **410**. As another example, the sequence may be the one or more second pressure ports **422**, followed by the second bendable portion **410**, followed by the second expandable member **420**. As another example, the sequence may be the second expandable member **420**, followed by the one or more second pressure ports **422**, followed by the second bendable portion **410**. As another example, the sequence may be the second expandable member **420**, followed by the second bendable portion **410**, followed by the one or more second pressure ports **422**.

The Magnetic Implant Assembly (e.g., Magnetic Implant Assembly **430**).

[0157] As illustrated in at least FIGS. 2 and 4E-I, the second main body assembly **400** includes a magnetic implant assembly (e.g., magnetic implant assembly **430**, magnetic implant body **430**, magnetic body **430**, or the like). The magnetic implant assembly **430** is securable to and unsecurable from the second main body **402** via the securing assembly **440**. As will be further described in the present disclosure, the magnetic implant assembly **430** may be formed, in whole or in part, as or using ferromagnetic or magnetic materials.

[0158] The magnetic implant assembly **430** may be configured in one or more configurations. In this regard, an example embodiment of a first magnetic implant assembly **430** of one system **100** (e.g., a first system **100** for delivering of the magnetic implant assembly **430** orally via entry through a mouth of the patient) may or may not be the same as an example embodiment of a second magnetic implant assembly **430** of another system **100** (e.g., a second system **100** for delivering of the magnetic implant assembly **430** rectally via entry through the rectum of the

patient). For example, as illustrated in FIGS. 6A, 6B, 7A, and 7B and as will be further described in the present disclosure, the second magnetic implant assembly **430'** may not include any protrusions **434** and/or indentations **435** and the first magnetic implant assembly **430** may include one or more protrusions **434** and/or one or more indentations **435**.

[0159] Example embodiments of the magnetic implant assembly **430** will now be described with reference to FIGS. 6A-D and 7A-D.

First Example Embodiment of the Magnetic Implant Assembly **430**.

[0160] As illustrated in the cross-sectional side views of FIG. 6A and FIG. 6B and the top views of FIG. 6C and FIG. 6D, a first magnetic implant assembly **430** may be formed as and/or include a flat and cylindrical body **432** (e.g., flat when viewed from the side and circular when viewed from the top). The body **432** is formed, in whole or in part, as or using a ferromagnetic or magnetic material.

[0161] The first magnetic implant assembly **430** includes a front wall **432a** (e.g., the lower wall **432a** of the upper magnetic implant assembly **430** illustrated in FIGS. 6A and 6B; the wall **432a** illustrated in FIGS. 6C and 6D). In example embodiments, the front wall **432a** is the wall that will be magnetically coupled to or facing the second magnetic implant assembly **430'**. The front wall **432a** may have a circular shape with a central axis formed through the center of the front wall **432a**. The front wall **432a** may have a radius of $R1$ (from the central axis), as illustrated in FIGS. 6C-D. In an example embodiment, the first magnetic implant assembly **430** may also include a hole, bore, or the like, through the center axis, as illustrated in at least FIG. 6D. In such embodiments, the hole may have a radius of $R4$ (from the central axis).

[0162] The first magnetic implant assembly **430** includes a rear wall **432b** opposite to the front wall **432a**. In example embodiments, the rear wall **432b** is not the wall that will be magnetically coupled to or facing the second magnetic implant assembly **430'**. The rear wall **432b** will have substantially the same shape and central axis as the front wall **432a**, and radius $R1$ (from the central axis).

[0163] The first magnetic implant assembly **430** includes a first exterior circumferential sidewall **432c** formed around the magnetic body **432**. The first exterior circumferential sidewall **432c** may define a thickness of the magnetic body **432**. The first exterior circumferential sidewall **432c** may be formed at the radius $R1$ from the central axis.

[0164] In an example embodiment, the first magnetic implant assembly **430** includes one or more protrusions **434** formed on the front wall **432a**. The one or more protrusions **434** may be formed in one or more of a plurality of shapes or forms. The one or more protrusions **434** may be formed using ferromagnetic or magnetic material. For example, as illustrated in FIGS. 6A-D, the protrusion **434** may be in the shape of a protruded ring having an exterior radius of $R2$ and an interior radius of $R3$. Alternatively or in addition, the first magnetic implant assembly **430** may include another protrusion (not shown) formed on the protrusion that has radii $R2$ and $R1$ (e.g., in a stepped manner), where the additional protrusion has an exterior radius between $R2$ and $R1$. One or more other protrusions may also be formed on the front wall **432a** and/or another protrusion **432** without departing from the teachings of the present disclosure. In example embodiments, one or more of the protrusions **434** may not necessarily be formed using the same ferromagnetic or magnetic material (and/or the same magnetic force) as the body **432**. For example, one or more of the protrusions **434** may be formed as or using a weaker magnet (or ferromagnetic or magnetic material having a weaker magnetic force) as compared to the ferromagnetic or magnetic material of the body **432**. As another example, one or more of the protrusions **434** may be formed as or using a stronger magnet (or ferromagnetic or magnetic material having a stronger magnetic force) as compared to the ferromagnetic or magnetic material of the first magnetic implant assembly **430**. As another example, one or more of the protrusions **434** may not be formed using (and/or may not include) a ferromagnetic or magnetic material, and instead may be formed using other materials such as plastic, silicon rubber, etc. Other configurations, magnetic coupling strengths, and/or materials/compositions are also contemplated without departing from the teachings of the present

disclosure.

[0165] In example embodiments where the first magnetic implant assembly **430** includes a circular or ring-shaped protrusion **434** formed on the front wall **432a**, the second magnetic implant assembly **430'** may be formed as and/or include a flat and cylindrical body **432'** (e.g., flat when viewed from the side and circular when viewed from the top).

[0166] The second magnetic implant assembly **430'** includes a front wall **432a'** (e.g., the upper wall **432a'** of the lower magnetic implant assembly **430'** illustrated in FIGS. 6A and 6B). In example embodiments, the front wall **432a'** is the wall that will be magnetically coupled to or facing the second magnetic implant assembly **430'**. The front wall **432a'** may have a circular shape with a central axis formed through the center of the front wall **432a'**. The front wall **432a'** may have a radius equal or not equal to R1 (from the central axis), but larger than radius R2. In an example embodiment, the second magnetic implant assembly **430'** may also include a hole, bore, or the like, through the center axis, similar to that of the first magnetic implant assembly **430**.

[0167] The second magnetic implant assembly **430'** includes a rear wall **432b'** opposite to the front wall **432a'**. In example embodiments, the rear wall **432b'** is not the wall that will be magnetically coupled to or facing the first magnetic implant assembly **430**. The rear wall **432b'** will have substantially the same shape and central axis as the front wall **432a'**, and radius (from the central axis).

[0168] The second magnetic implant assembly **430'** includes a second exterior circumferential sidewall **432c'** formed around the magnetic body **432'**. The second exterior circumferential sidewall **432c'** may define a thickness of the magnetic body **432'**. The thickness of the magnetic body **432'** may or may not be the same as the thickness of the magnetic body **432** of the first magnetic implant assembly **430**.

[0169] In an example embodiment, the front wall **432a'** of the second magnetic implant assembly **430'** does not include any protrusions like that of the first magnetic implant assembly **430**. It is recognized in the present disclosure that not having protrusions on the front wall **432a'** of the second magnetic implant assembly **430'** allows for a simple and aligned magnetic coupling with the front wall **432a** (i.e., with the protrusions **434**) of the first magnetic implant assembly **430** (as illustrated in FIG. 6B). It is to be understood, however, that the front wall **432a'** of the second magnetic implant assembly **430'** may also include one or more protrusions (not shown) similar to (e.g., different exterior and/or interior radiuses) or the same as (e.g., same exterior and/or interior radiuses) the protrusions **434** of the front wall **432** of the first magnetic implant assembly **430**.

[0170] It is recognized in the present disclosure that having an exterior portion of the first magnetic implant assembly **430** (i.e., the portion between radius R1 and radius R2) not being magnetically coupled to the second magnetic implant assembly **430'** (or being magnetically coupled to the second magnetic implant assembly **430'** with a lesser magnetic force due to the airgap between the exterior portion of the first magnetic implant assembly **430** and the second magnetic implant assembly **430'**) results in a force or pressure (F1) exerted on a first portion of the cavity wall of the patient (i.e., force between the exterior portion of the first magnetic implant assembly **430** between R1 and R2 and the second magnetic implant assembly **430'**) to be less than a force or pressure (F2) exerted on a second (adjacent) portion of the cavity wall of the patient (i.e., force between the protrusion **434** and the second magnetic implant assembly **430'**). In this regard, it is recognized in the present disclosure that the adjacent application of different forces or pressures, as described above and in the present disclosure, improves the healing (and/or enables better controlled healing) of the anastomosis and/or necrosis formed by the first and second magnetic implant assemblies **430**, **430'**.

[0171] It is to be noted in the present disclosure that the first magnetic implant assembly **430** (and/or the second magnetic implant assembly **430'**) may also include one or more indentations on the front wall **432a** (and **432a'**). For example, the section of the front wall **432a** between the exterior radius R2 and the radius R1 may be an indentation. As another example, the section of the

front wall **432a** between the interior radius **R3** and the central axis (or the radius **R4** for the example embodiment illustrated in FIG. 6D) may be an indentation.

Second Example Embodiment of the Magnetic Implant Assembly **430**.

[0172] As illustrated in the cross-sectional side views of FIG. 7A and FIG. 7B and the top views of FIG. 7C and FIG. 7D, a first magnetic implant assembly **430** may be formed as and/or include a flat and cylindrical body **432** (e.g., flat when viewed from the side and circular when viewed from the top). The body **432** is formed, in whole or in part, as or using a ferromagnetic or magnetic material.

[0173] The first magnetic implant assembly **430** includes a front wall **432a** (e.g., the lower wall **432a** of the upper magnetic implant assembly **430** illustrated in FIGS. 7A and 7B; the wall **432a** illustrated in FIGS. 7C and 7D). In example embodiments, the front wall **432a** is the wall that will be magnetically coupled to or facing the second magnetic implant assembly **430'**. The front wall **432a** may have a circular shape with a central axis formed through the center of the front wall **432a**. The front wall **432a** may have a radius of **R2** (from the central axis), as illustrated in FIGS. 7C-D. In an example embodiment, the first magnetic implant assembly **430** may also include a hole, bore, or the like, through the center axis, as illustrated in at least FIG. 7D. In such embodiments, the hole may have a radius of **R4** (from the central axis).

[0174] The first magnetic implant assembly **430** includes a rear wall **432b** opposite to the front wall **432a**. In example embodiments, the rear wall **432b** is not the wall that will be magnetically coupled to or facing the second magnetic implant assembly **430'**. The rear wall **432b** will have substantially the same shape and central axis as the front wall **432a**, and radius **R1** (from the central axis).

[0175] The first magnetic implant assembly **430** includes a first exterior circumferential sidewall **432c** formed around the magnetic body **432**. The first exterior circumferential sidewall **432c** may define a thickness of the magnetic body **432** (and protrusion **434**, as further described below). The first exterior circumferential sidewall **432c** may be formed at the radius **R2** from the central axis.

[0176] In an example embodiment, the first magnetic implant assembly **430** includes one or more protrusions **434** formed on the front wall **432a**. The one or more protrusions **434** may be formed in one or more of a plurality of shapes or forms. The one or more protrusions **434** may be formed using ferromagnetic or magnetic material. For example, as illustrated in FIGS. 7A-D, the protrusion **434** may be in the shape of a protruded ring having an exterior radius of **R2** and an interior radius of **R3**. One or more other protrusions may also be formed on the front wall **432a** without departing from the teachings of the present disclosure. In example embodiments, one or more of the protrusions **434** may not necessarily be formed using the same ferromagnetic or magnetic material (and/or the same magnetic force) as the body **432**. For example, one or more of the protrusions **434** may be formed as or using a weaker magnet (or ferromagnetic or magnetic material having a weaker magnetic force) as compared to the ferromagnetic or magnetic material of the body **432**. As another example, one or more of the protrusions **434** may be formed as or using a stronger magnet (or ferromagnetic or magnetic material having a stronger magnetic force) as compared to the ferromagnetic or magnetic material of the first magnetic implant assembly **430**. As another example, one or more of the protrusions **434** may not be formed using (and/or may not include) a ferromagnetic or magnetic material, and instead may be formed using other materials such as plastic, silicon rubber, etc. Other configurations, magnetic coupling strengths, and/or materials/compositions are also contemplated without departing from the teachings of the present disclosure.

[0177] As illustrated in FIGS. 7A-B, the first magnetic implant assembly **430** includes an exterior cylindrical or ring-shaped body **436**. The exterior body **436** is formed around and fixedly secured to the first exterior circumferential sidewall **432c** of the magnetic body **432**. The exterior body **436** includes a front exterior cylindrical or ring-shaped portion, at least a portion of which is adjacent to the front wall **432a**. The exterior body **436** may also include a rear exterior cylindrical or ring-shaped portion adjacent to the rear wall **432b**. In an example embodiment, the exterior body **436**

may or may not be a wholly or fully magnetic body and/or may or may not be magnetically coupled to the second magnetic implant assembly **430'** when the first magnetic implant assembly **430** is magnetically coupled to the second magnetic implant assembly **430'**. For example, the exterior body **436** may not be formed using (and/or may not include) a ferromagnetic or magnetic material, and instead may be formed using other materials such as plastic, silicon rubber, etc. As another example, the exterior body **436** may be formed as or using a weaker magnet (or ferromagnetic or magnetic material having a weaker magnetic force) as compared to the ferromagnetic or magnetic material of the body **432** and/or the protruding portion **434**. As another example, the exterior body **436** may be partially formed as or using one or more magnets and/or magnetic sections (or ferromagnetic or magnetic material) having a same, weaker, and/or stronger magnetic force as compared to the ferromagnetic or magnetic material of the body **432** and/or the protruding portion **434**. It is recognized in the present disclosure that having the exterior body **436** not being magnetically coupled to the second magnetic implant assembly **430'** (or being magnetically coupled to the second magnetic implant assembly **430'** with a lesser magnetic force) results in a force or pressure (**F1**) exerted on a first portion of the cavity wall of the patient (i.e., force between the exterior body **436** and the second magnetic implant assembly **430'**) to be less than a force or pressure (**F2**) exerted on a second (adjacent) portion of the cavity wall of the patient (i.e., force between the protrusion **434** and the second magnetic implant assembly **430'**). In this regard, it is recognized in the present disclosure that the adjacent application of different forces or pressures, as described above and in the present disclosure, improves the healing (and/or enables better controlled healing) of the anastomosis and/or necrosis formed by the first and second magnetic implant assemblies **430**, **430'**.

[0178] In example embodiments where the first magnetic implant assembly **430** includes a circular or ring-shaped protrusion **434** formed on the front wall **432a**, the second magnetic implant assembly **430'** may be formed as and/or include a flat and cylindrical body **432'** (e.g., flat when viewed from the side and circular when viewed from the top).

[0179] The second magnetic implant assembly **430'** includes a front wall **432a'** (e.g., the upper wall **432a'** of the lower magnetic implant assembly **430'** illustrated in FIGS. 7A and 7B). In example embodiments, the front wall **432a'** is the wall that will be magnetically coupled to or facing the second magnetic implant assembly **430'**. The front wall **432a'** may have a circular shape with a central axis formed through the center of the front wall **432a'**. The front wall **432a'** may have a radius equal or not equal to **R1** (from the central axis), but larger than radius **R2**. In an example embodiment, the second magnetic implant assembly **430'** may also include a hole, bore, or the like, through the center axis, similar to that of the first magnetic implant assembly **430**.

[0180] The second magnetic implant assembly **430'** includes a rear wall **432b'** opposite to the front wall **432a'**. In example embodiments, the rear wall **432b'** is not the wall that will be magnetically coupled to or facing the first magnetic implant assembly **430**. The rear wall **432b'** will have substantially the same shape and central axis as the front wall **432a'**, and radius (from the central axis).

[0181] The second magnetic implant assembly **430'** includes a second exterior circumferential sidewall **432c'** formed around the magnetic body **432'**. The second exterior circumferential sidewall **432c'** may define a thickness of the magnetic body **432'** and the protrusion **434**. The thickness of the magnetic body **432'** may or may not be the same as the thickness of the magnetic body **432** of the first magnetic implant assembly **430**.

[0182] In an example embodiment, the front wall **432a'** of the second magnetic implant assembly **430'** does not include any protrusions like that of the first magnetic implant assembly **430**. It is recognized in the present disclosure that not having protrusions on the front wall **432a'** of the second magnetic implant assembly **430'** allows for a simple and aligned magnetic coupling with the front wall **432a** (i.e., with the protrusions **434**) of the first magnetic implant assembly **430** (as illustrated in FIG. 7B). It is to be understood, however, that the front wall **432a'** of the second

magnetic implant assembly **430'** may also include one or more protrusions (not shown) similar to (e.g., different exterior and/or interior radiuses) or the same as (e.g., same exterior and/or interior radiuses) the protrusions **434** of the front wall **432** of the first magnetic implant assembly **430**.

[0183] It is to be noted in the present disclosure that the first magnetic implant assembly **430** (and/or the second magnetic implant assembly **430'**) may also include one or more indentations **435** on the front wall **432a** (and **432a'**). For example, the section **435** of the front wall **432a** between the interior radius R3 and the central axis (or the radius R4 for the example embodiment illustrated in FIG. 6D) may be an indentation.

[0184] It is to be understood in the present disclosure that an exterior body (e.g., similar to the exterior body **436** described above for the first magnetic implant assembly **430**) may be formed around and fixedly secured to the second exterior circumferential sidewall **432c'** of the second magnetic implant assembly **430'** in addition to or in replacement of the exterior body **436** for the first magnetic implant assembly **430** without departing from the teachings of the present disclosure. The Securing Assembly (e.g., Securing Assembly **440**).

[0185] As illustrated in at least FIGS. 2 and 5A-H, the second main body assembly **400** includes a securing assembly (e.g., securing assembly **440**). The securing assembly **440** is configured to secure the magnetic implant assembly **430** to and unsecure the magnetic implant assembly **430** from the second main body **402**.

[0186] The securing assembly **440** may be configured in one or more configurations. For example, as illustrated in FIGS. 5A-B and FIGS. 5E-F and as will be further described in the present disclosure, the securing assembly **440** may be configured in the form of a gripper **440**, or the like. As another example, as illustrated in FIGS. 5C-D and as will be further described in the present disclosure, the securing assembly **440** may be configured in the form of a quarter or half rotation screw lock **440**. As another example, as illustrated in FIGS. 5G-H and as will be further described in the present disclosure, the securing assembly **440** may be configured using a snare **440**, or the like.

[0187] Example embodiments of the securing assembly **440** will now be described with reference to FIGS. 5A-H.

First Example Embodiment of the Securing Assembly **440**.

[0188] As illustrated in the perspective view of FIG. 5A and cross-sectional view of FIG. 5B, a securing assembly **440** may be formed as and/or include a gripper **440**, or the like, for gripping at least a portion of the exterior circumferential sidewall **432c** of the magnetic implant assembly **430**. In an example embodiment, the magnetic implant assembly **430** may include a hole **430a** for use in receiving a protruding portion **402a** of the second main body **402**. In example embodiments, the gripper **440** may be sized so as to grip more than half of the circumference or perimeter of the magnetic implant assembly **430**.

[0189] In example embodiments, a wire or cable (not shown) may be secured at one end to a portion of the magnetic implant assembly **430** and provided through the hole **430a**, second main body **402**, and back to the controller (not shown) and/or surgeon console. Such a wire or cable may be useful in situations where the magnetic implant assembly **430** is accidentally, unintentionally, or mistakenly unsecured from the securing assembly **440**. Such a wire or cable may be cut or disconnected at or close to the magnetic implant assembly **430** upon the magnetic implant assembly **430** being magnetically coupled to another magnetic implant assembly **430'**.

Second Example Embodiment of the Securing Assembly **440**.

[0190] As illustrated in the perspective view of FIG. 5C and cross-sectional view of FIG. 5D, a securing assembly **440** may be formed as and/or include a screw lock **440**, or the like, for securing and unsecuring the magnetic implant assembly **430**. More specifically, the magnetic implant assembly **430** may include a quarter or half rotation (or other) threaded hole **430a** for use in receiving a screw-like protruding portion **402a** of the second main body **402**.

[0191] As described above and in the present disclosure, a wire or cable (not shown) may be

secured at one end to a portion of the magnetic implant assembly **430** and provided through the hole **430a**, second main body **402**, and back to the controller (not shown) and/or surgeon console (not shown).

Third Example Embodiment of the Securing Assembly **440**.

[0192] As illustrated in the perspective view of FIG. 5E and cross-sectional view of FIG. 5F, a securing assembly **440** may be formed as and/or include a gripper **440**, or the like, for gripping at least a portion of the front wall **432a** and rear wall **432b** (and/or at least a portion of a protrusion (not shown) similar to the protrusions illustrated in FIGS. 6A-D and 7A-D) of the magnetic implant assembly **430**. In an example embodiment, the magnetic implant assembly **430** may include a hole **430a** for use in receiving a protruding portion **402a** of the second main body **402**.

[0193] As described above and in the present disclosure, a wire or cable (not shown) may be secured at one end to a portion of the magnetic implant assembly **430** and provided through the hole **430a**, second main body **402**, and back to the controller (not shown) and/or surgeon console.

Fourth and Fifth Example Embodiments of the Securing Assembly.

[0194] As illustrated in the perspective view of FIG. 5G, the securing assembly may be formed as and/or include a snare, or the like, for securing and unsecuring the magnetic implant assembly **430**. More specifically, the magnetic implant assembly **430** includes a first cable **442** secured at one end to an actuatable member **444** and provided through the system **100** to a second end (i.e., at the end where the controller (not shown) or surgeon console (not shown) is located). The magnetic implant assembly **430** also includes a second cable **446** secured at both ends to the actuatable member **444**. The second cable **446** is fittedly provided in a groove, channel, or the like, formed around the exterior circumferential sidewall **432c** of the magnetic implant assembly **430** so as to prevent the second cable **446** from sliding around relative to and/or coming away from the exterior circumferential sidewall **432c**.

[0195] When the magnetic implant assembly **430** is to be secured to the second main body **402**, the first cable **442** is persistently pulled at the second end in such a way that the actuatable member **444** is persistently held in a first (secured) position (i.e., a furthest position from the magnetic implant assembly **430** (or a furthest position from the most distal point of the second end of the second main body **402**)). At the first (secured) position, the second cable **446** is persistently held in the groove of the exterior circumferential sidewall **432c**, and the magnetic implant assembly **430** is accordingly secured to the second main body **402**.

[0196] When the magnetic implant assembly **430** is to be unsecured from the second main body **402** (e.g., when the magnetic implant assembly **430** is to be magnetically coupled to another magnetic implant assembly **430'**), the first cable **442** is released or pushed from the second end in such a way that the actuatable member **444** is moved to a second (unsecured) position (i.e., a position that is closer to the magnetic implant assembly **430** (or a position that is closer to the most distal point of the second end of the second main body **402**)). At the second (unsecured) position, the second cable **446** is no longer persistently held in the groove of the exterior circumferential sidewall **432c** (i.e., becomes looser, becomes a bigger loop), and the magnetic implant assembly **430** is accordingly unsecured or unsecurable from the second main body **402**.

[0197] The securing assembly may be formed as and/or include other configurations of a snare, or the like, for securing and unsecuring the magnetic implant assembly **430**. For example, as illustrated in FIG. 5H, the magnetic implant assembly **430** includes a first cable **442** secured at one end to an actuatable member **444**, provided in a groove, channel, or the like, formed around the exterior circumferential sidewall **432c** of the magnetic implant assembly **430**, provided through the second main body **402**, and provided through the rest of the system **100** to a second end (i.e., at the end where the controller (not shown) or surgeon console (not shown) is located). The first cable **442** is fittedly provided in the groove formed around the exterior circumferential sidewall **432c** of the magnetic implant assembly **430** so as to prevent the second cable **446** from sliding around relative to and/or coming away from the exterior circumferential sidewall **432c**.

[0198] When the magnetic implant assembly **430** is to be secured to the second main body **402**, the first cable **442** is persistently pulled at the second end in such a way that the magnetic implant assembly **430** is persistently secured by the first cable **442**.

[0199] When the magnetic implant assembly **430** is to be unsecured from the second main body **402** (e.g., when the magnetic implant assembly **430** is to be magnetically coupled to another magnetic implant assembly **430'**), the first cable **442** is released or pushed from the second end in such a way that the first cable **442** is no longer persistently held in the groove of the exterior circumferential sidewall **432c** (i.e., becomes looser, becomes a bigger loop), and the magnetic implant assembly **430** is accordingly unsecured or unsecurable from the second main body **402**.

[0200] Other embodiments and/or configurations of snares are also contemplated in the present disclosure.

[0201] As described above and in the present disclosure, a wire or cable (not shown) may be secured at one end to a portion of the magnetic implant assembly **430** and provided through the hole **430a**, second main body **402**, and back to the controller (not shown) and/or surgeon console. Example Embodiments of a Method of Delivering a Magnetic Implant Assembly Rectally.

[0202] FIGS. **8A-8Q** illustrate an example embodiment of a method of delivering a magnetic implant assembly. The magnetic implant assembly may be or include one or more example embodiments of the magnetic implant assembly **430** (or second magnetic implant assembly **430'**) described in the present disclosure. The method may be performed using one or more example embodiments of the endoscopic anastomosis system **100** described in the present disclosure, which includes the first main body assembly **200**, the head assembly **300**, the second main body assembly **400**, and a first magnetic implant assembly **430**. The method illustrated in FIGS. **8A-8Q** is an example embodiment of a method of delivering a magnetic implant assembly rectally (i.e., via the rectum and anus).

[0203] As illustrated in FIG. **8A** and FIG. **8B**, an example embodiment of the method includes inserting the system **100** (which includes example embodiments of the first main body assembly **200**, the head assembly **300**, and the second main body assembly **400**, as described in the present disclosure) in the anus and through the rectum.

[0204] As illustrated in FIG. **8C** and FIG. **8D**, an example embodiment of the method includes bending or turning the system **100** (e.g., via the first bendable section **210** of the first main body assembly **200**) and further advancing the system **100** around one or more bends or turns and into the sigmoid colon.

[0205] As illustrated in FIG. **8E**, once the head assembly **300** is advanced around the bend or turn, the head assembly **300** (and first main body assembly **200**) is anchored or secured to a portion of the cavity wall of the patient (e.g., by controlling/configuring the first expandable member **320** to expand radially outwards while also controlling/configuring the first pressure port (not shown here, but as described for first pressure port **332**) to provide a negative pressure to encourage, bring in, suction inward, and/or collapse a portion of cavity wall of the patient towards the head assembly **300**).

[0206] As illustrated in FIG. **8F**, once the cavity wall of the patient is anchored or secured to the head assembly **300** and the first main body assembly **200** (as described above for FIG. **8E**), the bend or turn in the colon may be straightened by pulling the system **100** (i.e., pulling the first main body assembly **200**) back so as to concertina the colon.

[0207] As illustrated in FIG. **8G**, once the colon has been straightened, the system **100** can be unanchored or unsecured from the cavity wall of the colon (e.g., by unexpanding or deflating the first expandable member **320** and not applying negative pressure (or applying positive pressure) by the first pressure port).

[0208] As illustrated in FIG. **8H** and FIG. **8I**, an example embodiment of the method includes advancing the system **100** through the descending colon, splenic flexure, transverse colon, hepatic flexure, and ascending colon until the head assembly **300** approaches the ileocecal valve.

[0209] As illustrated in FIG. 8J, FIG. 8K, and FIG. 8L, an example embodiment of the method includes bending or turning the system **100** (e.g., via the first bendable section **210** of the first main body assembly **200**) and further advancing the system **100** into and through the ileocecal valve. [0210] As illustrated in FIG. 8M, once the head assembly **300** is advanced into the ileocecal valve and to the small bowel, the head assembly **300** (and first main body assembly **200**) is anchored or secured to a portion of the cavity wall (e.g., by controlling/configuring the first expandable member **320** to expand radially outwards while also controlling/configuring the first pressure port (not shown here, but as described for first pressure port **332**) to provide a negative pressure to encourage, bring in, suction inward, and/or collapse a portion of cavity wall of the patient towards the head assembly **300**).

[0211] As illustrated in FIG. 8N, once the cavity wall of the patient is anchored or secured to the head assembly **300** and first main body assembly **200** (as described above for FIG. 8M), the second main body assembly **400** (which includes example embodiments of the second main body **402**, the second expandable member **420**, and the magnetic implant assembly **430**, as described in the present disclosure) is advanced forward by sliding the second main body assembly **400** forward relative to the anchored first main body assembly **200**.

[0212] As illustrated in FIG. 8O, after the second main body assembly **400** has been advanced forward, the second main body assembly **400** may be anchored or secured to a portion of the cavity wall (e.g., by controlling/configuring the second expandable member **420** to expand radially outwards while also controlling/configuring the second pressure port (not shown here, but as described for second pressure port **442**) to provide a negative pressure to encourage, bring in, suction inward, and/or collapse a portion of cavity wall of the patient towards the second main body **402**).

[0213] As illustrated in FIG. 8P, once the cavity wall of the patient is anchored or secured to the second main body assembly **400** (as described above for FIG. 8O), the head assembly **300** and first main body assembly **200** may be unanchored or unsecured (e.g., by unexpanding or deflating the first expandable member **320** and not applying negative pressure (or applying positive pressure) by the first pressure port).

[0214] As illustrated in FIG. 8Q, after the head assembly **300** and the first main body assembly **200** have been unanchored or unsecured (as described above for FIG. 8P), the first main body assembly **200** and head assembly **300** are advanced forward toward the second expandable member **420** of the second main body assembly **400**.

[0215] As illustrated in FIG. 8R, the head assembly **300** and first main body assembly **200** are then anchored or secured to a portion of the cavity wall of the patient (e.g., by controlling/configuring the first expandable member **320** to expand radially outwards while also controlling/configuring the first pressure port to provide a negative pressure to encourage, bring in, suction inward, and/or collapse a portion of cavity wall of the patient towards the head assembly **300**). As explained for FIG. 8N, the second main body assembly **400** is then advanced forward.

[0216] As illustrated in FIG. 8S, the steps illustrated in FIGS. 8N-8R are repeated as necessary until the head assembly **300** approaches a distance of about 20-30 cm from the ileocecal valve. At this point, the system **100** is ready to deliver the magnetic implant assembly **430** to magnetically couple with another magnetic implant assembly **430'** delivered orally (i.e., via the mouth). Example Embodiments of a Method of Delivering a Magnetic Implant Assembly Orally.

[0217] FIGS. 9A-9I illustrate another example embodiment of a method of delivering a magnetic implant assembly. The magnetic implant assembly may be or include one or more example embodiments of the magnetic implant assembly **430'** (or **430**) described in the present disclosure. The method may be performed using one or more example embodiments of the endoscopic anastomosis system **100** described in the present disclosure, which includes the first main body assembly **200**, the head assembly **300**, the second main body assembly **400**, and a second magnetic implant assembly **430'**. The method illustrated in FIGS. 9A-9I is an example embodiment of a

method of delivering a magnetic implant assembly orally (i.e., via the mouth and esophagus).

[0218] As illustrated in FIG. 9A and FIG. 9B, an example embodiment of the method includes inserting the system **100** (which includes example embodiments of the first main body assembly **200**, the head assembly **300**, and the second main body assembly **400**, as described in the present disclosure) in the mouth, through the esophagus, and through the stomach. Once the system **100** has reached the pyloric sphincter, the system **100** is configured to bend or turn (e.g., via the first bendable section **210** of the first main body assembly **200**) into the pyloric sphincter.

[0219] As illustrated in FIG. 9C, an example embodiment of the method includes further advancing the system **100** to the duodenum.

[0220] As illustrated in FIG. 9D, once the head assembly **300** is advanced forward, the system **100** is anchored or secured to a portion of the cavity wall of the patient (e.g., by controlling/configuring the first expandable member **320** to expand radially outwards while also controlling/configuring the first pressure port (not shown here, but as described for first pressure port **332**) to provide a negative pressure to encourage, bring in, suction inward, and/or collapse a portion of cavity wall of the patient towards the head assembly **300**).

[0221] As illustrated in FIG. 9E, once the cavity wall of the patient is anchored or secured to the system **100** (as described above for FIG. 9D), the bend or turn may be straightened by pulling the system **100** (i.e., pulling the first main body assembly **200**) back so as to concertina the duodenum.

[0222] As illustrated in FIG. 9F, the second main body assembly **400** (which includes example embodiments of the second main body **402**, the second expandable member **420**, and the magnetic implant assembly **430**, as described in the present disclosure) is advanced forward by sliding the second main body assembly **400** forward relative to the anchored first main body assembly **200**.

[0223] As illustrated in FIG. 9G, after the second main body assembly **400** has been advanced forward, the second main body assembly **400** may be anchored or secured to a portion of the cavity wall (e.g., by controlling/configuring the second expandable member **420** to expand radially outwards while also controlling/configuring the second pressure port (not shown here, but as described for second pressure port **442**) to provide a negative pressure to encourage, bring in, suction inward, and/or collapse a portion of cavity wall of the patient towards the second main body **402**).

[0224] As illustrated in FIG. 9H, once the cavity wall of the patient is anchored or secured to the second main body assembly **400** (as described above for FIG. 9G), the head assembly **300** and first main body assembly **200** may be unanchored or unsecured (e.g., by unexpanding or deflating the first expandable member **320** and not applying negative pressure (or applying positive pressure) by the first pressure port).

[0225] As illustrated in FIG. 9I, after the head assembly **300** and the first main body assembly **200** have been unanchored or unsecured (as described above for FIG. 8P), the first main body assembly **200** and head assembly **300** are advanced forward toward the second expandable member **420** of the second main body assembly **400**. The steps illustrated in FIGS. 9D-9I are repeated as necessary until the head assembly **300** approaches the target location/position in the distal duodenum/jejunum (i.e., the location where the first magnetic implant assembly **430** has been delivered, as described in FIGS. 8A-S). The method then includes delivering the second magnetic implant assembly **430'** to magnetically couple with magnetic implant assembly **430** delivered rectally (as described in FIGS. 8A-S).

[0226] As shown in FIGS. 10-12D, exemplary embodiments of the present application provide an endoscopic magnetic anastomosis system that may include at least one endoscope assembly **500**. The endoscope assembly **500** includes an endoscope **51** and an adjustable snare mechanism. The endoscope **51** may include a flexible body or flexible tube having one or more internal channels (not shown). In an exemplary embodiment, the endoscope **51** has a first end and an opposite second end, and includes a magnetic implant assembly **511** disposed at the second end and a snare channel extending from the first end to the second end.

[0227] The adjustable snare mechanism includes a snare assembly **52** and a snare guide assembly **53**. The snare assembly **52** passes through the snare channel and includes a snare catheter **521** and a snare wire **522** passing through the snare catheter **521** for selectively tightening and releasing the magnetic implant assembly **511**.

[0228] In an exemplary embodiment, in an endoscopic magnetic anastomosis system, the endoscope assembly **500** includes an endoscope **51** and an adjustable snare mechanism. A first end of the endoscope **51** is a proximal end (i.e., an end located outside the human body), a second end of the endoscope **51** is a distal end (i.e., an end for entering the human body), and the second end of the endoscope **51** is provided with a magnetic implant assembly **511**. The adjustable snare mechanism is used to secure the magnetic implant assembly **511** at the second end of the endoscope **51** and can also release the magnetic implant assembly **511** in order to place the magnetic implant assembly **511** into the human body.

[0229] In an exemplary embodiment, as shown in FIG. **10**, the endoscopic magnetic anastomosis system may include two endoscope assemblies **500**, where one endoscope assembly **500** may deploy one magnetic implant assembly **511** to the proximal jejunum through the upper gastrointestinal tract, and the other endoscope assembly **500** may deploy another magnetic implant assembly **511** to the distal ileum through the lower gastrointestinal tract. The structure of the two endoscope assemblies **500** may be the same or different, for example, the endoscope assembly **500** through the upper gastrointestinal tract may not include the outer sheath **55** (see below for detailed description).

[0230] The endoscope **51** is also provided with a snare channel, the snare channel can extend from one end of the endoscope **51** (e.g., the distal end into the human body) to the other end (e.g., the proximal end located outside the human body), the snare assembly **52** in the adjustable snare mechanism may pass through the snare channel to secure and release the magnetic implant assembly **511**.

[0231] The snare assembly **52** of the adjustable snare mechanism includes a snare wire **522** and a snare catheter **521**, the snare wire **522** may pass through the snare catheter **521**, and both of the snare wire **522** and the snare catheter **521** may be bent and deformed to accommodate a curved lumen or intestinal tract in a human body. In an exemplary embodiment, the snare wire **522** may be made of stainless steel, the snare catheter **521** may be made of Teflon (polytetrafluoroethylene), or the materials of the snare wire **522** and the snare catheter **521** may be adjusted as desired.

[0232] In some exemplary embodiments, as shown in FIG. **12A** to FIG. **12D**, the adjustable snare mechanism further comprises a snare guide assembly **53**. The snare guide assembly **53** comprises a base body **531** and a motion mechanism movably installed on the base body **531**. The motion mechanism is connected to and is configured to drive at least one of the snare wire **522** and the snare catheter **521**, such that there is a relative movement between the snare wire **522** and the snare catheter **521** to allow the snare wire **522** to selectively extend and retract relative to the snare catheter **521**. In an exemplary embodiment, the snare guide assembly **53** may include a first movable member **532**, and a second movable member **533**. Both the first movable member **532** and the second movable member **533** are movably mounted on the base body **531**. The first movable member **532** is connected to the snare wire **522** and arranged to drive the snare wire **522** to move, and the second movable member **533** is connected to the snare catheter **521** and arranged to drive the snare catheter **521** to move such that when at least one of the first movable member **532** and the second movable member **533** is moved, there is the relative movement between the snare wire **522** and the snare catheter **521** that allows the snare wire **522** to extend and retract relative to the snare catheter **521**.

[0233] The adjustable snare mechanism includes a snare guide assembly **53** in addition to the snare assembly **52**. In the snare guide assembly **53**, both the first movable member **532** and the second movable member **533** are movably mounted on the base body **531**, and the first movable member **532** is connected with the snare wire **522**, and the second movable member **533** is connected with

the snare catheter **521**, so that the first movable member **532** can drive the snare wire **522** to move when the first movable member moves, and the second movable member **533** can drive the snare catheter **521** to move when the second movable member moves. Upon movement of at least one of the first movable member **532** and the second movable member **533**, there is a relative movement between the snare wire **522** and the snare catheter **521** such that the snare wire **522** can be retracted into and extended from the snare catheter **521**. During use, the snare wire **522** may first be retracted into the snare catheter **521** so that the snare assembly **52** passes through the snare channel of the endoscope **51** (as shown in the state (a1) in FIG. **12A** and FIG. **12B**); then, the snare wire **522** may be moved to extend out of the snare catheter **521** in order to install the magnetic implant assembly **511** into a portion of the snare wire **522** extending outside the snare catheter **521** (as shown in state (a2) in FIG. **12A**); after installation of the magnetic implant assembly **511**, the snare wire **522** may be retracted further into the snare catheter **521** so that the snare wire **522** tightens the magnetic implant assembly **511** (as shown in state (a3) in FIG. **12A** and FIG. **12C**); after the magnetic implant assembly **511** reaches a suitable position (set position) in the human body, the snare wire **522** may be further extended out of the snare catheter **521** to release the magnetic implant assembly **511**, releasing the magnetic implant assembly **511** in this suitable position (as shown in state (a4) in FIG. **12A** and FIG. **12D**).

[0234] In some exemplary embodiments, as shown in FIGS. **12A** to **12D**, the base body **531** is provided with a first slide rail **5313** and a second slide rail **5314** extending along the length direction of the base body **531**, the first movable member **532** is disposed to be slidable along the first slide rail **5313**, and the second movable member **533** is disposed to be slidable along the second slide rail **5314**.

[0235] The base body **531** of the snare guide assembly **53** is provided with a first slide rail **5313** and a second slide rail **5314** extending along the length direction thereof, and the first movable member **532** and the second movable member **533** can slide along the first slide rail **5313** and the second slide rail **5314** respectively, thereby driving the snare wire **522** and the snare catheter **521** to move, so that the snare wire **522** can be retracted into and extended out of the snare catheter **521**.

[0236] In some exemplary embodiments, as shown in FIGS. **12B** to **12D**, the first slide rail **5313** and the second slide rail **5314** are sequentially disposed along the length direction of the base body **531**, and one end of the second movable member **533** away from the first movable member **532** is connected to the first end (i.e., the proximal end) of the snare catheter **521**. The snare guide assembly **53** also includes a connecting member **534**, one end of the snare wire **522** passes out of the first end of the snare catheter **521** and is fixedly connected with one end of the connecting member **534**, and the other end of the connecting member **534** is connected with the first movable member **532**.

[0237] As shown in FIGS. **12B** to **12D**, both the first slide rail **5313** and the second slide rail **5314** extend along the length direction of the base body **531**, and the first slide rail **5313** is located on the left side of the second slide rail **5314**, the right end of the second movable member **533** is connected to the first end of the snare catheter **521**, and the second movable member **533** can drive the snare catheter **521** to move; the end portion of the snare wire **522** passes out of the first end of the snare catheter **521**, and is connected with the first movable member **532** through the connecting member **534**, so that the first movable member **532** can drive the snare wire **522** to move through the connecting member **534**. In an exemplary embodiment, the connector **534** may be a connecting crimp rod. For example, the connecting member **534** may be a hollow connecting tube, and the end portion of the snare wire **522** may extend into the connecting tube after passing out of the first end of the snare catheter **521**, and the connection between the connecting member **534** and the snare wire **522** can be achieved by crimping.

[0238] In some exemplary embodiments, as shown in FIGS. **12A** to **12D**, the base body **531** includes a first base body portion **5311** and a second base body portion **5312** connected to each other, the first slide rail **5313** includes a slide slot provided on the first base body portion **5311**, and

the first movable member **532** includes a slider cooperating with the slide slot; the second slide rail **5314** is provided on the outer surface of the second base body portion **5312**, and the second movable member **533** is sleeved on the outer surface of the second base body portion **5312**.
[0239] The base body **531** includes a first base body portion **5311** and a second base body portion **5312**, and the first base body portion **5311** is provided with a slide slot to form a first slide rail **5313**, and a slider of the first movable member **532** can cooperate with the slide slot so that the first movable member **532** slides along the first slide rail **5313**.

[0240] The outer surface of the second base body portion **5312** may form the second slide rail **5314**, or the outer surface of the second base body portion **5312** may be provided with the second slide rail **5314**, the second movable member **533** is sleeved on the outer surface of the second base body portion **5312**, and the second movable member **533** is in sliding fit with the second slide rail **5314** so that the second movable member **533** slides along the second slide rail **5314**. In an exemplary embodiment, the second movable member **533** may include a sheath segment **5331** and a fixed segment **5332**, the sheath segment **5331** may be sleeved on an outer surface of the second base body portion **5312** and may be slidable relative to the second base body portion **5312**; the fixed segment **5332** can be fixedly connected with the sleeve section **5331**, and the fixed segment **5332** is connected with the snare catheter **521** to realize the connection between the second movable member **533** and the snare catheter **521**.

[0241] The connecting member **534** may be disposed in the slide slot of the first base body portion **5311**, and the end portion of the snare wire **522** may be connected to the connecting member **534** after passing through the snare catheter **521**, the second movable member **533**, and the second base body portion **5312**.

[0242] In some exemplary embodiments, as shown in FIGS. **12A-12D**, the snare guide assembly **53** further includes a releasable locking mechanism **54** mounted to the first movable member **532**, the releasable locking mechanism **54** is arranged to be in locking fit with the base body **531** to secure the first movable member **532** to the base body **531**, and to disengage from being in locking fit with the base body **531** to enable the first movable member **532** to move relative to the base body **531**.

[0243] In the snare guide assembly **53**, the releasable locking mechanism **54** has a locked state and a released state. As shown in the state (a3) of FIG. **12A**, when the releasable locking mechanism **54** is in the locked state, it can be in locking fit with the base body **531** (such as the first base body portion **5311**), and at this time, the releasable locking mechanism **54** is fixed with the base body **531**, so that the first movable member **532** is fixed with the base body **531** to prevent the snare wire **522** from moving, and so that the snare wire **522** can be maintained in the state of tightening the magnetic implant assembly **511** to firmly secure the magnetic implant assembly **511**. As shown in the states (a1), (a2) and (a4) in FIG. **12A**, when the releasable locking mechanism **54** is in the released state, the releasable locking mechanism may be disengaged from being in locking fit with the base body **531** (e.g., the first base body portion **5311**), and at this time, the releasable locking mechanism **54** is disengaged from being in fixation with the base body **531**, so that the first movable member **532** can move relative to the base body **531**, allowing the first movable member **532** to drive the snare wire **522** to move, so that the snare wire **522** can be retracted into the snare catheter **521**, and the magnetic implant assembly **511** can be installed and released.

[0244] In some exemplary embodiments, as shown in FIGS. **12A to 12D**, the releasable locking mechanism **54** includes a rotary locking mechanism including a knob **541** and a clamping member **542**, the knob **541** is connected to the clamping member **542**, and the knob **541** is configured to be rotatable in two opposite directions to drive the clamping member **542** to clamp and release the base body **531** accordingly.

[0245] The releasable locking mechanism **54** may be a rotary locking mechanism mounted to the first movable member **532**, and the knob **541** thereof may be connected to the clamping member **542**, and when the knob **541** is rotated, the knob can drive the clamping member **542** to move. Specifically, when the knob **541** is rotated in one direction (for example, the counterclockwise

direction or the clockwise direction indicated by the curved arrow in the state (a4) in FIG. 12A), the knob 541 can drive the clamping member 542 to release the base body 531 (for example, the first base body portion 5311) so that the releasable locking mechanism 54 and the first movable member 532 move relative to the base body 531; when the knob 541 is rotated in the opposite direction (e.g., the clockwise direction or the counterclockwise direction indicated by the curved arrow in the state (a3) of FIG. 12A), the knob 541 can drive the clamping member 542 to clamp the base body 531 (e.g., the first base body portion 5311), so that the releasable locking mechanism 54 and the first movable member 532 are clamped and fixed to the base body 531, preventing the releasable locking mechanism 54 and the first movable member 532 from moving relative to the base body 531.

[0246] During use, as shown in the state (a1) of FIG. 12A, the second movable member 533 can be slid along the base body 531 (as shown in the direction shown by the straight arrow in the state (a1) of FIG. 12A) first, so that the second movable member 533 moves in a direction away from the first base body portion 5311 and drives the snare catheter 521 to move, so that the snare wire 522 can be gradually retracted into the snare catheter 521, which facilitates the snare assembly 52 to pass through the snare channel of the endoscope 51; then, as shown in the state (a2) of FIG. 12A, the second movable member 533 can be slid in the opposite direction (as shown in the direction shown by the straight arrow in the state (a2) of FIG. 12A) to move the second movable member 533 in the direction close to the first base body portion 5311 and drive the snare catheter 521 to move so that the snare wire 522 can gradually extend out of the snare catheter 521, at this time, the snare wire 522 is in a released state, so as to install the magnetic implant assembly 511 into the portion of the snare wire 522 extending out of the snare catheter 521; after installing the magnetic implant assembly 511, as shown in the state (a3) in FIG. 12A, the first movable member 532 can be slid along the base body 531 (as shown in the direction indicated by the straight arrow in the state (a3) in FIG. 12A) to move the first movable member 532 in the direction away from the second base body portion 5312, and drive the snare wire 522 to move so that the snare wire 522 can gradually retract into the snare catheter 521 so that the snare wire 522 tightens the magnetic implant assembly 511, and the knob 541 of the releasable locking mechanism 54 is rotated (in the direction indicated by the curve arrow in the state (a 3) in FIG. 12A) to lock the first movable member 532 with the base body 531, the snare wire 522 always remains tightened and fixes the state of the magnetic implant assembly 511 during intubation of the endoscope 51 and delivery of the snare guide assembly 53. When the magnetic implant assembly 511 needs to be released after reaching a suitable position in the human body, as shown in the state (a4) of FIG. 12A, the knob 541 of the releasable locking mechanism 54 can be rotated in an opposite direction (as shown by the curved arrow in the state (a4) of FIG. 12A) to unlock the first movable member 532 from the base body 531, then the first movable member 532 can be moved in a direction close to the second base body portion 5312 (as indicated by the straight arrow in state (a4) in FIG. 12A), and drive the snare wire 522 to move so that the snare wire 522 can gradually extend out of the snare catheter 521 to release the magnetic implant assembly 511. At this time, the magnetic implant assembly 511 can be released in the appropriate position. When the snare wire 522 releases the magnetic implant assembly 511 to release the magnetic implant assembly 511, the connecting member 534 may remain within the snare catheter 521 (as shown in FIG. 12D), preventing the connecting member 534 from pinching any tissue in the human body.

[0247] In some exemplary embodiments, the snare guide assembly 53 further includes a unlockable locking device mounted to the second movable member 533, the unlockable locking device is arranged to be in locking fit with the base body 531 to secure the second movable member 533 to the base body 531, and to disengage from being in locking fit with the base body 531 to enable the second movable member 533 to move relative to the base body 531.

[0248] In the snare guide assembly 53, the unlockable locking device may have a locked state and a released state. When the unlockable locking device is in the locking state, the unlockable locking

device can be in locking fit with the base body **531** (such as the second base body portion **5312**), and at this time, the unlockable locking device is fixed with the base body **531**, so that the second movable member **533** is fixed with the base body **531** to prevent the snare catheter **521** from moving; when the unlockable locking device is in the released state, the unlockable locking device can be disengaged from being in locking fit with the base body **531** (such as the second base body portion **5312**), and at this time, the unlockable locking device is disengaged from being in fixation with the base body **531**, so that the second movable member **533** can move relative to the base body **531**, so that the second movable member **533** drives the snare catheter **521** to move. In an exemplary embodiment, the structure of the unlockable locking device may be the same as or different from the structure of the releasable locking mechanism **54**.

[0249] It should be understood that the second movable member **533** and the base body **531** may be locked or unlocked by the unlockable locking device, or may be locked or released by other means. For example, in other exemplary embodiments, as shown in FIGS. **12A** to **12D**, the second movable member **533** is in friction fit with the base body **531**, so that the second movable member **533** can remain fixed relative to the base body **531** under the action of the frictional force of the base body **531**, and can also be moved relative to the base body **531** against the frictional force of the base body **531**. Specifically, in a case where the second movable member **533** is sleeved over the second base body portion **5312** of the base body **531**, the inner diameter of the second movable member **533** may be smaller than the outer diameter of the second base body portion **5312**, so that the second movable member **533** is in friction fit (e.g., interference fit) with the second base body portion **5312**, and therefore, there is a frictional force between the second movable member **533** and the second base body portion **5312** due to extrusion, and the second movable member **533** may be fixed relative to the second base body portion **5312** under the action of the frictional force; the second movable member **533** may also overcome the frictional force and slide relative to the second base body portion **5312** under the action of an external force (e.g., under the urging action of a human hand).

[0250] In some exemplary embodiments, as shown in FIGS. **13A-13G**, the endoscope assembly **500** further includes an outer sheath **55**, the outer sheath **55** may include a body tube having a first end (i.e., a proximal end) and an opposite second end (i.e., a distal end), and a tube-locking mechanism **552** mounted at the first end of the body tube, the body tube has an endoscope channel **5519** extending from the first end to the second end of the body tube for passage of the endoscope **51**, and the tube-locking mechanism **552** is arranged to be in locking fit with the endoscope **51** to secure the endoscope **51** to the body tube and also disengage from being in locking fit with the endoscope **51** to allow the endoscope **51** to slide and rotate relative to the body tube.

[0251] In the outer sheath **55**, the body tube thereof is provided with an endoscope channel **5519**, the endoscope **51** can pass through the endoscope channel **5519**, and the endoscope **51** and the endoscope channel **5519** can be in gap fit so that the endoscope **51** can slide or rotate within the endoscope channel **5519**. The tube-locking mechanism **552** may be installed at one end of the body tube (for example, the proximal end located outside the human body), and the tube-locking mechanism **552** may have a locked state and a released state. As shown in FIG. **13A**, when the tube-locking mechanism **552** is in the locked state, it can be in locking fit with the endoscope **51**, and at this time, the tube-locking mechanism **552** is fixed with the endoscope **51**, so that the body tube and the endoscope **51** are fixed to prevent the body tube and the endoscope **51** from moving relatively, so as to drive the endoscope **51** to move or rotate by applying a thrust or torque force to the body tube, so that the endoscope **51** and the body tube are inserted into the human body together; as shown in FIG. **13B**, when the tube-locking mechanism **552** is in the released state, the tube-locking mechanism can be disengaged from being in locking fit with the endoscope **51**, and at this time, the tube-locking mechanism **552** is unfixed from the endoscope **51** so that the endoscope **51** can be moved relative to the body tube so as to move or rotate the endoscope **51** by applying a thrust or torque force to the endoscope **51** so that the endoscope **51** can be further inserted into the

human body relative to the body tube.

[0252] By providing the tube-locking mechanism 552, the outer sheath 55 can be engaged to the endoscope 51 by the tube-locking mechanism 552 in the locked state so as to transmit a driving torque applied to the outer sheath 55 to the endoscope 51, and a 1:1 torque transmission from the outer sheath 55 to the endoscope 51 can be realized so that the outer sheath 55 and the endoscope 51 are moved or rotated together to be inserted into the human body; the tube-locking mechanism 552 in the released state may be used to disengage the outer sheath 55 from the endoscope 51 such that the endoscope 51 may move or rotate relative to the outer sheath 55.

[0253] In some exemplary embodiments, as shown in FIGS. 13A-13C and 13E, the tube-locking mechanism 552 includes a locking seat 5521, a rotatable member 5522, and a plurality of locking blocks 5524. The locking seat 5521 is mounted at the first end of the body tube. The rotatable member 5522 is rotatably mounted to the locking seat 5521, and is provided with a spiral driving part 5523, the spiral center line of the spiral driving part 5523 coincides with the rotation center line of the rotatable member 5522. The plurality of locking blocks 5524 are movably mounted to the locking seat 5521, and are configured to be in driving fit with the spiral driving part so as to be translatable in the radial direction of the spiral driving part 5523. End portions of the plurality of locking blocks 5524 adjacent to each other collectively define an opening 5525 for the endoscope 51 to pass through. The size of the opening 5525 is changed with the radial translation of the locking blocks 5524.

[0254] The plurality of locking blocks 5524 are configured such that when the rotatable member 5522 is rotated in the first direction (as shown by the curved arrow in the state (b2) of FIG. 13E), the plurality of locking blocks 5524 are translated radially inward toward the spiral center line of the spiral driving part 5523 (as shown by the straight arrow in the state (b2) of FIG. 13E) to clamping the endoscope 51 by narrowing the opening 5525. The driving fit of the plurality of locking blocks 5524 with the spiral driving part 5523 also causes the plurality of locking blocks 5524 to be translated radially outward away from the spiral center line of the spiral driving part 5523 (as shown by the straight arrow in the state (b1) in FIG. 13E) when the rotatable member 5522 is rotated in a second direction opposite to the first direction (as shown by the curved arrow in the state (b1) in FIG. 13E) to enlarge the opening 5525 and release the endoscope 51.

[0255] In the tube-locking mechanism 552, the rotatable member 5522 is rotatably mounted to the locking seat 5521, the plurality of locking blocks 5524 are translationally mounted to the locking seat 5521, and the locking seat 5521 is mounted to the first end of the body tube, so that the tube-locking mechanism 552 is mounted to the first end of the body tube. The rotatable member 5522 is provided with a spiral driving part 5523, the spiral center line of the spiral driving part 5523 coincides with the rotating center line of the rotatable member 5522, and the plurality of locking blocks 5524 are in driving fit with the spiral driving part 5523, and can be translated along a side close to or away from the spiral center line of the spiral driving part 5523 under the driving of the rotatable member 5522. In an exemplary embodiment, when the rotatable member 5522 is rotated in a first direction (clockwise or counterclockwise direction as shown by the curved arrow in the state (b2) in FIG. 13E), the plurality of locking blocks 5524 can be translated radially inward toward the spiral center line of the spiral driving part 5523 under the push of the spiral driving part 5523 of the rotatable member 5522 (as shown by the straight arrow in the state (b2) in FIG. 13E), so that the cross-sectional area of the opening 5525 formed between the plurality of locking blocks 5524 for the endoscope 51 to pass through is reduced, so that the plurality of locking blocks 5524 clamp the endoscope 51, the locking fit is achieved between the tube-locking mechanism 552 and the endoscope 51. When the rotatable member 5522 is rotated in a second direction opposite to the first direction (counterclockwise or clockwise direction as indicated by the curved arrow in the state (b1) in FIG. 13E), the plurality of locking blocks 5524 can be translated radially outward away from the spiral center line of the spiral driving part 5523 under the push of the spiral driving part 5523 of the rotatable member 5522 (as indicated by the straight arrow in the state (b1) in FIG.

13E), so that the cross-sectional area of the opening 5525 formed between the plurality of locking blocks 5524 through which the endoscope 51 passes is increased, so that the plurality of locking blocks 5524 releases the endoscope 51 and it is achieved that the tube-locking mechanism 552 is disengaged from being in locking fit with the endoscope 51.

[0256] In some exemplary embodiments, as shown in FIG. 13E, at least two locking blocks 5524 may be provided, such as four, etc. The plurality of locking blocks 5524 may be uniformly arranged along the circumferential direction of the spiral driving part 5523.

[0257] In some exemplary embodiments, as shown in FIGS. 13A to 13C, the outer surface of the rotatable member 5522 is provided with a non-slip structure (e.g., a non-slip protrusion) to rotate the rotatable member 5522.

[0258] In some exemplary embodiments, as shown in FIGS. 13A-13C and 13F, the outer sheath 55 further includes an expandable member 554, the expandable member 554 is sleeved over an outer surface of the body tube at a second end of the body tube, the body tube provided with a gas channel 5511 extending between an inner surface and an outer surface of the body tube in an axial direction of the body tube, the gas channel 5511 is in communication with the expandable member 554 at the second end of the body tube, and the gas channel 5511 is in communication with an external pressure source 553 at the first end of the body tube so that the expandable member 554 expands outwardly in a radial direction of the body tube from a non-expanded state to an expanded state or retracts from the expanded state to the non-expanded state in response to action of the external pressure source 553.

[0259] In the outer sheath 55, the expandable member 554 is sleeved over the outer surface of the body tube at the second end of the body tube (e.g., the distal end into the human body). The expandable member 554 may include a balloon, and the body tube is provided with a gas channel 5511 that communicates the external pressure source 553 with the expandable member 554 so that the external pressure source 553 supplies air to the expandable member 554 through the gas channel 5511 so that the expandable member 554 can be expanded outward along the radial direction of the body tube to the lumen wall of the human body so that the expandable member 554 is anchored or gripped to the lumen wall of the human body, achieving fixation of the outer sheath 55 to the human body.

[0260] The external pressure source 553 may not only provide positive pressure to supply air to the expandable member 554 to bring the expandable member 554 in the expanded configuration; the external pressure source 553 may also provide negative pressure to extract gas from the expandable member 554 such that the expandable member 554 may contract inwardly along the radial direction of the body tube while the expandable member 554 is in a normal or unexpanded configuration.

[0261] In some exemplary embodiments, the endoscope assembly 500 further includes the external pressure source 553 and the external pressure source 553 is configured to provide a fixed volume of gas to the expandable member 554 to expand the expandable member 554 outwardly in a radial direction of the body tube.

[0262] When the expandable member 554 is expanded, the external pressure source 553 may provide a fixed volume of gas to the expandable member 554 to control the expanded volume of the expandable member 554. Compared to providing a fixed pressure gas to control the expansion of the expandable member 554 through pressure, the embodiments of the present application provides an external pressure source 553, making the expansion control of the expandable member 554 simpler, easier, and safer (such as when the expandable member 554 leaks gas).

[0263] In some exemplary embodiments, as shown in FIG. 13H, the external pressure source 553 includes a housing 5531, a first linear actuator 5532 mounted within the housing 5531, a syringe 5533 mounted within the housing 5531, and two position sensors 5537 mounted within the housing 5531. The first linear actuator 5532 has a drive end configured to move between a first position (as shown in state (c2) in FIG. 13H) and a second position (as shown in state (c1) in FIG. 13H). The syringe 5533 includes a syringe barrel 5534 and a syringe plunger 5535 movably mounted to the

syringe barrel **5534**, the syringe barrel **5534** has an injection port **5536** in communication with the gas channel **5511**, the drive end of the first linear actuator **5532** is connected to the syringe plunger **5535** and configured to drive the syringe plunger **5535** to reciprocate relative to the syringe barrel **5534** to supply gas to the expandable member **554** or draw gas from the expandable member **554** through the injection port **5536**. Two position sensors **5537** are provided for detecting the position of the drive end of the first linear actuator **5532**.

[0264] In the external pressure source **553**, the drive end of the first linear actuator **5532** can reciprocate and telescopically move, and can drive the syringe plunger **5535** of the syringe **5533** connected thereto to reciprocate. When the syringe plunger **5535** of the syringe **5533** is moved in one direction (e.g., to the right to transition from the state (c1) in FIG. 13H to the state (c2) in FIG. 13H), the syringe barrel **5534** of the syringe **5533** can supply gas to the expandable member **554** through the injection port **5536** so that the expandable member **554** can be expanded outward in a radial direction of the body tube; when the syringe plunger **5535** of the syringe **5533** is moved in the opposite direction (e.g., to the left to transition from the state (c2) in FIG. 13H to the state (c1) in FIG. 13H), the syringe barrel **5534** of the syringe **5533** may draw gas from the expandable member **554** through the injection port **5536** so that the expandable member **554** may be retracted inward in a radial direction of the body tube.

[0265] The two position sensors **5537** may be used to detect the position of the drive end of the first linear actuator **5532**, causing the drive end of the first linear actuator **5532** to move between the first position and the second position. For example, when the drive end of the first linear actuator **5532** is moved to the first position (in which the syringe **5533** supplies a fixed volume of gas to the expandable member **554**), a sensor is actuated, and then the drive end of the first linear actuator **5532** can be controlled to stop moving and stop driving the syringe plunger **5535** to continue inflating the expandable member **554**; when the drive end of the first linear actuator **5532** is moved to the second position (during which the syringe **5533** draws air from the expandable member **554**), another sensor is actuated, and the drive end of the first linear actuator **5532** can be controlled to stop moving and return to the original position.

[0266] In some exemplary embodiments, as shown in FIGS. 13A-13C, 13F-13G, the outer sheath **55** further includes a first seal **556** and a second seal **555**, the first seal **556** is disposed at a first end of the body tube (e.g., a proximal end of the body tube outside the human body) between the body tube and the outer surface of the endoscope **51**, and the second seal **555** is disposed at a second end of the body tube (e.g., a distal end of the body tube into the human body) between the body tube and the outer surface of the endoscope **51**.

[0267] In some exemplary embodiments, as shown in FIGS. 13A to 13D, 13F to 13G, the gap between the inner surface of the body tube and the outer surface of the endoscope **51** forms a suction channel **5512**, the body tube has one or more suction openings **5513** provided in the circumferential direction of the body tube at the second end of the body tube and a suction connector **5514** provided at the first end of the body tube, the suction openings **5513** and the suction connector **5514** are located between the second seal **555** and the first seal **556**, two ends of the suction channel **5512** is in communication with the suction opening **5513** and the suction connector **5514**, respectively. The suction connector **5514** is set to connect with a suction apparatus used to provide negative pressure. In an exemplary embodiment, the suction connector **5514** may be a suction interface.

[0268] Two ends of the suction channel **5512** formed by the gap between the inner surface of the body tube and the outer surface of the endoscope **51** can be sealed by the first seal **556** and the second seal **555**, respectively, and the suction opening **5513** and the suction connector **5514** located between the first seal **556** and the second seal **555** are opened on the body tube, so that the suction opening **5513** can be in communication with the suction channel **5512** on the one hand, and can be in communication with the lumen or intestinal tract of the human body when one end of the endoscope **51** and the outer sheath **55** is inserted into the lumen or intestinal tract of the human

body on the other hand; the suction connector **5514** is in communication with the suction channel **5512** on the one hand, and may be in communication with the suction apparatus on the other hand, so that the suction apparatus can extract gas from the lumen or intestinal tract of the human body through the suction connector **5514**, the suction channel **5512**, and the suction opening **5513** (the flow direction of the gas is shown by dashed arrows in FIGS. **13F** and **13G**), so that the lumen wall of the lumen or intestinal tract is tightly adsorbed to the outer surface of the body tube under the action of negative pressure, to realize the fixation of the body tube of the outer sheath **55** to the lumen wall of the human body.

[0269] The expandable member **554** is expanded and anchored to the lumen wall of the human body and the lumen wall of the human body is adsorbed and fixed to the body tube of the outer sheath **55** by suction negative pressure, thereby enhancing the fixing effect of the outer sheath **55** and the lumen wall of the human body, and reducing the size of the expandable member **554**. Compared to the expandable member **554** fixed by anchoring only (e.g., a balloon having an expanded outer diameter of up to 60 mm), the expandable member **554** (e.g., a balloon having an expanded outer diameter of up to 30 mm) according to an embodiment of the present application is small in size and provides a secure anchoring without slipping and over-inflating in the lumen or intestinal tract of the human body. Furthermore, using the expandable member **554** according to an embodiment of the present application can save installation time and cost as compared to the expandable member **554** that is only fixed by anchoring.

[0270] In some exemplary embodiments, as shown in FIGS. **13F** and **13G**, the second seal **555** is located on a side close to the suction opening **5513** and the first seal **556** is located on a side close to the suction connector **5514**. In an exemplary embodiment, the second seal **555** and the suction opening **5513** may be located at one end of the body tube (e.g., the distal end of the body tube into the human body), and the first seal **556** and the suction connector **5514** may be located at the other end of the body tube (e.g., the proximal end of the body tube outside the human body).

[0271] In some exemplary embodiments, as shown in FIG. **13F**, a side of the second seal **555** away from the first seal **556** is provided with a conical part **5551**, and the tapered end of the conical part **5551** away from the first seal **556** is in seal fit with the outer surface of the endoscope **51**; the second seal **555** is further provided with a second sealing rib **5552**, the second sealing rib **5552** is closer to the first seal **556** than the conical part **5551**, and the second sealing rib **5552** is in seal fit with the outer surface of the endoscope **51**.

[0272] The side of the second seal **555** away from the first seal **556** is provided with a conical part **5551**, and the outer diameter of the conical part **5551** gradually decreases along the direction away from the first seal **556**, so as to avoid the lumen wall, intestinal wall or mucosal folds of the human body being caught by the outer sheath **55** during the advancement of the endoscope **51** and the outer sheath **55** in the human body.

[0273] The second seal **555** includes not only the conical part **5551**, but also a second sealing rib **5552** located at one end of the conical part **5551** close to the first seal **556**. When the second seal **555** is in seal fit with the outer surface of the endoscope **51**, one end of the conical part **5551** away from the first seal **556** can be in seal fit with the outer surface of the endoscope **51** on the one hand, and the second sealing rib **5552** can be in seal fit with the outer surface of the endoscope **51** on the other hand, thereby achieving double sealing with good sealing effect, so as to block the body fluid in the human body from entering the suction channel **5512** between the inner surface of the body tube of the outer sheath **55** and the outer surface of the endoscope **51**.

[0274] In some exemplary embodiments, as shown in FIG. **13G**, the first seal **556** is provided with a first sealing rib **5561**, the first sealing rib **5561** is in seal fit with the outer surface of the endoscope **51** to ensure the sealing effect between the inner surface of the body tube of the outer sheath **55** and the outer surface of the endoscope **51** to prevent the body fluid in the human body from leaking to the outside of the body, and can maintain the negative pressure around the suction opening **5513** by preventing the air outside the body from entering the suction channel **5512**.

between the inner surface of the body tube of the outer sheath **55** and the outer surface of the endoscope **51**, ensuring the adsorption and fixation effect between the lumen wall of the human body and the body tube of the outer sheath **55**.

[0275] In some exemplary embodiments, the second seal **555** may be a conical silicone rubber seal and the first seal **556** may be a silicone rubber seal. It should be understood that the first seal **556** and the second seal **555** may also be of other materials.

[0276] In some exemplary embodiments, as shown in FIG. **13I**, the body tube is of a multi-layer structure and includes a spiral band layer **5515**, a mesh tube layer **5516**, an outer polymer layer **5517**, and an inner polymer layer **5518**. In an exemplary embodiment, the spiral band layer **5515** is disposed between the outer polymer layer **5517** and the inner polymer layer **5518**, and the mesh tube layer **5516** is disposed between the outer polymer layer **5517** and the spiral band layer **5515**, the outer polymer layer **5517** may overlay the mesh tube layer **5516** and the mesh tube layer **5516** may overlay the inner polymer layer **5518**.

[0277] The multilayer structure of the body tube of the outer sheath **55** may include an inner polymer layer **5518**, a spiral band layer **5515**, a mesh tube layer **5516** and an outer polymer layer **5517** arranged sequentially from the inside to the outside. The outer polymer layer **5517** may be made of TPU (thermoplastic polyurethane elastomer) material or other material, and the surface of the outer polymer layer **5517** may or may not be provided with a coating, and if provided with a coating, it may be, for example, a hydrophilic coating, a Teflon coating or other coating. The inner polymer layer **5518** may be made of TPU (thermoplastic polyurethane elastomer) material or other material, and the surface of the inner polymer layer **5518** may or may not be provided with a coating, and if provided with a coating, it may be, for example, a hydrophilic coating, a Teflon coating or other coating, the mesh tube layer **5516** may be made of stainless steel or other material, and the spiral band layer **5515** may be made of stainless steel or other material. By this arrangement, the body tube of the outer sheath **55** is provided with a certain structural strength in order to transmit force to the endoscope **51** through the outer sheath **55**, and the body tube of the outer sheath **55** can be bent and deformed for delivery in the human body.

[0278] The outer sheath **55** according to an embodiment of the present application is equipped with an expandable member **554** for anchoring and fixing, and can be fixed by suction through the suction channel **5512**; its body tube is of a multilayer structure reinforced by a spiral band layer **5515** and a mesh tube layer **5516** such that the body tube is flexible but can provide a 1:1 torque transmission from the outer sheath **55** to the endoscope **51**; the tube-locking mechanism **552** can realize the locking fit/unlocking fit between the body tube and the endoscope **51**; delivery of the endoscope **51** in the human body can be achieved by movement of the outer sheath **55** and the endoscope **51** together and separate movement of the endoscope **51**; the second seal **555** can block the body fluid in the human body from entering the suction channel **5512** between the outer sheath **55** and the endoscope **51**, and the conical part **5551** of the second seal **555** also prevents the lumen wall, intestinal wall or mucosal folds of the human body from being caught by the outer sheath **55** during the advancement of the endoscope **51** and the outer sheath **55** in the human body; the first seal **556** can prevent body fluid in the human body from leaking out of the human body, and the negative pressure around the suction opening **5513** can be maintained by preventing air outside the human body from entering the suction channel **5512** between the outer sheath **55** and the endoscope **51**.

[0279] During use, the expandable member **554** may be first inflated and simultaneously a suction force is applied by a suction apparatus to secure the outer sheath **55** to the intestinal or luminal wall of the human body; then, the tube-locking mechanism **552** can be released so that the body tube of the outer sheath **55** and the endoscope **51** are disengaged from being in locking fit; subsequently, the body tube of the outer sheath **55** can be held, and then the endoscope **51** can be pushed to move, so that the endoscope **51** can be avoided from being looped.

[0280] In some exemplary embodiments, the endoscope assembly **500** further includes an image

capturing assembly disposed at the second end of the endoscope **51**, and the endoscope magnetic anastomosis system further includes at least one video processor console. The video processor console includes a video processor **56**, as shown in FIGS. **15A** and **15B**. The image capturing assembly is electrically connected to the video processor **56** by a cable **57** so as to transmit information in the human body captured by the image capturing assembly to the video processor **56** for processing and display with the video processor **56**.

[0281] The video processor **56** has a main connector socket **561**, and one end of the cable **57** electrically connected to the video processor **56** is provided with a main connector **571** for plug-in connection with the main connector socket **561**. A locking structure is provided between the main connector socket **561** and the main connector **571** to lock the main connector socket **561** and the main connector **571** in plug-in connection. In an exemplary embodiment, the main connector **571** may be in a form of a plug.

[0282] A locking structure is provided between the main connector **571** of the cable **57** and the main connector socket **561** of the video processor **56**, which can lock the main connector socket **561** and the main connector **571** in plug-in connection to prevent the main connector **571** from being separated from the main connector socket **561** during the delivery of the endoscope **51** in the human body.

[0283] In some exemplary embodiments, as shown in FIGS. **15C-15I**, the main connector socket **561** includes a base **5611** and a rotatable locking ring **5612**, the locking ring **5612** is rotatably mounted to the base **5611** to enable the main connector socket **561** to be switched between an initial state (shown in (e1) in FIGS. **15D** and **15F**) and a locked state (shown in (e2) in FIGS. **15E** and **15F**). The main connector socket **561** can be switched between the initial state and the locked state by rotation of the locking ring **5612**.

[0284] In some exemplary embodiments, the locking structure provided between the main connector **571** and the main connector socket **561** comprises a positioning key **5711** that is provided on one of the locking ring **5612** and the main connector **571**, and a locking slide slot **5613** that is provided in the other of the locking ring **5612** and the main connector **571** and extends in the circumferential direction. One end of the locking slide slot **5613** is an insertion end **5614**. The positioning key **5711** is configured to be inserted into the locking slide slot **5613** from the insertion end **5614** when the main connector **571** is in plug-in connection with the main connector socket **561** in the initial state, and is configured to slide relative to the locking slide slot **5613** during the rotation of the locking ring **5612**, so that the positioning key **5711** and the insertion end **5614** of the locking slide slot **5613** are misaligned and thus the main connector socket **561** switches to the locking state.

[0285] In some exemplary embodiments, as shown in FIGS. **15C** to **15G**, one or more positioning keys **5711** may be provided. For example, as shown in FIG. **15G**, two positioning keys **5711** may be provided. The two positioning keys **5711** may be respectively provided on two sides of the main connector **571**. Correspondingly, one or more locking slide slots **5613** may be provided to cooperate with one or more positioning keys **5711** in a one-to-one manner. For example, two locking slide slots **5613** may be provided. These two locking slide slots **5613** may be provided on the locking ring **5612**, may extend along the circumferential direction of the locking ring **5612**, and may respectively cooperate with the two positioning keys **5711**. When the main connector **571** is in plug-in connection with the main connector socket **561** in the initial state, the positioning keys **5711** can be inserted into the locking slide slot **5613** from the insertion end **5614** of the locking slide slot **5613**; then, by rotating the locking ring **5612**, the positioning keys **5711** can slide in the locking slide slot **5613**, and then the positioning keys **5711** and the insertion end **5614** of the locking slide slot **5613** are misaligned, so as to prevent the positioning keys **5711** from coming out of the insertion end **5614** of the locking slide slot **5613**, and further prevent the main connector **571** from being separated from the main connector socket **561**. At this time, the main connector socket **561** is switched to the locked state, and the lock fixation of the main connector **571** and thus the

main connector socket **561** is realized.

[0286] It should be understood that the positioning key **5711** may be provided on the locking ring **5612**, and the locking slide slot **5613** may be provided on the main connector **571** and extend along the circumferential direction of the main connector **571**, and the lock fixation of the main connector **571** and the main connector socket **561** can also be achieved. It should also be understood that the main connector **571** may also be provided to include a base **5611** and a locking ring **5612**, with a positioning key **5711** provided in one of the locking ring **5612** and the main connector socket **561**, and a circumferentially extending locking slide slot **5613** provided in the other of the locking ring **5612** and the main connector socket **561**.

[0287] In some exemplary embodiments, as shown in FIGS. **15D** and **15E**, the locking slide slot **5613** is a spiral slot, and the positioning key **5711** is configured to, during rotation of the locking ring **5612**, be pressed by the spiral groove to drive the main connector **571** to be further deeply inserted into the interior of the main connector socket **561**.

[0288] The locking slide slot **5613** is a spiral slot, in the direction away from the insertion end **5614** of the locking slide slot **5613**, the locking slide slot **5613** is screwed toward the side of the main connector socket **561** away from the main connector **571** so that during the rotation of the locking ring **5612**, the groove wall of the locking slide slot **5613** and the positioning key **5711** can be pressed against each other, and under the action of the pressing force, the main connector **571** may be moved toward a side of the main connector socket **561** away from the main connector **571** (as shown in FIG. **15I**, the main connector **571** is moved to the left from the position indicated in state (f1) to the position indicated in state (f2) by a distance of d), so that the main connector **571** more tightly plug into the main connector socket **561** to ensure a good connection between the main connector **571** and the main connector socket **561** and to prevent accidental pull-out of the main connector **571**.

[0289] In some exemplary embodiments, where the positioning key **5711** is provided on the main connector **571** and the locking slide slot **5613** is provided in the locking ring **5612**, as shown in FIGS. **15C** to **15E**, the base **5611** is provided with an insertion slot **5615**, the insertion slot **5615** is configured to be in communication with the insertion end **5614** of the locking slide slot **5613** when the main connector socket **561** is in the initial state (as shown in FIG. **15D**) and disconnect from the insertion end **5614** of the locking slide slot **5613** when the locking ring **5612** is rotated to switch the main connector socket **561** to the locked state (as shown in FIG. **15E**).

[0290] The locking ring **5612** is provided with a locking slide slot **5613**, and the base **5611** is provided with an insertion slot **5615**, and when the main connector socket **561** is in the initial state, the insertion slot **5615** can be aligned with and in communication with the insertion end **5614** of the locking slide slot **5613**, so that the positioning key **5711** passes through the insertion slot **5615** and the insertion end **5614** of the locking slide slot **5613** sequentially and then enters the locking slide slot **5613**; during the rotation of the locking ring **5612**, the insertion slot **5615** is misaligned and disconnected from the insertion end **5614** of the locking slide slot **5613** to prevent the positioning key **5711** in the locking slide slot **5613** from coming out of the insertion slot **5615**.

[0291] In some exemplary embodiments, as shown in FIGS. **15D-15F**, **15H**, and **15I**, the base **5611** is provided with a rotatable latch **5617**, the locking ring **5612** is provided with a limit slot, and the latch **5617** is configured such that when the main connector socket **561** is in the initial state, the first portion **5618** of the latch **5617** is located within the limit slot and the second portion **5619** of the latch **5617** is located outside the limit slot, as shown in FIGS. **15D** and **15H**, to circumferentially secure the locking ring **5612** to the base **5611** and to maintain the main connector socket **561** in the initial state. The latch **5617** is further configured such that when the main connector **571** is in plug-in connection with the main connector socket **561** in the initial state, the second portion **5619** of the latch **5617** located outside the limit slot can rotate under the urging of the main connector **571** as shown in the state (f1) in FIG. **15I**, thereby causing the latch **5617** to rotate out of the limit slot and enabling the locking ring **5612** to rotate.

[0292] The base **5611** is provided with a latch **5617**, and the locking ring **5612** is provided with a rotatable limit slot, and the locking ring **5612** can be rotated or fixed relative to the base **5611** by cooperation between the latch **5617** and the limit slot. Specifically, when the main connector socket **561** is in the initial state, as shown in FIGS. **15D** and **15H**, the first portion **5618** of the latch **5617** is located in the limit slot, and the second portion **5619** of the latch **5617** protrudes from the limit slot. By cooperation between the latch **5617** and the limit slot, the locking ring **5612** and the base **5611** can be fixed in the circumferential direction to prevent the locking ring **5612** from rotating, so that the main connector socket **561** can be maintained in the initial state, and the plug-in connection between the main connector **571** and the main connector socket **561** can be facilitated. When the main connector **571** is in plug-in connection with the main connector socket **561** in the initial state, the second portion **5619** of the latch **5617** protruding from the limit slot can be rotated under the urging action of the main connector **571** as shown in the state (f1) in FIG. **15I**, so that the latch **5617** entirely comes out of the limit slot. At this time, the locking ring **5612** can be rotated relative to the base **5611**, so that the base **5611** switches to the locked state and locks the main connector **571** (as shown in the state (f2) in FIG. **15I**), preventing the main connector **571** from accidentally separating from the main connector socket **561**.

[0293] In some exemplary embodiments, as shown in FIGS. **15H** and **15I**, the main connector **571** is provided with an avoidance groove **5712**, the avoidance groove **5712** is configured to, during plug-in connection between the main connector **571** and the main connector socket **561**, receive the first portion **5618** of the latch **5617** disengaged from the limit slot (as shown in FIG. **15I**), and further configured to drive the latch **5617** to rotate when the main connector **571** is separated from the main connector socket **561** to reset the latch **5617** so that the first portion **5618** is positioned within the limit slot and the second portion **5619** is positioned outside the limit slot, to circumferentially secure the locking ring **5612** to the base **5611**.

[0294] The main connector **571** is provided with an avoidance groove **5712**, and when the main connector **571** is in plug-in connection with the main connector socket **561** in the initial state, the latch **5617** as a whole is gradually rotated to come out of the limit slot (switching from the state shown in FIG. **15H** to the state (f1) shown in FIG. **15I**), and the avoidance groove **5712** can accommodate a portion of the latch **5617** disengaged (protruded) from the limit slot so as not to hinder the rotation of the latch **5617**. In the process of gradually separating the main connector **571** and the main connector socket **561** in the plug-in state, the groove wall of the avoidance groove **5712** can push the latch **5617** and drive the latch **5617** to rotate in the opposite direction, so that the latch **5617** is gradually reset to a state in which a part of the latch is located in the limit slot of the base **5611** and the other part of the latch protrudes from the limit slot (as shown in FIG. **15H**). At this time, the latch **5617** cooperates with the limit slot to limit and secure the locking ring **5612** and the base **5611** in the circumferential direction, and the main connector socket **561** is in the initial state as a whole.

[0295] In some exemplary embodiments, as shown in FIGS. **15H** and **15I**, the center of rotation of the latch **5617** is located in the middle of the latch **5617** and offset from the center of gravity of the latch **5617** so that the latch **5617** can maintain the first portion **5618** within the limit slot and the second portion **5619** outside the limit slot under its own weight (as shown in FIG. **15H**).

[0296] The center of rotation of the latch **5617** may be located in the middle of the latch **5617** and offset from the center of gravity of the latch **5617**, so that the latch **5617** can automatically reset and maintain the first portion **5618** in the limit slot and the second portion **5619** outside the limit slot under the action of its own weight, so that the latch **5617** can automatically fix the locking ring **5612** and the base **5611** in the circumferential direction after the main connector **571** is separated from the main connector socket **561**.

[0297] When the main connector **571** and the main connector socket **561** according to an embodiment of the present application are connected, the main connector **571** can be inserted into the main connector socket **561** first (the main connector **571** can be inserted in the direction

indicated by the straight arrow in the state (d1) in FIG. 15B), and before insertion, the positioning key 5711 of the main connector 571 can be aligned with the insertion slot 5615 on the base 5611 of the main connector socket 561, and as the main connector 571 is gradually inserted into the main connector socket 561, the positioning key 5711 is gradually inserted into the locking slide slot 5613 of the locking ring 5612; then, the locking ring 5612 of the main connector socket 561 is toggled or rotated downward (the locking ring 5612 can be toggled or rotated in the direction indicated by the curved arrow in the state (d2) in FIG. 15B) to slide the positioning key 5711 in the locking slide slot 5613 to lock the main connector 571 and the main connector socket 561, and the locking slide slot 5613 presses the positioning key 5711 inward to move the positioning key 5711 and the main connector 571 toward the side of the main connector socket 561 away from the main connector 571 to achieve a tight connection between the main connector 571 and the main connector socket 561 (as shown in state (d3) in FIG. 15B) to ensure a good video signal connection between the video processor 56 and the cable 57, and to prevent the main plug from being accidentally pulled out. Finally, the video processor 56 may be ready for operation.

[0298] In some exemplary embodiments, the endoscope 51 includes a head assembly 58 disposed at a second end of the endoscope 51. As shown in FIGS. 14A to 14D, the head assembly 58 includes a head assembly body 581, one end surface of the head assembly body 581 is provided with a support part 582 and a limit part 583 which are disposed opposite to each other, for clamping the magnetic implant assembly 511 between the limit part 583 and the support part 582. The bottom of the magnetic implant assembly 511 is supported on the support part 582, and the top of the magnetic implant assembly 511 abuts against on the limit part 583. In an exemplary embodiment, the limit part 583 can be designed as a tongue-like structure (as shown in FIGS. 14A to 14D).

[0299] As shown in FIGS. 14E and 14F, a head assembly 58 is provided at the second end of the endoscope 51 (for example, the distal end into the human body), and a support part 582 is provided on one end surface of the head assembly body 581 of the head assembly 58, and the end surface of the head assembly body 581 and the support part 582 can cooperate to form a mounting base for the magnetic implant assembly 511, and the magnetic implant assembly 511 can be supported by the support part 582 and abut against the end surface of the head assembly body 581. As a result of this installation, the magnetic implant assembly 511 may be tilted (as shown in the direction of the arrow in FIG. 14F) by impacting the luminal wall, intestinal wall, or mucosal folds in the human body.

[0300] Therefore, in the embodiment of the present application, as shown in FIGS. 14A to 14D, a limit part 583 is further provided on the end surface of the head assembly body 581, and the latch 5617 and the support part 582 are oppositely disposed along the height direction of the magnetic implant assembly 511, so that the limit part 583 can contact the end surface of the magnetic implant assembly 511 on a side away from the support part 582, and restrain and limit the magnetic implant assembly 511 to prevent the magnetic implant assembly 511 from tilting when it impacts the lumen wall, intestinal wall or mucosal fold in the human body.

[0301] In some exemplary embodiments, as shown in FIGS. 16A to 16C, the magnetic implant assembly 511 of the endoscope 51 includes an enclosure 5111 and a magnet 5117, the magnet 5117 is disposed in the enclosure 5111, the enclosure 5111 includes a first end surface 5115 and a second end surface 5116 which are disposed opposite to each other, and the first end surface 5115 and the second end surface 5116 are both in an arc shape. In an exemplary embodiment, as shown in FIG. 14D, the second end surface 5116 of the enclosure 5111 may be supported on the support part 582 of the head assembly body 581, the first end surface 5115 of the enclosure 5111 may be in contact with a limit part 583 provided on the head assembly body 581, and a portion of the limit part 583 that is in contact with the first end surface 5115 of the enclosure 5111 is a curved surface, and the curved surface is adapted to the shape of the first end surface 5115 so as to prevent the magnetic implant assembly 511 from tilting by the limit part 583.

[0302] In some exemplary embodiments, as shown in FIGS. 16A-16C, the circumferential side surface of the enclosure 5111 of the magnetic implant assembly 511 is provided with a fixing ring groove 5113, and the snare wire 522 of the snare assembly 52 may be received within the fixing ring groove 5113 so that the snare wire 522 secures the magnetic implant assembly 511 tightly. The circumferential side surface of the enclosure 5111 of the magnetic implant assembly 511 is further provided with a recess 5112 for housing the tip of the snare catheter 521 of the snare assembly 52.

[0303] In some exemplary embodiments, as shown in FIGS. 16A-16C, the radii of curvature of the arc-shaped first end surface 5115 and the arc-shaped second end surface 5116 of the enclosure 5111 are different. For example, the radius of curvature of the first end surface 5115 is smaller than the radius of curvature of the second end surface 5116, and a groove 5114 is provided at the center of the first end surface 5115.

[0304] As shown in FIG. 16D, when the magnetic implant assemblies 511 of the two endoscopes 51 enter the human body and are magnetically coupled, the first end surface 5115 of the magnetic implant assembly 511 of the one endoscope 51 and the second end surface 5116 of the magnetic implant assembly 511 of the other endoscope 51 are opposite to each other and clamp the lumen wall or intestinal wall of the human body. Since the radius of curvature of the first end surface 5115 of the enclosure 5111 of the magnetic implant assembly 511 is less than the radius of curvature of the second end surface 5116, the distance s1 between a portion on the first end surface 5115 of the magnetic implant assembly 511 of one endoscope 51 surrounding the groove 5114 and the second end surface 5116 of the magnetic implant assembly 511 of another endoscope 51 is less than the distance s2 between an edge portion on the first end surface 5115 of the magnetic implant assembly 511 of the endoscope 51 away from the groove 5114 and the second end surface 5116 of the magnetic implant assembly 511 of another endoscope 51. Furthermore, the force F1 exerted by the portion of the first end surface 5115 of the magnetic implant assembly 511 of the endoscope 51 surrounding the groove 5114 on the lumen wall or the intestinal wall of the human body is greater than the force F2 exerted by the edge portion of the first end surface 5115 of the magnetic implant assembly 511 of the endoscope 51 away from the groove 5114 on the lumen wall or the intestinal wall of the human body. The adjacent application of different forces F1, F2 may improve the anastomosis formed by the magnetic implant assemblies 511 of the two endoscopes 51 and/or the healing of necrosis of the luminal or intestinal wall of the human body (and/or enable better controlled healing of the luminal or intestinal wall of the human body).

[0305] In some exemplary embodiments, as shown in FIG. 16C, the magnet 5117 of the magnetic implant assembly 511 is solid disc-shaped and is a permanent magnet.

[0306] In some exemplary embodiments, as shown in FIG. 10 and FIG. 16D, the at least one endoscopic assembly of the endoscopic magnetic anastomosis system comprises two endoscopic assemblies. The magnetic implant assemblies of the two endoscopic assemblies are respectively a first magnetic assembly configured to be located within a first luminal tissue region and a second magnetic implant assembly configured to be located within a second luminal tissue region (as shown in FIG. 10). The structures of the first magnetic implant assembly and the second magnetic implant assembly may be set to be the same. For example, the structures of each of the first magnetic implant assembly and the second magnetic implant assembly can be as shown in FIGS. 16A-16C.

[0307] An enclosure (i.e., a first enclosure) of the first magnetic implant assembly is provided with a fixing ring groove (i.e., a first fixing ring groove) on a circumferential side surface thereof to receive the snare wire of the snare assembly of one of the two endoscopic assemblies. An enclosure (i.e., a second enclosure) of the second magnetic implant assembly is provided with a fixing ring groove (i.e., a second fixing ring groove) on a circumferential side surface thereof to receive the snare wire of the snare assembly of the other of the two endoscopic assemblies.

[0308] The first enclosure of the first magnetic implant assembly has a first engagement surface, and the second enclosure of the second magnetic implant assembly has a second engagement

surface. The first engagement surface and the second engagement surface are configured, when the first magnetic implant assembly and the second magnetic implant assembly are magnetically anastomosed with the first and second luminal tissue regions interposed therebetween (as shown in FIG. 10), to face each other and be capable of exerting a non-uniform compressive force on the first luminal tissue region and the second luminal tissue region, enabling better controlled healing of the luminal or intestinal wall of the human body.

[0309] The first engagement surface is a concave curved surface with a first radius of curvature, and the second engagement surface is a convex curved surface with a second radius of curvature. The radius of curvature of the convex surface is smaller than that of the concave surface. The second end surface of the first enclosure of the first magnetic implant assembly can form the first engagement surface, and the first end surface of the second enclosure of the second magnetic implant assembly can form the second engagement surface.

[0310] The convex curved surface and the concave curved surface are configured such that when the first magnetic implant assembly and the second magnetic implant assembly are magnetically anastomosed, the convex curved surface protrudes towards the concave curved surface, and the axial distance between the convex curved surface and the concave curved surface along the central line direction of the first magnetic implant assembly increases as the radial distance from the central line of the first magnetic implant assembly increases, so as to exert the non-uniform compressive force on the first luminal tissue region and the second luminal tissue region.

[0311] In some exemplary embodiments, a magnet (i.e., a first magnet) disposed within the first enclosure of the first magnetic implant assembly and a magnet (i.e., a second magnet) disposed within the second enclosure of the second magnetic implant assembly are both solid disc-shaped permanent magnet. Thereby, when the first magnetic implant assembly and the second magnetic implant assembly are magnetically anastomosed, the center line of the first magnetic implant assembly and the center line of the second magnetic implant assembly can be automatically adjusted to an aligned state even when there is a certain deviation between them, with the magnetic poles of the ends of the first magnet and the second magnet adjacent to each other being opposite.

[0312] FIG. 16F is a schematic diagram of magnetic field interaction between two magnetic implant assemblies 511 (whose magnets 5117 are in a shape of a solid disc as shown in FIG. 16C) according to an embodiment of the present application, FIG. 16G is a schematic diagram of magnetic field interaction between two magnetic implant assemblies 511 (whose magnets 5117 are in a shape of a hollow ring as shown in FIG. 16E) in some cases, FIG. 16H is a schematic diagram of a relationship between the magnetic field force and distance between two magnetic implant assemblies 511 (whose magnets 5117 are in the shape of a solid disc as shown in FIG. 16C) according to an embodiment of the present application (where the horizontal axis represents the distance between the two magnetic implant assemblies, and the vertical axis represents the transverse magnetic field force between the two magnetic implant assemblies). FIG. 16I is a schematic diagram of a relationship between the magnetic field force and distance between two magnetic implant assemblies 511 (whose magnets 5117 are in a shape of a hollow ring as shown in FIG. 16E) (where the horizontal axis represents the distance between the two magnetic implant assemblies, and the vertical axis represents the transverse magnetic field force between the two magnetic implant assemblies) in some cases. According to FIGS. 16F to 16I, compared to setting the magnet 5117 of the magnetic implant assembly 511 in the hollow annular shape shown in FIG. 16E, the magnet 5117 of the magnetic implant assembly 511 is disposed in the solid disc shape shown in FIG. 16C so that the central axes of two such magnetic implant assemblies 511 can automatically adjust to an aligned state even when they deviate by a certain distance, allowing the magnetic implant assembly 511 to have unique self-alignment performance, which can better improve the alignment performance of the magnetic implant assemblies 511 of the two endoscopes 51 so that the magnetic implant assemblies 511 of the two endoscopes 51 can be better aligned to the formed anastomosis.

[0313] In some exemplary embodiments, as shown in FIG. 16C, the outer diameter D of the magnetic implant assembly **511** is 8 mm-30 mm, such as 15 mm-23 mm; the height H of the outer surface of the magnetic implant assembly **511** is 4 mm-8 mm.

[0314] In some exemplary embodiments, the width of the magnetic implant assembly **511** is larger than the width of the snare channel of the endoscope **51**. For example, the outer diameter D of the magnetic implant assembly **511** (i.e., the width of the magnetic implant assembly **511**), in the embodiment shown in FIG. 16C, is larger than the diameter of the snare channel (i.e., the width of the snare channel, not shown), preventing the magnetic implant assembly **511** from entering or passing through the snare channel.

[0315] In some exemplary embodiments, the magnet **5117** of the magnetic implant assembly **511** may be a strong neodymium magnet, and the enclosure **5111** of the magnetic implant assembly **511** may be a biocompatible enclosure **5111** (e.g., polycarbonate or other material).

[0316] In some exemplary embodiments, the endoscopic magnetic anastomosis system further includes a magnet detector **600** and the magnet detector **600** is configured to detect the position of the magnetic implant assembly **511** of the endoscope **51** within the human body.

[0317] The magnet detector **600** can detect the position of the magnetic implant assembly **511** of the endoscope **51** in the human body, so that the medical worker can know whether the magnetic implant assembly **511** is in place, and when the magnetic implant assembly **511** is not in place, the magnetic implant assembly **511** can be conveniently located and moved to the appropriate position.

[0318] In some exemplary embodiments, as shown in FIGS. 17A-17F, the magnet detector **600** includes a detector body **61**, a first printed circuit board **62**, a second printed circuit board **63**, and a processing unit. The first printed circuit board **62** is provided at the first end of the detector body **61**, and a first group of magnetometers **621** is provided thereon. The second printed circuit board **63** is provided at a second end of the detector main body **61** opposite to the first end of the detector main body **61**, and a second group of magnetometers **631** is provided thereon. The processing unit is arranged to be able to receive measurement data from the first group of magnetometers **621** and the second group of magnetometers **631**, and to determine the position of the magnetic implant assembly **511** of the endoscope **51** in the human body from the data.

[0319] The magnet detector **600** may be a portable magnet detector **600**, and the detector body **61** thereof may be provided with a grip part for taking and carrying the magnet detector; the first printed circuit board **62** and the second printed circuit board **63** are respectively provided at two ends (e.g., upper and lower ends) of the detector body **61**, and the first group of magnetometers **621** and the second group of magnetometers **631** are respectively provided on the first printed circuit board **62** and the second printed circuit board **63**, the first group of magnetometers **621** and the second group of magnetometers **631** can be used to measure an external magnetic field (e.g., the strength and direction of the magnetic field can be measured). The processing unit can be mounted to the detector body **61** and receive measurement data from the first group of magnetometers **621** and the second group of magnetometers **631**, in order to determine the position of the magnetic implant assembly **511** of the endoscope **51** in the human body based on this data.

[0320] In some exemplary embodiments, as shown in FIG. 17D, the first group of magnetometers **621** includes four magnetometers distributed in a square array, and the center of the square array coincides with the center of the first printed circuit board **62**. As shown in FIG. 17F, the second group of magnetometers **631** includes four magnetometers distributed in a square array, and the center of the square array coincides with the center of the second printed circuit board **63**. In an exemplary embodiment, the spacing between the first printed circuit board **62** and the second printed circuit board **63** may be 15 cm, the four magnetometers in the first group of magnetometers **621** may be 2.5 cm apart from each other in both transverse and longitudinal directions, and the four magnetometers in the second group of magnetometers **631** may be 2.5 cm apart from each other in both transverse and longitudinal directions.

[0321] In the magnet detector **600** according to an embodiment of the present application, the first

printed circuit board (upper layer PCB) **62** and the second printed circuit board (lower layer PCB) **63**, spaced 15 cm vertically, may be horizontally arranged and parallel to each other. The first printed circuit board **62** has a first group of magnetometers **621** comprising four magnetometers that are positioned symmetrically in a square formation, with equal spacing of 2.5 cm along both width and length of the square. The second printed circuit board **63** has a second group of magnetometers **631** comprising four magnetometers that are positioned symmetrically in a square formation, with equal spacing of 2.5 cm along both width and length of the square.

[0322] The magnet detector **600** includes multiple layers of magnetometers arranged in a grid pattern (e.g., two layers of magnetometers: a first group of magnetometers **621** and a second group of magnetometers **631** disposed on the first printed circuit board **62** and the second printed circuit board **63**, respectively), allowing for long range spatial magnetic field measurements. The magnet detector **600** can be used to detect the position of the magnet **5117** on a side of the second printed circuit board **63** away from the first printed circuit board **62**, for example, it can be used to detect the magnet **5117** below it, for example, the magnet detector **600** can provide accurate sensing of the magnet **5117** within a range of 3 cm to 15 cm beneath the second printed circuit board **63**. When in use, the magnet detector **600** can be moved over a human body (e. g., the abdomen of the human body), to detect the position of the magnetic implant assembly **511** in the human body.

[0323] The magnet detector **600** has a calibration function to eliminate background noise. The magnet detector **600** undergoes an initial zero-point calibration, setting all magnetometers to a baseline value of zero. The magnetic field above the first printed circuit board **62** acts as a background noise reference, and the first group of magnetometers **621** can continuously measure magnetic field disturbances from external sources; the background noise reading is then subtracted from a lower magnetic field measurement (detected by the second group of magnetometers **631**) including both the background noise and any target magnetic signal to obtain the target magnetic signal.

[0324] The magnet detector **600** can perform signal preprocessing for long-distance detection, and the first group of magnetometers **621** and the second group of magnetometers **631** can perform noise reduction using a high-resolution magnetic field measurement magnetometer MMC56x3.

[0325] When processing the measurement data of the first group of magnetometers **621** and the second group of magnetometers **631**, a weighted average method may be used. The signals from the first group of magnetometers **621** and the second group of magnetometers **631** are combined using a weighted average, where the more reliable or noise-free measurements contribute more significantly to the final result.

[0326] The processing unit may use artificial intelligence (AI)-powered signals so as to: enable accurate detection even when the signal strength is low (e.g., in the range of 10 cm-15 cm below the second printed circuit board **63**); and learn to differentiate between real magnetic signals and interference signals using spatial data across multiple magnetometers.

[0327] The processing unit may integrate an AI-neural network processing function.

[0328] Inputs to the neural network structure: The neural network structure receives preprocessed magnetic field data (x, y, z axes) from all the first group of magnetometers **621** and the second group of magnetometers **631** as a multi-dimensional feature vector. Each of the first group of magnetometers **621** and the second group of magnetometers **631** contributes to the perception of the magnetic environment by the neural network structure.

[0329] Output of the neural network structure: The neural network structure uses a sigmoid activation function in the output layer to produce a probability between 0 and 1. This represents the likelihood of a magnet target being present.

[0330] The execution pipeline of the neural network processing is: reading the data of the first group of magnetometers **621** and the second group of magnetometers **631**.
fwdarw.preprocessing.fwdarw.running deep neural network (DNN)
inference.fwdarw.outputting the detection result on LED.

[0331] Edge AI optimization capabilities for neural network processing: the model is quantized (INT 8) and stripped of unnecessary complexity using TensorFlow Lite for microcontrollers, allowing it to fit within the MCU's limited memory and computation budget.

[0332] In some exemplary embodiments, the magnet detector **600** further includes an indicator **622**, and the indicator **622** is electrically connected to the processing unit and configured to indicate whether the magnetic implant assembly **511** of the endoscope **51** has been detected. In an exemplary embodiment, as shown in FIGS. **17B** and **17C**, the indicator **622** is an LED mounted on the first printed circuit board **62**, and the first end of the detector body **61** is further provided with a transparent cover **64** that covers the first printed circuit board **62**.

[0333] The detection status of the magnet detector **600** is visually indicated by the LED on the first printed circuit board **62** at the top, which indicates whether the magnetic implant assembly **511** in the human body has been detected by whether the LED is on. For example, the LED is lit to indicate that the magnetic implant assembly **511** in the human body is detected, and the LED is not lit to indicate that the magnetic implant assembly **511** in the human body is not detected.

[0334] In other exemplary embodiments, as shown in FIGS. **18A** and **18B**, the magnet detector **600** includes a mounting substrate **65**, at least one sensor module **66**, and a processing module **67**. Each sensor module **66** may include eight magnetic sensors **661**. The eight magnetic sensors **661** are regularly arranged in four rows and each row has two magnetic sensors **661**. The two magnetic sensors **661** are a first magnetic sensor **661** and a second magnetic sensor **661**, respectively, and the two magnetic sensors **661** in each row are in a misaligned arrangement with the two magnetic sensors **661** in an adjacent row. The inertial measurement unit (IMU) **5118** may be embedded in the magnetic implant assembly **511** of the endoscope **51**, and the processing module **67** is arranged to receive data from the sensor module **66** and data of the IMU **5118** embedded in the magnetic implant assembly **511**, and determine the position of the magnetic implant assembly **511** of the endoscope **51** in the human body based on the data.

[0335] In an exemplary embodiment, the number of sensor modules **66** is between 1-32. In an exemplary embodiment, the spacing **S3** between two magnetic sensors **661** located in a same row is 5 cm, a transverse misalignment spacing between the first magnetic sensor **661** in each row and the first magnetic sensor **661** in the adjacent row is 2.5 cm, a transverse misalignment spacing between the second magnetic sensor **661** in each row and the second magnetic sensor **661** in the adjacent row is 2.5 cm, and the longitudinal spacing between the magnetic sensors **661** in any two adjacent rows is 2.5 cm, so that the spacing **S4** between the two rows of magnetic sensors **661** spaced by one row is 5 cm.

[0336] The magnet detector **600** has a modular sensor matrix including N (e.g., 1-32) sensor module(s) **66** with a dimension of 10 cm×10 cm, each sensor module houses eight magnetic sensors **661**, the magnetic sensors **661** are arranged in a 4×4 checkerboard pattern and positioned at alternating nodes of the grid. Each magnetic sensor **661** outputs a 3D magnetic field vector (B.sub.x, B.sub.y, B.sub.z). It in total generates an 8N×3 data matrices per sampling interval.

[0337] The IMU **5118** may be embedded in the magnetic implant assembly **511** of the endoscope **51**. In an exemplary embodiment having two endoscopes **51**, each of the magnetic implant assemblies **511** of two endoscopes **51** is embedded with an IMU **5118**. Each IMU **5118** can measure 3D acceleration (A.sub.x, A.sub.y, A.sub.z) and 3D angular velocity (W.sub.x, W.sub.y, W.sub.z).

[0338] The processing module **67** may host a trained AI model to process the sensor matrix data and the input data of the IMU **5118**, and may outputs real-time (x, y) coordinates of the magnetic implant assemblies **511** relative to the sensor matrix.

[0339] The magnet detector **600** according to an embodiment of the present application is a modular real-time magnetic positioning system with a configurable sensor module **66** and AI-driven tracking. This system introduces a novel magnetic localization architecture combining a modular sensor matrix, a magnetic implant assembly **511** (the magnet **5117** of which is a permanent

magnet) equipped with an IMU **5118**, and an AI processing unit (processing module **67**) to achieve real-time positional tracking. When in use, the magnet detector **600** may be positioned underneath or on the back side of the human body to detect the position of the magnetic implant assembly **511** in the human body above or in front thereof.

[0340] The configurable sensor matrix consists of a user-deployable array of 1-32 sensor modules **66**, and each sensor module **66** is embedded with **8** magnetic sensors **661** to measure the 3D magnetic field, thus enabling adaptable spatial coverage.

[0341] The AI processing unit enables sensor fusion: 3D magnetic field data (from the sensor matrix) and 6-degree-of-freedom inertial data (from the IMU **5118**) can be integrated to enhance positional and dynamic tracking.

[0342] AI model of AI processing unit: the deep learning model is trained to predict (x, y) positions of a pair of magnetic implant assemblies **511** of the two endoscope assemblies **500** relative to the sensor matrix in real time, leveraging both magnetic and inertial inputs.

[0343] According to the AI model architecture shown in FIG. **18B**, it can be seen that:

[0344] The input data of the AI processing unit includes: [0345] 1) Data from the sensor matrix: (B.sub.x,1, B.sub.y,1, B.sub.z,1), (B.sub.x,2, B.sub.y,2, B.sub.z,2), . . . (B.sub.x,n, B.sub.y,n, B.sub.z,n); [0346] 2) Data from one IMU **5118**: (A.sub.x, A.sub.y, A.sub.z), (W.sub.x, W.sub.y, W.sub.z); and [0347] 3) Data from another IMU **5118**: (A.sub.x, A.sub.y, A.sub.z), (W.sub.x, W.sub.y, W.sub.z).

[0348] The output data of the AI processing unit includes the (x, y) position of the magnetic implant assembly **511** relative to the sensor matrix.

[0349] In some exemplary embodiments, the endoscope magnetic anastomosis system further includes a magnetic navigation console **700**, and the magnetic navigation console **700** is configured to drive the magnetic implant assembly **511** of the endoscope **51** to move to a set position in the human body by the action of a magnetic field.

[0350] When the endoscope **51** is delivered in the human body, if the position of the magnetic implant assembly **511** does not reach the appropriate position (set position), the magnetic navigation console **700** can be used to drive the magnetic implant assembly **511** to move in the human body so that the magnetic implant assembly **511** reaches the appropriate position.

[0351] In some exemplary embodiments, as shown in FIGS. **19A-19B**, the magnetic navigation console **700** includes a mounting bracket **71**, a first magnetic actuator **72**, and a second magnetic actuator **73**.

[0352] The mounting bracket **71** includes a movable first mounting arm **711** and a movable second mounting arm **712**. The first magnetic actuator **72** is mounted to the first mounting arm **711**, and is arranged to drive the magnetic implant assembly **511** of one endoscope assembly **500** of the two endoscope assemblies **500** to move in the human body by the action of a magnetic field. The second magnetic actuator **73** is mounted to the second mounting arm **712**, and is arranged to drive the magnetic implant assembly **511** of another endoscope assembly **500** of the two endoscope assemblies **500** to move in the human body by the action of a magnetic field. The first magnetic actuator **72** and the second magnetic actuator **73** are disposed to be movable to a state in which they overlap in the vertical direction (as shown in the state (g3) in FIG. **19C**) so as to move the magnetic implant assemblies **511** of the two endoscopes **51** to a state where they overlap in the vertical direction in the human body (as shown in FIG. **16D**). In an exemplary embodiment, the second magnetic actuator **73** is disposed such that the magnetic field strength generated by the second magnetic actuator is greater than the magnetic field strength generated by the first magnetic actuator **72**, and the first magnetic actuator **72** is located below the second magnetic actuator **73** when the first magnetic actuator **72** and the second magnetic actuator **73** overlap in the vertical direction.

[0353] The magnetic navigation console **700** is a dual magnetic actuator system having a first magnetic actuator **72** and a second magnetic actuator **73**, the first magnetic actuator **72** and the

second magnetic actuator **73** are supported by a first mounting arm **711** and a second mounting arm **712**, respectively, and the first mounting arm **711** and the second mounting arm **712** are movable so that the first magnetic actuator **72** and the second magnetic actuator **73** are movable, thereby driving the magnetic implant assemblies **511** of the two endoscopes **51** to move within the human body. The first magnetic actuator **72** and the second magnetic actuator **73** can move to a state where they vertically overlap, which may remotely bring the two magnetic implant assemblies **511** within the human body close enough for mating.

[0354] Therefore, when the first magnetic actuator **72** and the second magnetic actuator **73** overlap in the vertical direction, the first magnetic actuator **72** is located below the second magnetic actuator **73**, so that the first magnetic actuator **72** is closer to the magnetic implant assembly **511** driven by the first magnetic actuator, and the second magnetic actuator **73** is away from the magnetic implant assembly **511** driven by the first magnetic actuator. However, since the magnetic field strength generated by the second magnetic actuator **73** is greater than that generated by the first magnetic actuator **72**, the first magnetic actuator **72** can still drive the corresponding magnetic implant assembly **511** to move by action of magnetic field, and the second magnetic actuator **73** can still drive the corresponding magnetic implant assembly **511** to move by action of magnetic field.

[0355] The second magnetic actuator **73** can be sized larger than the first magnetic actuator **72** such that the magnetic field strength generated by the second magnetic actuator **73** is greater than the magnetic field strength generated by the first magnetic actuator **72**. Of course, the magnetic field strength generated by the second magnetic actuator **73** may be made greater than the magnetic field strength generated by the first magnetic actuator **72** not only by the size, but also by material selection, structural setting, and the like.

[0356] In some exemplary embodiments, as shown in FIG. **19E**, the first magnetic actuator **72** includes a permanent magnet **721**, and the permanent magnet **721** of the first magnetic actuator **72** includes a conical section **7211** having a lower end cross-sectional dimension smaller than an upper end cross-sectional dimension.

[0357] The permanent magnet **721** of the first magnetic actuator **72** includes a conical section **7211**, and the provision of the conical section **7211** can be used to establish a magnetic flux concentration (as shown in the rectangular frame B in FIG. **19F**), which helps the first magnetic actuator **72** to drive the corresponding magnetic implant assembly **511** to move within the human body, thereby facilitating the two magnetic implant assemblies **511** in the human body to be sufficiently close to each other to cooperate.

[0358] In some exemplary embodiments, as shown in FIG. **19E**, the permanent magnet **721** of the first magnetic actuator **72** further includes a cylindrical section **7212** extends upward from the upper end of the conical section **7211**, and the cross-sectional dimension of the cylindrical section **7212** is equal to the cross-sectional dimension of the upper end of the conical section **7211**. The permanent magnet **721** of the first magnetic actuator **72** further includes a cylindrical section **7212** extends upward from the upper end of the conical section **7211**, and the arrangement of the cylindrical section **7212** enables the first magnetic actuator **72** to have a larger dimension as a whole, thereby enabling the first magnetic actuator **72** to generate a greater magnetic field strength. As can be seen from FIG. **19F**, compared with the magnetic field generated by the cylindrical section **7212** (as shown in the rectangular frame A in FIG. **19F**), the magnetic field generated by the conical section **7211** can achieve magnetic flux concentration (as shown in the rectangular frame B in FIG. **19F**).

[0359] In some exemplary embodiments, as shown in FIG. **19E**, the diameter D1 of the bottom surface of the conical section **7211** is 20 mm-60 mm, the diameter of the upper end of conical section **7211** is equal to the diameter D2 of cylindrical section **7212** and ranges from 60 mm-100 mm, and the sum H2 of the height of the conical section **7211** and the height H1 of the cylindrical section **7212** is 10 mm-50 mm. For example, in one example, the diameter D1 of the bottom

surface of the conical section **7211** is 40 mm, the diameter of the upper end surface of the conical section **7211** and the diameter **D2** of the cylindrical section **7212** are both 80 mm, the sum **H2** of the heights of the conical section **7211** and the cylindrical section **7212** is 30 mm, the height of the conical section **7211** may be 20 mm, and the height **H1** of the cylindrical section **7212** may be 10 mm.

[0360] It should be understood that the size of the permanent magnet **721** of the first magnetic actuator **72** is not limited to the above range, and may be adjusted according to actual needs.

[0361] In some exemplary embodiments, the second magnetic actuator **73** includes a permanent magnet; alternatively, the second magnetic actuator **73** includes an electromagnetic coil; alternatively, the second magnetic actuator **73** includes a permanent magnet and an electromagnetic coil disposed above or below the permanent magnet.

[0362] The second magnetic actuator **73** may drive the magnetic implant assembly **511** to move within the human body by a magnetic field generated by a permanent magnet and/or an electromagnetic coil.

[0363] In some exemplary embodiments, as shown in FIGS. **19A** to **19B**, the first mounting arm **711** is longitudinally slidably mounted to the mounting bracket **71** and hinged with the mounting bracket **71**, and the first mounting arm **711** includes a plurality of connecting arms **7111** hinged sequentially; the second mounting arm **712** is longitudinally slidably mounted to the mounting bracket **71** and hinged with the mounting bracket **71**. The second mounting arm **712** includes a plurality of connecting arms **7121** hinged sequentially.

[0364] The first mounting arm **711** and the second mounting arm **712** are slidably mounted to the mounting bracket **71** in the longitudinal direction (vertical direction) so that the first magnetic actuator **72** and the second magnetic actuator **73** mounted to the first mounting arm **711** and the second mounting arm **712** can move up and down in the vertical direction; the first mounting arm **711** includes a plurality of connecting arms **7111** hinged sequentially, the plurality of connecting arms **7111** of the first mounting arm **711** are rotatable along a longitudinal axis such that the first magnetic actuator **72** mounted to the first mounting arm **711** is movable in a horizontal plane; the second mounting arm **712** includes a plurality of connecting arms **7121** hinged sequentially, the plurality of connecting arms **7121** of the second mounting arm **712** are rotatable along a longitudinal axis such that the second magnetic actuator **73** mounted to the second mounting arm **712** is movable in a horizontal plane.

[0365] The structure of the plurality of connecting arms **7121** of the first mounting arm **711** and the plurality of connecting arms **7121** of the second mounting arm **712** is configured such that the first magnetic actuator **72** and the second magnetic actuator **73** are movable in a three-dimensional space so that the first magnetic actuator **72** and the second magnetic actuator **73** move to a vertically overlapping state.

[0366] In some exemplary embodiments, since the first magnetic actuator **72** is located below the second magnetic actuator **73** when the first magnetic actuator **72** and the second magnetic actuator **73** overlap in the vertical direction, the anti-friction coating **722** (as shown in FIG. **19C**) may be provided at the upper end of the first magnetic actuator **72**, and/or the anti-friction coating may be provided at the lower end of the second magnetic actuator **73**.

[0367] By providing the anti-friction coating **722** at the upper end of the first magnetic actuator **72** and/or providing the anti-friction coating at the lower end of the second magnetic actuator **73**, the friction force between the first magnetic actuator **72** and the second magnetic actuator **73** can be reduced during the movement of the first magnetic actuator **72** and the second magnetic actuator **73** to a vertically overlapping state or during the separation from the vertically overlapping state.

[0368] In some exemplary embodiments, the thickness of the anti-friction coating **722** at the upper end of the first magnetic actuator **72** may be 5 mm-20 mm, and the material of the anti-friction coating may be Teflon (polytetrafluoroethylene)/acetal; and/or the thickness of the anti-friction coating at the lower end of the second magnetic actuator **73** may be 5 mm to 20 mm, and the

material of the anti-friction coating may be Teflon (polytetrafluoroethylene)/acetal.

[0369] It should be understood that the thickness and material of the anti-friction coating **722** at the upper end of the first magnetic actuator **72** and the thickness and material of the anti-friction coating **722** at the lower end of the second magnetic actuator **73** are not limited to the above ranges, and may be adjusted according to actual needs.

[0370] In some exemplary embodiments, as shown in FIGS. **19A-19D**, the magnetic navigation console **700** further includes a push rod mechanism **74**, the push rod mechanism **74** is configured to extend and retract and capable of pushing one of the first magnetic actuator **72** and the second magnetic actuator **73** to move so that the first magnetic actuator **72** and the second magnetic actuator **73** are away from each other. In an exemplary embodiment, the push rod mechanism **74** may be mounted to the first mounting arm **711** and may urge the second magnetic actuator **73** to move. It should be understood that the push rod mechanism **74** may also be mounted to the second mounting arm **712** and may urge the first magnetic actuator **72** in motion.

[0371] During the movement of the first magnetic actuator **72** and the second magnetic actuator **73** to the vertically overlapping state (as moved from the state (g1) in FIG. **19C** through the state (g2) and to the state (g3)), the push rod mechanism **74** may be contracted (as changed from the state (h1) to the state (h2) in FIG. **19D**) so as not to hinder the movement of the first magnetic actuator **72** and the second magnetic actuator **73** to the vertically overlapping state; during separation of the first magnetic actuator **72** and the second magnetic actuator **73** from the vertically overlapping state (as moved from the state (g3) in FIG. **19C** through the state (g2) and to the state (g1)), the push rod mechanism **74** may be extended (as changed from the state (h2) to the state (h1) in FIG. **19D**) to push one of the first magnetic actuator **72** and the second magnetic actuator **73** so that the first magnetic actuator **72** and the second magnetic actuator **73** are away from each other.

[0372] In some exemplary embodiments, as shown in FIG. **19D**, the push rod mechanism **74** includes a second linear actuator **741** and a push rod **742**, the second linear actuator **741** is mounted to an end of the first mounting arm **711** close to the first magnetic actuator **72**, and the push rod **742** is connected to a drive end of the second linear actuator **741** and configured to extend and retract under the driving of the second linear actuator **741**.

[0373] The drive end of the second linear actuator **741** is linearly telescopically movable and can drive the push rod **742** to extend and retract so that during the movement of the first magnetic actuator **72** and the second magnetic actuator **73** to the vertically overlapping state, the drive end of the second linear actuator **741** can drive the push rod **742** to retract (as shown by the state (h2) in FIG. **19D**) to avoid the second magnetic actuator **73**; during the separation of the first magnetic actuator **72** and the second magnetic actuator **73** from the vertically overlapping state, the drive end of the second linear actuator **741** may drive the extension of the push rod **742** (as shown by state (h1) in FIG. **19D**) to push one of the first magnetic actuator **72** and the second magnetic actuator **73** so that the first magnetic actuator **72** and the second magnetic actuator **73** are away from each other.

[0374] The magnetic navigation console **700** according to an embodiment of the present application is an innovative motorized dual magnetic actuator system for remotely navigating magnetic implant assemblies **511** (e.g., two magnetic implant assemblies **511**) within a human body. The dual magnetic actuator system primarily includes a small auxiliary magnetic actuator (a first magnetic actuator **72**) and a large main magnetic actuator (a second magnetic actuator **73**) which are respectively supported by a first mounting arm **711** and a second mounting arm **712** (e.g., two dual planar articulating arms), allowing vertical overlapping of the first magnetic actuator **72** and the second magnetic actuator **73** which can remotely bring the two magnetic implant assemblies **511** close enough for mating. The dual magnetic actuator system also includes a motorized pusher (a push rod mechanism **74**) for pushing apart the first magnetic actuator **72** and the second magnetic actuator **73**. The permanent magnet **721** of the first magnetic actuator **72** includes a conical section **7211** for establishing a magnetic flux concentration to help to mate the

two magnetic implant assemblies **511**. The second magnetic actuator **73** may include a permanent magnet and an electromagnetic coil, and the electromagnetic force of the second magnetic actuator **73** may be output continuously and in pulse mode (100V, 60 A). The magnetic navigation console **700** can be designed in a compact single cart design for enhancing mobility and ease of use. [0375] In summary, as shown in FIG. **11**, the endoscopic magnetic anastomosis system according to an embodiment of the present application may include two endoscope assemblies **500**, two video processors **56** and an external pressure source **553** used in conjunction with the two endoscope assemblies **500**, and may also include a magnet detector **600** and a magnetic navigation console **700**.

[0376] An embodiment of the present application also provides an adjustable snare mechanism, as shown in FIGS. **12A** to **12D**, including a snare assembly **52** and a snare guide assembly **53**, wherein the snare assembly **52** includes a snare catheter **521** and a snare wire **522** passing through the snare catheter **521**; the snare guide assembly **53** includes a base body **531**, a first movable member **532** and a second movable member **533**, the first movable member **532** and the second movable member **533** are both movably mounted on the base body **531**, and the first movable member **532** is connected with the snare wire **522** and arranged to drive the snare wire **522** to move; the second movable member **533** is connected to the snare catheter **521** and is arranged to drive the snare catheter **521** to move; the snare wire **522** is disposed to be movable relative to the snare catheter **521** upon movement of at least one of the first movable member **532** and the second movable member **533** to cause the snare wire **522** to be retracted into the snare catheter **521** and extended out of the snare catheter **521**.

[0377] The adjustable snare mechanism according to an embodiment of the present application may be the adjustable snare mechanism of the endoscopic magnetic anastomosis system described above, and may have some or all of the features of the adjustable snare mechanism described above.

[0378] An embodiment of the present application also provides an outer sheath **55**, as shown in FIGS. **13A** to **13I**, including a body tube having a first end and an opposite second end, and a tube-locking mechanism **552** mounted at the first end of the body tube. The body tube has an endoscope channel **5519** extending from the first end of the body tube to the second end of the body tube for passage of the endoscope **51**. The tube-locking mechanism **552** is configured to be capable of being in locking fit with the endoscope **51** passing through the endoscope channel **5519** to fix the endoscope **51** to the body tube, and to disengage from being in locking fit with the endoscope **51** passing through the endoscope channel **5519** to enable the endoscope **51** to slide and rotate relative to the body tube.

[0379] The outer sheath **55** according to an embodiment of the present application may be the outer sheath **55** of the endoscopic magnetic anastomosis system described above, and may have some or all of the characteristics of the outer sheath **55** described above.

[0380] An embodiment of the present application also provides a plug-in connection structure, as shown in FIGS. **15A** to **15I**, including a main connector socket **561** and a main connector **571**, the main connector **571** is in plug-in connection with the main connector **571**, and a locking structure is provided between the main connector socket **561** and the main connector **571** to lock the main connector socket **561** and the main connector **571** in plug-in connection.

[0381] The main connector socket **561** includes a base **5611** and a rotatable locking ring **5612**, and the locking ring **5612** is rotatably mounted to the base **5611** to enable the main connector socket **561** to be switched between an initial state and a locked state.

[0382] One of the locking ring **5612** and the main connector **571** is provided with a positioning key **5711**, the other of the locking ring **5612** and the main connector **571** is provided with a locking slide slot **5613** extending in the circumferential direction, one end of the locking slide slot **5613** is an insertion end **5614**, and the positioning key **5711** is configured to be inserted into the locking slide slot **5613** from the insertion end **5614** when the main connector **571** is in plug-in connection

with the main connector socket **561** in the initial state, and is configured to slide in the locking slide slot **5613** during the rotation of the locking ring **5612**, so that the positioning key **5711** and the insertion end **5614** of the locking slide slot **5613** are misaligned and the main connector socket **561** switches to the locking state.

[0383] The plug-in connection structure according to an embodiment of the present application may be a plug-in connection structure between the video processor **56** and the cable **57** of the endoscopic magnetic anastomosis system described above, and may have some or all of the features of the outer sheath **55** described above. Of course, the plug-in connection structure can also be used for plug-in connection between the main connector **571** and the main connector socket **561** of other devices.

[0384] An embodiment of the present application also provides a magnet detector **600**, as shown in FIGS. **17A** to **17F**, the magnet detector **600** is configured to detect the position of the magnet **5117**, and the magnet detector **600** includes a detector body **61**, a first printed circuit board **62**, a second printed circuit board **63**, and a processing unit. The first printed circuit board **62** is provided at the first end of the detector body **61**, and a first group of magnetometers **621** is provided thereon. The second printed circuit board **63** is provided at a second end of the detector body **61** opposite to the first end, and a second group of magnetometers **631** is provided thereon. The processing unit is arranged to be able to receive measurement data from the first group of magnetometers **621** and the second group of magnetometers **631** and determine the position of the magnet **5117** based on the data.

[0385] The magnet detector **600** according to an embodiment of the present application may be the magnet detector **600** of the endoscopic magnetic anastomosis system described above, and may have some or all of the characteristics of the magnet detector **600** described above. Of course, the magnet detector **600** may also be used to drive the location of other magnets other than the magnetic implant assembly **511** of the endoscopic magnetic anastomosis system.

[0386] An embodiment of the present application also provides another magnet detector **600**, as shown in FIGS. **18A** and **18B**, the magnet detector **600** is arranged to be able to detect the position of the magnet **5117**, and the magnet detector **600** includes a mounting substrate **65**, at least one sensor module **66**, and a processing module **67**. Each sensor module **66** includes eight magnetic sensors **661**, the eight magnetic sensors **661** are regularly distributed in four rows with two magnetic sensors **661** per row. The two magnetic sensors **661** in each row are in a misaligned arrangement with the two magnetic sensors **661** in an adjacent row. The processing module **67** is arranged to receive data from the sensor module **66** and data of the IMU **5118** embedded in the magnet **5117**, and determine the position of the magnet **5117** based on the data.

[0387] The magnet detector **600** according to an embodiment of the present application may be the magnet detector **600** of the endoscopic magnetic anastomosis system described above, and may have some or all of the characteristics of the magnet detector **600** described above. Of course, the magnet detector **600** may also be used to drive the location of other magnets other than the magnetic implant assembly **511** of the endoscopic magnetic anastomosis system.

[0388] An embodiment of the present application also provides a magnetic navigation console **700**, as shown in FIGS. **19A** to **19F**, the magnetic navigation console **700** is arranged to drive the magnet **5117** to move by the action of a magnetic field, the magnetic navigation console **700** includes a mounting bracket **71**, a first magnetic actuator **72**, and a second magnetic actuator **73**, and the mounting bracket **71** includes a first mounting arm **711** and a second mounting arm **712** which are movable; the first magnetic actuator **72** is mounted to the first mounting arm **711**; the second magnetic actuator **73** is mounted to the second mounting arm **712**, and the first magnetic actuator **72** and the second magnetic actuator **73** are disposed to be movable to a state where they overlap in the vertical direction.

[0389] The magnetic navigation console **700** according to an embodiment of the present application may be the magnetic navigation console **700** of the endoscopic magnetic anastomosis

system described above, and may have some or all of the features of the magnetic navigation console **700** described above. Of course, the magnetic navigation console **700** may also be used to drive the movement of other magnets other than the magnetic implant assembly **511** of the endoscopic magnetic anastomosis system.

[0390] Various embodiments are described herein, but the description is exemplary and not restrictive, and it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible within the scope of the embodiments described herein. Although many possible combinations of features are shown in the drawings and discussed in the detailed description, many other combinations of the disclosed features are also possible. Unless specifically limited, any feature or element of any embodiment may be used in combination with, or may be substituted for, any other feature or element of any other embodiment.

[0391] The present application includes and contemplates combinations with features and elements known to those of ordinary skill in the art. The embodiments, features, and elements already disclosed herein may also be combined with any conventional features or elements to form the unique inventive schemes defined by the claims. Any feature or element of any embodiment may also be combined with features or elements from other inventive schemes to form another unique inventive schemes defined by the claims. Accordingly, it should be understood that any of the features shown and/or discussed in this application may be implemented alone or in any suitable combination. Accordingly, the embodiments are not subject to any other restrictions other than those in accordance with the appended claims and their equivalent substitutions. Furthermore, various modifications and changes can be made within the scope of protection of the appended claims.

[0392] Furthermore, when describing representative embodiments, the specification may have presented a method and/or process as a particular sequence of steps. However, to the extent that the method or process does not depend on the specific order of steps described herein, the method or process should not be limited to the specific order of steps described. As will be understood by those of ordinary skills in the art, other order of steps is also possible. Accordingly, a particular order of steps set forth in the specification should not be construed as limitations on the claims. Furthermore, the claims for the method and/or process should not be limited to the steps which are performed in the written order. Those skilled in the art can readily understand that these orders can be changed and the changed orders still remain within the spirit and scope of the embodiments of the present application.

[0393] While various embodiments in accordance with the disclosed principles have been described above, it should be understood that they have been presented by way of example only, and are not limiting. Thus, the breadth and scope of the example embodiments described in the present disclosure should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the claims and their equivalents issuing from this disclosure. Furthermore, the above advantages and features are provided in described embodiments, but shall not limit the application of such issued claims to processes and structures accomplishing any or all of the above advantages.

[0394] For example, “assembly,” “device,” “portion,” “segment,” “member,” “body,” or other similar terms should generally be construed broadly to include one part or more than one part attached or connected together.

[0395] Various terms used herein have special meanings within the present technical field. Whether a particular term should be construed as such a “term of art” depends on the context in which that term is used. “Connected,” “connecting,” “attached,” “attaching,” “anchored,” “anchoring,” “in communication with,” “communicating with,” “associated with,” “associating with,” or other similar terms should generally be construed broadly to include situations where attachments, connections, and anchoring are direct between referenced elements or through one or more intermediaries between the referenced elements. These and other terms are to be construed in light

of the context in which they are used in the present disclosure and as one of ordinary skill in the art would understand those terms in the disclosed context. The above definitions are not exclusive of other meanings that might be imparted to those terms based on the disclosed context.

[0396] As referred to in the present disclosure, a computing device, controller, manipulator, master input device, a processor, and/or a system may be a virtual machine, computer, node, instance, host, and/or device in a networked or non-networked computing environment. A networked computing environment may be a collection of devices connected by communication channels that facilitate communications between devices and allow devices to share resources. Also as referred to in the present disclosure, a computing device may be a device deployed to execute a program operating as a socket listener and may include software instances.

[0397] Resources may encompass any type of resource for running instances including hardware (such as servers, clients, mainframe computers, networks, network storage, data sources, memory, central processing unit time, scientific instruments, and other computing devices), as well as software, software licenses, available network services, and other non-hardware resources, or a combination thereof.

[0398] A networked computing environment may include, but is not limited to, computing grid systems, distributed computing environments, cloud computing environment, etc. Such networked computing environments include hardware and software infrastructures configured to form a virtual organization comprised of multiple resources that may be in geographically disperse locations.

[0399] Furthermore, the coverage of the present application and any patents issuing from the present application may extend to one or more communications protocols, including TCP/IP.

[0400] Words of comparison, measurement, and timing such as “at the time,” “equivalent,” “during,” “complete,” and the like should be understood to mean “substantially at the time,” “substantially equivalent,” “substantially during,” “substantially complete,” etc., where “substantially” means that such comparisons, measurements, and timings are practicable to accomplish the implicitly or expressly stated desired result.

[0401] Additionally, the section headings herein are provided for consistency with the suggestions under 37 C.F.R. 1.77 or otherwise to provide organizational cues. These headings shall not limit or characterize the invention(s) set out in any claims that may issue from this disclosure. Specifically, a description of a technology in the “Background” is not to be construed as an admission that technology is prior art to any invention(s) in this disclosure. Furthermore, any reference in this disclosure to “invention” in the singular should not be used to argue that there is only a single point of novelty in this disclosure. Multiple inventions may be set forth according to the limitations of the multiple claims issuing from this disclosure, and such claims accordingly define the invention(s), and their equivalents, that are protected thereby. In all instances, the scope of such claims shall be considered on their own merits in light of this disclosure, but should not be constrained by the headings herein.

Claims

1. An endoscopic magnetic anastomosis system, comprising at least one endoscopic assembly comprising: an endoscope having a first end and an opposite second end, the endoscope comprising a snare channel extending from the first end to the second end and a magnetic implant assembly disposed at the second end; and an adjustable snare mechanism, comprising a snare assembly passing through the snare channel, and comprising a snare catheter and a snare wire passing through the snare catheter for selectively tightening and releasing the magnetic implant assembly.
2. The endoscopic magnetic anastomosis system according to claim 1, wherein the adjustable snare mechanism further comprises a snare guide assembly that comprises a base body and a motion mechanism movably installed on the base body, and wherein the motion mechanism is connected to and is configured to drive at least one of the snare wire and the snare catheter, such that there is a

relative movement between the snare wire and the snare catheter to allow the snare wire to selectively extend and retract relative to the snare catheter.

3. The endoscopic magnetic anastomosis system according to claim 2, wherein the motion mechanism comprises a first movable member and a second movable member, wherein the first movable member is connected to the snare wire and configured to be able to drive the snare wire to move and the second movable member is connected to the snare catheter and configured to be able to drive the snare catheter to move, such that there is the relative movement between the snare wire and the snare catheter when at least one of the first movable member and the second movable member moves.

4. The endoscopic magnetic anastomosis system according to claim 3, wherein the base body of the snare guide assembly is provided with a first slide rail and a second slide rail extending along a length direction of the base body, the first movable member is configured to be able to slide along the first slide rail, and the second movable member is configured to be able to slide along the second slide rail.

5. The endoscopic magnetic anastomosis system according to claim 3, wherein the snare guide assembly further comprises a releasable locking mechanism mounted to the first movable member, the releasable locking mechanism is configured to be able to be in locking fit with the base body to secure the first movable member to the base body and to also be able to disengage from being in locking fit with the base body to enable the first movable member to move relative to the base body; and/or wherein the snare guide assembly further comprises a unlockable locking device mounted to the second movable member, the unlockable locking device is configured to be able to be in locking fit with the base body to secure the second movable member to the base body, and also be able to disengage from being in locking fit with the base body to enable the second movable member to move relative to the base body; or wherein the second movable member is in friction fit with the base body such that the second movable member can remain fixed relative to the base body under the action of the frictional force between the second movable member and the base body, and also can overcome the friction force to move relative to the base body under action of an external force.

6. The endoscopic magnetic anastomosis system according to claim 5, wherein the releasable locking mechanism comprises a rotary locking mechanism comprising a knob and a clamping member, wherein the knob is connected to the clamping member, and wherein the knob is configured to be rotatable in two opposite directions to drive the clamping member to clamp and release the base body accordingly.

7. The endoscopic magnetic anastomosis system according to claim 1, wherein the endoscopic assembly further comprises: an outer sheath comprising a body tube having a first end and an opposite second end and a tube-locking mechanism mounted at the first end of the body tube, the body tube further has an endoscopic channel extending from the first end of the body tube to the second end of the body tube for passage of the endoscope, the tube-locking mechanism is configured to be able to be in locking fit with the endoscope to secure the endoscope with the body tube and also be able to disengage from being in locking fit with the endoscope to enable the endoscope to slide and rotate relative to the body tube.

8. The endoscopic magnetic anastomosis system according to claim 7, wherein the tube-locking mechanism comprises: a locking seat mounted at the first end of the body tube; a rotatable member rotatably mounted to the locking seat and provided with a spiral driving part, wherein a spiral center line of the spiral driving part coincides with a rotation center line of the rotatable member; and a plurality of locking blocks movably mounted to the locking seat, and configured to be in driving fit with the spiral driving part so as to be translatable in a radial direction of the spiral driving part as the spiral driving part rotates, end portions of the plurality of locking blocks adjacent to each other collectively defining an opening for the endoscope to pass through; wherein the plurality of locking blocks are configured such that when the rotatable member rotates in a first

direction, the plurality of locking blocks translate radially inwardly toward the spiral center line of the spiral driving part to narrow the opening to clamp the endoscope, and when the rotatable member rotates in a second direction opposite to the first direction, the plurality of locking blocks translate radially outwardly away from the spiral center line of the spiral driving part to enlarge the opening to release the endoscope.

9. The endoscopic magnetic anastomosis system according to claim 7, wherein the outer sheath further comprises: a first seal disposed at the first end of the body tube to form a seal between the body tube and an outer surface of the endoscope; and a second seal disposed at the second end of the body tube to form a seal between the body tube and the outer surface of the endoscope; wherein a gap between the inner surface of the body tube and the outer surface of the endoscope forms a suction channel, the body tube has one or more suction openings provided at the second end of the body tube in a circumferential direction of the body tube, and a suction connector provided at the first end of the body tube, the suction openings and the suction connector are located between the first seal and the second seal, two ends of the suction channel are in communication with the suction openings and the suction connector, respectively, and the suction connector is configured to be connected with a suction apparatus for providing a negative pressure.

10. The endoscopic magnetic anastomosis system according to claim 9, wherein the second seal is disposed proximate the suction openings and the first seal is disposed proximate the suction connector; and wherein the first seal has a first sealing rib in seal fit with the outer surface of the endoscope; one end of the second seal away from the first seal is provided with a conical part, a tapered end of the conical part away from the first seal being in seal fit with the outer surface of the endoscope; and the second seal is further provided with a second sealing rib closer to the first seal than the conical part, the second sealing rib being in seal fit with the outer surface of the endoscope.

11. The endoscopic magnetic anastomosis system according to claim 7, wherein the outer sheath further comprises: an expandable member sleeved on an outer surface of the body tube at the second end of the body tube, the body tube is provided with a gas channel extending between an inner surface and the outer surface of the body tube in an axial direction of the body tube, the gas channel is in communication with the expandable member at the second end of the body tube, and is in communication with an external pressure source at the first end of the body tube, so that the expandable member expands outwardly in a radial direction of the body tube from a non-expanded state to an expanded state or retracts from the expanded state to the non-expanded state in response to action of the external pressure source.

12. The endoscopic magnetic anastomosis system according to claim 11, wherein the endoscopic assembly further comprises: the external pressure source configured to be able to provide a fixed volume of gas to the expandable member to expand the expandable member outwardly in a radial direction of the body tube.

13. The endoscopic magnetic anastomosis system according to claim 12, wherein the external pressure source comprises: a housing; a first linear actuator mounted within the housing, the first linear actuator having a drive end configured to be movable between a first position and a second position; a syringe mounted within the housing, the syringe comprising a syringe barrel and a syringe plunger movably mounted in the syringe barrel, the syringe barrel has an injection port in communication with the gas channel, the drive end of the first linear actuator is connected with the syringe plunger and configured to be able to drive the syringe plunger to reciprocate relative to the syringe barrel to respectively supply gas to or draw gas from the expandable member through the injection port; and two position sensors mounted within the housing, configured to detect a position of the drive end of the first linear actuator.

14. The endoscopic magnetic anastomosis system according to claim 1, wherein the endoscope further comprises a head assembly disposed at the second end of the endoscope, the head assembly comprises a head assembly body, one end surface of the head assembly body is provided with a

support part and a limit part which are disposed opposite to each other, for clamping the magnetic implant assembly between the limit part and the support part, a bottom portion of the magnetic implant assembly is supported on the support part, and a top portion of the magnetic implant assembly abuts against the limit part.

15. The endoscopic magnetic anastomosis system according to claim 1, wherein the endoscopic assembly further comprises an image capturing assembly disposed at the second end of the endoscope, wherein the endoscopic magnetic anastomosis system further comprises a video processor electrically connected to the image capturing assembly via a cable, and a main connector provided at one end of the cable for electrically connected to the video processor, and wherein the video processor has a main connector socket for plug-in connection with the main connector, and a locking structure is provided between the main connector socket and the main connector to lock the main connector socket and the main connector which are in plug-in connection.

16. The endoscopic magnetic anastomosis system according to claim 15, wherein the main connector socket comprises a base and a rotatable locking ring, the locking ring rotatably mounted to the base to enable the main connector socket to be switched between an initial state and a locked state; and wherein the locking structure comprises a positioning key that is provided on one of the locking ring and the main connector, and a locking slide slot that is provided in the other of the locking ring and the main connector and extends in a circumferential direction, one end of the locking slide slot is an insertion end, and the positioning key is configured to be able to be inserted into the locking slide slot from the insertion end when the main connector is in plug-in connection with the main connector socket in the initial state, and is arranged to be able to slide relative to the locking slide slot during the rotation of the locking ring, so that the positioning key and the insertion end of the locking slide slot are misaligned and thus the main connector socket switches to the locking state.

17. The endoscopic magnetic anastomosis system according to claim 1, wherein the at least one endoscopic assembly comprises two endoscopic assemblies, the magnetic implant assemblies of the two endoscopic assemblies are respectively a first magnetic assembly configured to be located within a first luminal tissue region and a second magnetic implant assembly configured to be located within a second luminal tissue region; wherein the first magnetic implant assembly comprises a first enclosure and a first magnet disposed within the first enclosure, the first enclosure being provided with a first fixing ring groove on a circumferential side surface of the first enclosure to receive the snare wire of the snare assembly of one of the two endoscopic assemblies, and the first enclosure having a first engagement surface; wherein the second magnetic implant assembly comprises a second enclosure and a second magnet disposed within the second enclosure, the second enclosure being provided with a second fixing ring groove on a circumferential side surface of the second enclosure to receive the snare wire of the snare assembly of the other of the two endoscopic assemblies, and the second enclosure having a second engagement surface; wherein the first engagement surface and the second engagement surface are configured, when the first magnetic implant assembly and the second magnetic implant assembly are magnetically anastomosed with the first and second luminal issue regions interposed therebetween, to face each other and be capable of exerting a non-uniform compressive force on the first luminal tissue region and the second luminal tissue region; and wherein the first and second magnets are both solid disc-shaped permanent magnet such that when the first magnetic implant assembly and the second magnetic implant assembly are magnetically anastomosed, the center line of the first magnetic implant assembly and the center line of the second magnetic implant assembly can be automatically adjusted to an aligned state even when there is a certain deviation between them.

18. The endoscopic magnetic anastomosis system according to claim 17, wherein the first engagement surface is a concave curved surface with a first radius of curvature, and the second engagement surface is a convex curved surface with a second radius of curvature, the radius of curvature of the convex surface being smaller than that of the concave surface; and wherein the

convex curved surface and the concave curved surface are configured such that when the first magnetic implant assembly and the second magnetic implant assembly are magnetically anastomosed, the convex curved surface protrudes towards the concave curved surface, and the axial distance between the convex curved surface and the concave curved surface along the central line direction of the first magnetic implant assembly increases as the radial distance from the central line of the first magnetic implant assembly increases, so as to exert the non-uniform compressive force on the first luminal tissue region and the second luminal tissue region.

19. The endoscopic magnetic anastomosis system according to claim 1, further comprising: a magnet detector configured to be able to detect a position of the magnetic implant assembly of the endoscope within a human body.

20. The endoscopic magnetic anastomosis system according to claim 19, wherein the magnet detector comprises: a detector body; a first printed circuit board provided on a first end of the detector body and provided with a first group of magnetometers; a second printed circuit board provided on a second end of the detector body opposite to the first end of the detector body, and provided with a second group of magnetometers; and a processing unit configured to receive measurement data from the first group of magnetometers and the second group of magnetometers and to determine a position of the magnetic implant assembly of the endoscope in a human body based on the received data.

21. The endoscopic magnetic anastomosis system according to claim 20, wherein, the first group of magnetometers comprises four magnetometers distributed in a square array, and a center of the square array coincides with a center of the first printed circuit board; the second group of magnetometers comprises four magnetometers distributed in a square array, and a center of the square array coincides with a center of the second printed circuit board.

22. The endoscopic magnetic anastomosis system according to claim 19, wherein an IMU is embedded within the magnetic implant assembly of the endoscope, and wherein the magnet detector comprises: a mounting substrate; at least one sensor module, each sensor module comprises eight magnetic sensors, the eight magnetic sensors are regularly distributed in four rows and each row has two magnetic sensors, the two magnetic sensors are a first magnetic sensor and a second magnetic sensor, respectively, and two magnetic sensors in each row are in a misaligned arrangement with two magnetic sensors in an adjacent row; and a processing module configured to be able to receive data from the sensor module and data of the IMU embedded within the magnetic implant assembly, and to determine a position of the magnetic implant assembly of the endoscope within a human body based on the received data.

23. The endoscopic magnetic anastomosis system according to claim 22, wherein the number of the at least one sensor module is between 1-32.

24. The endoscopic magnetic anastomosis system according to claim 1, further comprising: a magnetic navigation console configured to move the magnetic implant assembly of the endoscope to a set position within a human body under action of a magnetic field.

25. The endoscopic magnetic anastomosis system according to claim 24, wherein the at least one endoscopic assembly comprises two endoscopic assemblies, and the magnetic navigation console comprises: a mounting bracket comprising a movable first mounting arm and a movable second mounting arm; a first magnetic actuator mounted to the first mounting arm and configured to enable the magnetic implant assembly of one of the two endoscope assemblies to move within a human body under action of a magnetic field; and a second magnetic actuator mounted to the second mounting arm and configured to enable the magnetic implant assembly of the other of the two endoscope assemblies to move within a human body under action of a magnetic field, wherein the first magnetic actuator and the second magnetic actuator are configured to be able to move to a state where they overlap in a vertical direction so that magnetic implant assemblies of two endoscopes are able to move to a state where they overlap in a vertical direction within a human body.

- 26.** The endoscopic magnetic anastomosis system according to claim 25, wherein the first magnetic actuator comprises a permanent magnet, and the permanent magnet of the first magnetic actuator comprises a conical section, and wherein a lower end cross-sectional dimension of the conical section is smaller than an upper end cross-sectional dimension of the conical section.
- 27.** The endoscopic magnetic anastomosis system according to claim 26, wherein, the permanent magnet of the first magnetic actuator further comprises a cylindrical section extending upward from the upper end of the conical section, and a cross-sectional dimension of the cylindrical section is equal to the upper end cross-sectional dimension of the conical section.
- 28.** The endoscopic magnetic anastomosis system according to claim 25, wherein, the second magnetic actuator comprises a permanent magnet; or the second magnetic actuator comprises an electromagnetic coil; or the second magnetic actuator comprises a permanent magnet and an electromagnetic coil disposed above or below the permanent magnet.
- 29.** The endoscopic magnetic anastomosis system according to claim 25, wherein the first mounting arm is longitudinally slidably mounted to and hinged with the mounting bracket, the first mounting arm comprising a plurality of connecting arms hinged sequentially; and/or the second mounting arm is longitudinally slidably mounted to the mounting bracket and hinged with the mounting bracket, the second mounting arm comprising a plurality of connecting arms hinged sequentially; and/or wherein the second magnetic actuator is arranged so that an intensity of a magnetic field generated by the second magnetic actuator is greater than an intensity of a magnetic field generated by the first magnetic actuator, and the first magnetic actuator is positioned below the second magnetic actuator when the first magnetic actuator and the second magnetic actuator overlap in a vertical direction; and wherein at least one of an upper end of the first magnetic actuator and a lower end of the second magnetic actuator is provided with an anti-friction coating.
- 30.** The endoscopic magnetic anastomosis system according to claim 25, wherein the magnetic navigation console further comprises: a push rod mechanism configured to be able to extend and retract, and capable of pushing one of the first magnetic actuator and the second magnetic actuator to move the first magnetic actuator and the second magnetic actuator away from each other.
- 31.** The endoscopic magnetic anastomosis system according to claim 1, wherein the at least one endoscopic assembly comprises two endoscope assemblies that are identical or different in structure.
- 32.** An outer sheath comprising: a body tube having a first end and an opposite second end; and a tube-locking mechanism mounted at the first end of the body tube, wherein the body tube further has an endoscopic channel extending from the first end of the body tube to the second end of the body tube for passage of an endoscope, the tube-locking mechanism is configured to be able to be in locking fit with the endoscope passing through the endoscopic channel to secure the endoscope with the body tube, and to disengage from being in locking fit with the endoscope passing through the endoscopic channel to enable the endoscope to slide and rotate relative to the body tube.
- 33.** A magnet detector for detecting a position of a magnet, the magnet detector comprising: a detector body; a first printed circuit board provided on a first end of the detector body and provided with a first group of magnetometers; a second printed circuit board provided on a second end of the detector body opposite to the first end, and provided with a second group of magnetometers; and a processing unit configured to be able to receive measurement data from the first group of magnetometers and the second group of magnetometers and to determine a position of the magnet based on the received data.
- 34.** A magnetic navigation console for driving a magnet to move under action of a magnetic field, the magnetic navigation console comprising: a mounting bracket comprising a movable first mounting arm and a movable second mounting arm; a first magnetic actuator mounted to the first mounting arm; and a second magnetic actuator mounted to the second mounting arm, the first magnetic actuator and the second magnetic actuator are configured to be able to move to overlap in a vertical direction.

